

# Conditional moment conditions to reproduce statistical inference from synthetic data

Ruijie Li

## Abstract

Synthetic data are typically generated to match the joint distribution of the original dataset. That match does not reproduce the statistical inference an analyst draws from it. Generators are trained and evaluated for distributional fidelity, which constrains neither the coefficients and standard errors a generalised linear model returns from the release nor the inferences drawn from them. Preserving those quantities is inferential fidelity, and explanatory models require it. Inferential fidelity requires conditional moment conditions, matching the moments of the response given the predictors on a support-respecting scale. For a canonical link, the first conditional moment suffices. For a non-canonical link, the bias decays factorially in the number of matched moments, an order we show cannot be reduced. No divergence criterion can substitute, since a rare subgroup can carry a large coefficient at negligible distributional weight. Our main result is constructive. A sufficient statistics projection reproduces every coefficient, standard error, and  $p$ -value of a canonical-link model, and of all its submodels at once, from only  $p + 1$  constraints, an exact release for pre-registered analyses such as clinical trials and regulatory submissions. A public-covariate variant extends the coefficient exactness to a second analyst who fits a different family, and the projection composes with a  $\mu$ -Gaussian differentially private release. We further obtain the empirical Bayes shrinkage rate for sparse strata, and confirm the theory across eleven simulation scenarios.

## 1 Introduction

Synthetic data generators based on generative adversarial networks (GANs), variational autoencoders (VAEs), and related deep learning architectures are designed and evaluated for *predictive* tasks, in which the synthetic data should yield similar predictions to the original when used to train a supervised learning model. This objective, which we call *distributional fidelity*, amounts to matching the joint distribution  $F(\mathbf{X}, Y)$  under some divergence measure.

Explanatory modelling differs from prediction [47]. In an explanatory model, the quantities of interest are not predictions  $\hat{Y}$  but estimated parameters  $\hat{\beta}$ , their standard errors, significance levels, and confidence intervals. A synthetic dataset  $\{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^n$  is useful for explanatory purposes only if statistical analyses performed on it yield the same inferential conclusions as those performed on the original data  $\{(\mathbf{X}_i, Y_i)\}_{i=1}^n$ . We call this property *inferential fidelity*.

Distributional fidelity does not imply inferential fidelity. Regression coefficients are functionals of the *conditional* distribution  $F(Y | \mathbf{X})$ , not of the joint or marginal distributions, and a generator that

matches marginal moments to any finite order can still fail to preserve the conditional structure that drives regression estimates. This failure is structural, not architecture-specific.

We formalise the distinction between distributional and inferential fidelity for the class of generalised linear models [32]. We derive conditions on the synthetic data generating mechanism under which the quasi-likelihood estimator [53] applied to synthetic data is consistent for the same parameter as the estimator applied to the original data. Our main results are:

- (i) For canonical link functions, preserving the covariate distribution and the first conditional moment of the response itself, with no support transform, is sufficient for consistency (Theorem 4). This covers the logit link for logistic regression and the log link for Poisson regression.
- (ii) For non-canonical links with a support-respecting transformation  $h$ , matching conditional moments of  $h(Y)$  given  $\mathbf{X}$  to order  $k$  yields a bias of order  $\nu_{k+1}/(k+1)!$  (Theorem 6). Here  $\nu_{k+1}$  is the larger of the two laws'  $(k+1)$ -th conditional absolute central moments of  $h(Y)$  given  $\mathbf{X}$ . The moment order cannot be reduced, since matching  $k-1$  moments can leave non-vanishing bias (Theorem 10).
- (iii) No distributional divergence criterion, including those optimised by GANs and VAEs, guarantees inferential fidelity (Proposition 13).
- (iv) When the analyst's model is unknown to the data custodian, matching the first two within-stratum moments of all variables on a support-respecting scale bounds the bias for every choice of response at once by the conditional skewness (Theorem 15). That bias vanishes when every transform is the identity, and inferential fidelity then holds in every direction.
- (v) When the analyst's model is known and uses a canonical link, a sufficient statistics projection achieves *exact* inferential fidelity across all  $2^p - 1$  submodels from  $p+1$  constraints (Theorem 17).

The moment order in (ii) cannot be reduced. The tightness result (Theorem 10) shows that matching  $k-1$  moments can leave non-vanishing bias whenever the inverse support transform has a non-zero  $k$ -th derivative. The direction-agnostic extension (iv) relies on the elliptical conditional structure of the Gaussian copula. Departures from ellipticity introduce an additive error bounded by the conditional-mean deviation from the elliptical formulas, whose tail-driven part scales with the tail dependence coefficient (Section ). These results characterise the explanatory use case. For purely predictive tasks, distributional fidelity remains appropriate (Remark 23). A simulation study across eleven scenarios and two real datasets confirms the hierarchy the theory predicts (Section 6), and a  $\mu$ -Gaussian differentially private variant reconciles the projection with formal privacy (Section 5.4).

## Related work

The idea of synthetic data for disclosure limitation originates with Rubin [45], who proposed generating synthetic datasets from the posterior predictive distribution. Raghunathan, Reiter and Rubin [42] developed the framework for fully synthetic data with valid inference via multiple imputation combining rules. Partially synthetic data was proposed by Little [31], with combining rules derived by Reiter [43]. Valid combining-rule inference requires the synthesis model to be *congenial*

to the analyst’s model [34]. Reiter [44] introduced synthesis based on classification and regression trees. Nowok, Raab and Dibben [38] provide the R implementation `synthpop`. The closely related multiple imputation by chained equations framework [7] shares the same conditional specification approach. These methods provide valid inference by construction but are tied to specific parametric or semiparametric synthesis models.

Machine learning generators for tabular data, including Conditional Tabular GAN (CTGAN) and Tabular VAE (TVAE) [57], optimise distributional divergence measures and provide no congeniality guarantee. Wilde et al. [56] show that Bayesian inference from synthetic data is valid only if the generator equals the true data-generating process. Ghalebikesabi et al. [22] show that differentially private synthetic data, generated for example by differentially private stochastic gradient descent (DP-SGD) [2] or PATE-GAN [27], introduce systematic bias in downstream estimators. For linear regression this bias takes the form of the classical errors-in-variables attenuation [20, Section 1.1.1].

Empirical work since 2023 confirms that adversarial and variational latent-space generators produce inflated type-I error rates when used for naive inference, because synthetic standard errors ignore the generative uncertainty [13]. A reanalysis of 128 treatment comparisons from 115 phase 3 oncology trials under sixteen generators reports naive synthetic analyses declaring falsely significant treatment effects in up to half of the trials generated by deep architectures, with the excess variability rising from parametric through tree-based to deep generators [11]. Denoising diffusion generators for tabular data [30] share the same distributional training objective and fall under the same no-guarantee result. Debiasing corrections based on targeted learning improve coverage but do not guarantee it [12].

The copula representation of joint distributions dates to Sklar [49], with a textbook treatment in Nelsen [35, Section 2.3]. Semiparametric copula estimation was developed for continuous margins by Genest, Ghoudi and Rivest [21] and extended to mixed continuous and discrete data by Hoff [25]. The Cornish–Fisher expansion for quantile approximation via cumulants dates to Cornish and Fisher [10]. See Hall [24] for the connection to Edgeworth expansions. Our contribution is to connect these tools to the quasi-likelihood framework [32, 53] and derive the conditional moment conditions that bridge synthetic data generation and downstream inference.

The theoretical results of Sections 3–5 are generator-agnostic, stating conditions on the synthetic distribution itself rather than on the mechanism that produces it. The failure of distributional objectives is structural rather than a deficiency of any particular architecture, since the quasi-likelihood estimating equations depend on conditional moments that distributional objectives do not control. Section 5 connects the resulting sparse-strata problem to the James–Stein simultaneous estimation paradigm, showing that the optimal bias–variance trade-off for stratum-specific conditional moments follows from the classical shrinkage theory of James and Stein [26] and Efron and Morris [18]. The copula enters in two roles: as a constructive proof that the moment-matching conditions are achievable, and as a generator that is competitive in practice and that, when combined with sufficient statistics projection (Section 5.3), provides exact inferential fidelity. The simulation study (Section 6) compares this implementation against four machine learning generators, including a tuned variant to confirm that hyperparameter optimisation does not resolve the structural limitation.

## 2 Framework

This section fixes the objects the paper studies: the quasi-likelihood estimator whose behaviour on synthetic data is at issue, the two fidelity notions the paper contrasts, and the support transforms on which all moment conditions are stated.

### 2.1 Data and model

Let  $(\mathbf{X}_i, Y_i)$ ,  $i = 1, \dots, n$ , be independent observations from a distribution  $P$  on  $\mathcal{X} \times \mathcal{Y}$ , where  $\mathbf{X}_i = (\mathbf{X}_{i,\text{cat}}, \mathbf{X}_{i,\text{cnt}}) \in \mathcal{X}$  is a  $p$ -dimensional covariate vector comprising categorical and continuous components, and  $Y_i \in \mathcal{Y} \subseteq \mathbb{R}$  is the response. We consider the quasi-likelihood framework, in which the analyst specifies a link function  $g$ , a variance function  $V$ , and a linear predictor  $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ , where  $\mathbf{x}_i$  is the model matrix row for observation  $i$ . The conditional mean is  $\mu_i = g^{-1}(\eta_i)$ .

The quasi-likelihood estimator  $\hat{\boldsymbol{\beta}}_n$  solves the estimating equation

$$U_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \frac{\partial \mu_i}{\partial \eta_i} \frac{1}{V(\mu_i)} (Y_i - \mu_i) \mathbf{x}_i = \mathbf{0}, \quad (1)$$

where  $\partial \mu_i / \partial \eta_i = 1/g'(\mu_i)$ . Under standard regularity conditions [33, 36, 50],  $\hat{\boldsymbol{\beta}}_n$  is consistent for the parameter  $\boldsymbol{\beta}_0$  satisfying  $E[U_1(\boldsymbol{\beta}_0)] = \mathbf{0}$  and is asymptotically normal:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, A^{-1}BA^{-1}),$$

where  $A = E[-\partial U_1 / \partial \boldsymbol{\beta}^\top]$  and  $B = E[U_1 U_1^\top]$  are the sensitivity and variability matrices evaluated at  $\boldsymbol{\beta}_0$ .

### 2.2 Synthetic data

A synthetic data generator produces observations  $(\mathbf{X}_i^*, Y_i^*)$ ,  $i = 1, \dots, n$ , from a distribution  $P^*$  constructed from the original sample. The *synthetic data estimator*  $\hat{\boldsymbol{\beta}}_n^*$  solves

$$U_n^*(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \frac{\partial \mu_i^*}{\partial \eta_i^*} \frac{1}{V(\mu_i^*)} (Y_i^* - \mu_i^*) \mathbf{x}_i^* = \mathbf{0}, \quad (2)$$

where the same link  $g$ , variance function  $V$ , and model matrix construction are applied to the synthetic data.

**Definition 1** (Distributional fidelity). *A synthetic distribution  $P^*$  has distributional fidelity if the empirical joint distribution  $\hat{F}_n^*$  of the synthetic sample converges weakly to the data-generating distribution  $F(\mathbf{X}, Y)$  as  $n \rightarrow \infty$ . For  $k \in \{1, \dots, p+1\}$ ,  $P^*$  has  $k$ -wise distributional fidelity if weak convergence holds for every  $k$ -dimensional marginal of  $\hat{F}_n^*$ .*

**Definition 2** (Inferential fidelity). *A synthetic distribution  $P^*$  has inferential fidelity for a generalised linear model  $(g, V)$  if  $\hat{\boldsymbol{\beta}}_n^* \xrightarrow{P} \boldsymbol{\beta}_0$  as  $n \rightarrow \infty$ . It has exact inferential fidelity if, additionally,  $\sqrt{n}(\hat{\boldsymbol{\beta}}_n^* - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, A^{-1}BA^{-1})$ .*

**Remark 1** (Fidelity lattice). *Definitions 1 and 2 are endpoints of a lattice of fidelity targets indexed by the class of functionals  $P^*$  preserves.  $k$ -wise distributional fidelity preserves functionals of at most  $k$  coordinates. Inferential fidelity preserves the functionals that determine  $\hat{\beta}_n$  under  $(g, V)$ , the conditional moments of the response entering the expected score Equation (2). For a canonical link the score depends on the data only through the sufficient statistics. Machine learning generators are trained and evaluated against the distributional side (Section 4). The sufficient statistics projection (Section 5.3) attains inferential fidelity exactly without controlling  $\hat{F}_n^*$  beyond the  $p + 1$  projection constraints.*

## 2.3 Support-respecting transformations

When the response  $Y$  has restricted support  $\mathcal{Y} \subsetneq \mathbb{R}$ , generating synthetic values on  $\mathbb{R}$  and mapping back can violate the support constraint. We address this by working on a transformed scale.

**Definition 3** (Support transform). *A support transform for  $\mathcal{Y}$  is a measurable injection  $h : \mathcal{Y} \rightarrow \mathbb{R}$  together with a measurable map  $h^{-1} : \mathbb{R} \rightarrow \mathcal{Y}$  satisfying  $h^{-1}\{h(y)\} = y$  for every  $y \in \mathcal{Y}$ . The canonical choices are:*

| $\mathcal{Y}$  | $h$                               | $h^{-1}$                            |
|----------------|-----------------------------------|-------------------------------------|
| $\mathbb{R}$   | identity                          | identity                            |
| $(0, \infty)$  | log                               | exp                                 |
| $(0, 1)$       | logit                             | expit                               |
| $\mathbb{N}_0$ | $y \mapsto \log(y + \frac{1}{2})$ | $z \mapsto \lfloor \exp(z) \rfloor$ |

*On the three continuous supports  $h$  is a bijection onto  $\mathbb{R}$  and  $h^{-1}$  is its two-sided inverse.*

## 3 Main results

We begin with a regularity condition that applies throughout this section.

**Assumption 1.** *The parameter space  $\Theta$  is compact,  $\beta_0$  is in the interior of  $\Theta$ , the model matrix  $\mathbf{x}$  has bounded entries, and  $\mathbb{E}[\|U_1(\beta)\|^2]$  is uniformly bounded on  $\Theta$ . The functions  $g^{-1}$ ,  $V$ , and their derivatives are continuous on the range of  $\eta = \mathbf{x}^\top \beta$ , with  $V(\mu) > 0$  for all  $\mu$  in the range of  $g^{-1}$ . The same conditions are required to hold under  $P^*$ .*

If the synthetic distribution preserves the covariate distribution and the full conditional  $F(Y | \mathbf{X})$ , then  $P^* = P$  and the standard  $M$ -estimator theory [36] applies directly. That case is immediate, and the substantive question is what happens when only partial information about  $F(Y | \mathbf{X})$  is preserved. We address this in two stages: canonical links (Theorem 4), where a single conditional moment suffices, and non-canonical links (Theorem 6), where the bias depends on the unmatched moment order. Four results then complete the characterisation: the same moment conditions control the full regression summary (Proposition 8), the mean squared error decomposes into truncation bias and sampling variance (Theorem 9), the moment order in Theorem 6 cannot be reduced (Theorem 10), and marginal moment matching to any order fails (Theorem 12).

### 3.1 Canonical link functions

**Theorem 4** (Canonical links). *Suppose Assumption 1 holds and  $g$  is the canonical link, so that  $V(\mu) \cdot g'(\mu) = 1$ . If the synthetic distribution satisfies*

$$(i) \ P^*(\mathbf{X} \in \cdot) = P(\mathbf{X} \in \cdot), \text{ and}$$

$$(ii) \ E^*[Y \mid \mathbf{X} = \mathbf{x}] = E[Y \mid \mathbf{X} = \mathbf{x}] \text{ for } P\text{-almost every } \mathbf{x},$$

*then  $E^*[U_1^*(\beta_0)] = \mathbf{0}$  and  $\hat{\beta}_n^* \xrightarrow{P} \beta_0$ .*

*Proof.* For the canonical link,  $\partial\mu/\partial\eta = V(\mu)$ , so the score (1) simplifies to

$$U_1(\beta) = (Y - \mu) \mathbf{x}.$$

Taking expectations under  $P^*$  at  $\beta = \beta_0$  gives

$$E^*[U_1^*(\beta_0)] = E^*[(Y^* - \mu_0(\mathbf{X}^*)) \mathbf{X}^*] = E^*[E^*[Y^* - \mu_0(\mathbf{X}^*) \mid \mathbf{X}^*] \mathbf{X}^*].$$

By condition (ii),  $E^*[Y^* \mid \mathbf{X}^* = \mathbf{x}] = E[Y \mid \mathbf{X} = \mathbf{x}] = \mu_0(\mathbf{x})$ , so the inner expectation vanishes. Hence  $E^*[U_1^*(\beta_0)] = \mathbf{0}$ , and consistency follows from the standard argument for unbiased estimating equations under Assumption 1.  $\square$

**Remark 2** (First moment suffices for canonical links). *Matching the single conditional moment  $E[Y \mid \mathbf{X}]$  suffices for consistency because the canonical link makes the score linear in  $Y$ .*

### 3.2 Non-canonical links: partial moment matching

When the link is not canonical, the score involves nonlinear functions of  $Y$ , and matching  $E[Y \mid \mathbf{X}]$  alone does not guarantee  $E^*[U_1^*(\beta_0)] = \mathbf{0}$ . We now quantify the bias introduced by matching only finitely many conditional moments.

Let  $h : \mathcal{Y} \rightarrow \mathbb{R}$  be a support transform (Definition 3), and define  $Z = h(Y)$ . Write  $\kappa_j(\mathbf{x}) = E[(Z - E[Z \mid \mathbf{X} = \mathbf{x}])^j \mid \mathbf{X} = \mathbf{x}]$  for the  $j$ -th conditional central moment of  $Z$  given  $\mathbf{X} = \mathbf{x}$ .

**Definition 5** (Moment-matched synthetic distribution). *The synthetic distribution  $P^*$  matches conditional moments of  $h(Y)$  to order  $k$  if, for  $P$ -almost every  $\mathbf{x}$ :*

$$(i) \ P^*(\mathbf{X} \in \cdot) = P(\mathbf{X} \in \cdot),$$

$$(ii) \ E^*[Z^{*j} \mid \mathbf{X}^* = \mathbf{x}] = E[Z^j \mid \mathbf{X} = \mathbf{x}] \text{ for } j = 1, \dots, k,$$

where  $Z^* = h(Y^*)$ .

**Assumption 2.** *There exists  $K > 0$  such that  $E[|Z|^{k+1} \mid \mathbf{X} = \mathbf{x}] \leq K$  for  $P$ -almost every  $\mathbf{x}$ , and the moment generating function  $M(\mathbf{x}, t) = E[e^{tZ} \mid \mathbf{X} = \mathbf{x}]$  exists in a neighbourhood of  $t = 0$  for  $P$ -almost every  $\mathbf{x}$ . The same bounds are required to hold under  $P^*$ .*

**Theorem 6** (Partial moment matching). *Suppose Assumptions 1 and 2 hold and  $P^*$  matches conditional moments of  $h(Y)$  to order  $k \geq 2$ . Define the score bias*

$$b_k(\beta_0) = E^*[U_1^*(\beta_0)] - E[U_1(\beta_0)].$$

Then

$$\|b_k(\beta_0)\| \leq C \cdot E\left[\frac{\nu_{k+1}(\mathbf{X})}{(k+1)!} \cdot w(\mathbf{X}, \beta_0)\right], \quad (3)$$

where  $C$  depends on  $h, g, V$ , and the model matrix,  $\nu_{k+1}(\mathbf{x})$  is the larger of the  $(k+1)$ -th conditional absolute central moments of the transformed response under the two laws,

$$\nu_{k+1}(\mathbf{x}) = \max(E[|Z - E(Z | \mathbf{X} = \mathbf{x})|^{k+1} | \mathbf{X} = \mathbf{x}], E^*[|Z^* - E^*(Z^* | \mathbf{X}^* = \mathbf{x})|^{k+1} | \mathbf{X}^* = \mathbf{x}]),$$

and

$$w(\mathbf{x}, \beta) = \left\| \frac{1}{g'(\mu) V(\mu)} \mathbf{x} \right\|$$

is the score weight, evaluated at  $\mu = g^{-1}(\mathbf{x}^\top \beta)$ . For inverse transforms whose  $(k+1)$ th derivative is unbounded, the derivative factor remains inside the expectation, as in Corollary 7(ii) for the exponential transform, with the boundary analysis in Section .

Consequently,  $\hat{\beta}_n^* = \beta_0 + A^{-1}b_k(\beta_0) + o_p(1)$ , so the synthetic data estimator is consistent up to a bias controlled by the unmatched conditional moments.

*Proof.* The score evaluated at  $\beta_0$  involves  $E^*[Y^* | \mathbf{X}^* = \mathbf{x}]$  through the term

$$E^*\left[\frac{Y^* - \mu_0(\mathbf{x})}{V(\mu_0(\mathbf{x}))g'(\mu_0(\mathbf{x}))} \mid \mathbf{X}^* = \mathbf{x}\right].$$

Since  $Y = h^{-1}(Z)$ , we have

$$E[Y | \mathbf{X} = \mathbf{x}] = E[h^{-1}(Z) | \mathbf{X} = \mathbf{x}]. \quad (4)$$

We expand  $h^{-1}$  around  $E[Z | \mathbf{X} = \mathbf{x}]$ . Write  $\mu_Z(\mathbf{x}) = E[Z | \mathbf{X} = \mathbf{x}]$  and  $\tilde{Z} = Z - \mu_Z(\mathbf{x})$ . Then

$$h^{-1}(Z) = h^{-1}(\mu_Z + \tilde{Z}) = \sum_{j=0}^k \frac{(h^{-1})^{(j)}(\mu_Z)}{j!} \tilde{Z}^j + R_k(\tilde{Z}),$$

where  $R_k(\tilde{Z}) = \frac{(h^{-1})^{(k+1)}(\xi)}{(k+1)!} \tilde{Z}^{k+1}$  for some  $\xi$  between  $\mu_Z$  and  $Z$ , by Taylor's theorem.

Taking conditional expectations:

$$E[Y | \mathbf{x}] = \sum_{j=0}^k \frac{(h^{-1})^{(j)}(\mu_Z(\mathbf{x}))}{j!} \kappa_j(\mathbf{x}) + E[R_k(\tilde{Z}) | \mathbf{x}],$$

where  $\kappa_0 = 1$ ,  $\kappa_1 = 0$  (centred), and  $\kappa_j(\mathbf{x}) = E[\tilde{Z}^j | \mathbf{x}]$  for  $j \geq 2$ .

Since  $P^*$  matches moments to order  $k$ , the sum is identical under  $P$  and  $P^*$ . The discrepancy is

$$E^*[Y^* | \mathbf{x}] - E[Y | \mathbf{x}] = E^*[R_k^* | \mathbf{x}] - E[R_k | \mathbf{x}].$$

By the Lagrange remainder and Assumption 2:

$$|E^*[R_k^* | \mathbf{x}] - E[R_k | \mathbf{x}]| \leq \frac{\sup_{\xi} |(h^{-1})^{(k+1)}(\xi)|}{(k+1)!} (E^*[|\tilde{Z}^*|^{k+1} | \mathbf{x}] + E[|\tilde{Z}|^{k+1} | \mathbf{x}]).$$

When  $(h^{-1})^{(k+1)}$  is bounded the supremum is finite and the display stands as written. For the exponential transform the factor remains inside the conditional expectation, where the moment generating function condition of Assumption 2 controls it (Section ). Each of the two terms is at most  $\nu_{k+1}(\mathbf{x})$ , the larger of the two conditional absolute central moments, and the resulting factor of two is absorbed into  $C$ . The bound (3) follows by substituting into the score expectation and applying the triangle inequality.

For consistency, the bias enters through the expansion  $\hat{\beta}_n^* = \beta_0 + A^{-1}n^{-1} \sum_i U_i^*(\beta_0) + o_p(n^{-1/2})$ , giving  $\hat{\beta}_n^* \rightarrow \beta_0 + A^{-1}b_k(\beta_0)$  in probability.  $\square$

**Corollary 7** (Sufficient orders for common families). *Under the conditions of Theorem 6, with the canonical support transform  $h$  from Definition 3:*

- (i) *Gaussian with identity link:  $h = \text{id}$ ,  $h^{-1} = \text{id}$ , all derivatives of order  $\geq 2$  vanish. The bias is exactly zero for  $k \geq 1$ .*
- (ii) *Gamma or inverse Gaussian with log link:  $h = \log$ ,  $h^{-1} = \exp$ ,  $(h^{-1})^{(j)} = \exp$ . The bias bound is  $\|b_k\| \leq C E[e^{\mu z(\mathbf{X})} \nu_{k+1}(\mathbf{X})]/(k+1)!$ .*
- (iii) *Poisson with log link (canonical): by Theorem 4,  $k = 1$  on the natural scale suffices, regardless of  $h$ .*
- (iv) *Binomial with logit link: the link is canonical, so  $k = 1$  suffices by Theorem 4. In practice, binary  $Y$  is handled via latent normal thresholds, which preserve the first conditional moment exactly.*
- (v) *Binomial with non-canonical link (e.g., probit, log):  $h = \text{logit}$ ,  $h^{-1} = \text{expit}$ . Since  $\text{expit}''(z) = \text{expit}(z)(1 - \text{expit}(z))(1 - 2 \text{expit}(z))$ , the bias from the second-order term is bounded by  $1/8$  times  $|\kappa_2^* - \kappa_2|$ , so  $k = 2$  on the logit scale provides strong control.*

The factorial decay of the bias bound in  $k$  (the  $1/(k+1)!$  factor) shows that even modest moment orders provide strong bias control. In practice,  $k = 2$  (matching conditional mean and variance on the  $h$ -scale) suffices because the conditional distribution of  $h(Y)$  given  $\mathbf{X}$  is approximately Gaussian by the central limit theorem when  $Y$  is an aggregate of small effects.

**Remark 3** (Multimodal conditional distributions). *The Cornish–Fisher (CF) quantile approximation assumes approximate unimodality of  $h(Y) | \mathbf{X}$ . Multimodality does not affect the bias bound of Theorem 6 or the projection (Theorem 17), but makes the copula generator unreliable. Stratification that resolves mixture components is needed before applying it.*

The MGF condition in Assumption 2 is mild, since it holds automatically for standard exponential families, since  $b(\theta) = \log E[e^{\theta Y}]$  is finite on the natural parameter space [32, Section 2.2.2], and the support transform lightens the tails of heavy-tailed responses with log-concave survival (Pareto  $Y$



gives exponential  $Z = \log Y$ ). A fractional moment relaxation of Assumption 2 and the boundary behaviour of the Taylor remainder are given in Section .

### 3.3 Sufficient conditions for regression summary statistics

Theorem 6 controls the point estimate  $\hat{\beta}_n^*$ . The practitioner also needs standard errors, test statistics, and significance levels. The same conditional moment conditions control all of these, so that  $k = 2$  on the  $h$ -scale is sufficient for the entire inferential output.

**Proposition 8** (Sufficient conditions for summary statistics). *Under Assumptions 1 and 2, let  $P^*$  match conditional moments of  $h(Y)$  to order  $k \geq 2$ . Let  $\hat{\phi}_n$  and  $\hat{\phi}_n^*$  denote the dispersion estimates from the original and synthetic data, and  $\widehat{SE}_n, \widehat{SE}_n^*$  the model-based standard errors. Then:*

- (a) (Coefficients.)  $\|\hat{\beta}_n^* - \beta_0\| \leq \|A^{-1}\| \cdot C \mathbb{E}[\nu_{k+1}(\mathbf{X}) w(\mathbf{X})]/(k+1)! + O_p(n^{-1/2})$ , by Theorem 6.
- (b) (Dispersion.) The Pearson dispersion estimate  $\hat{\phi}_n = (n-p)^{-1} \sum_i (Y_i - \hat{\mu}_i)^2 / V(\hat{\mu}_i)$  satisfies

$$|\hat{\phi}_n^* - \hat{\phi}_n| \leq D \cdot \frac{\mathbb{E}[\nu_{k+1}(\mathbf{X}) w^{(2)}(\mathbf{X})]}{(k+1)!} + O_p(n^{-1/2}),$$

where  $D$  depends on  $h, g, V$ , and  $w^{(2)}$  is the second-order score weight.

- (c) (Standard errors.) Since  $\widehat{SE}_n^* = \sqrt{\hat{\phi}_n^*} \sqrt{\text{diag}\{(\mathbf{X}^\top \hat{W}^* \mathbf{X})^{-1}\}}$ , parts (a) and (b) jointly yield  $|\widehat{SE}_n^* - \widehat{SE}_n| = O(\nu_{k+1}/(k+1)!) + O_p(n^{-1/2})$ .
- (d) (Test statistics and significance.) With  $t_j^* = \hat{\beta}_{n,j}^* / \widehat{SE}_{n,j}^*$ , parts (a)–(c) give

$$|t_j^* - t_j| = O\left(\frac{\nu_{k+1}}{(k+1)!}\right) + O_p(n^{-1/2}).$$

*Significance is preserved whenever the moment-truncation error is smaller than the original test statistic's margin from the critical value.*

*Proof.* Part (a) restates Theorem 6. For part (b), the dispersion estimate involves  $\mathbb{E}[(Y^* - \mu^*)^2 / V(\mu^*) \mid \mathbf{X} = \mathbf{x}]$ . Writing  $Y = h^{-1}(Z)$  and expanding  $h^{-1}$  to order  $k$ , the squared residual involves  $\mathbb{E}[(h^{-1}(Z) - \mu)^2 \mid \mathbf{x}]$ . Since  $P^*$  matches the first  $k$  moments of  $Z$  given  $\mathbf{x}$ , a Taylor expansion analogous to the proof of Theorem 6 yields the stated bound with a second-order weight. When  $h = \text{id}$  (Gaussian family), the second moment is matched exactly for  $k \geq 2$ , giving  $\hat{\phi}_n^* = \hat{\phi}_n + O_p(n^{-1/2})$ .

For part (c), the standard error is a smooth function of  $(\hat{\beta}_n^*, \hat{\phi}_n^*)$ . Since both converge at the stated rates, the delta method gives the combined rate. Part (d) follows because  $t_j^* = \hat{\beta}_j^* / \widehat{SE}_j^*$  is a ratio of the quantities in (a) and (c).  $\square$

**Remark 4** (Second order controls the full summary). *Proposition 8 implies that  $k = 2$  on the  $h$ -scale is the minimal order that controls the full regression summary, since  $k = 1$  preserves coefficients (for canonical links) but not standard errors, because the dispersion estimate requires the second conditional moment. For the Gaussian family with identity link, parts (a)–(d) are exact at  $k = 2$  (zero residual bias). For non-Gaussian families, the residual error decays as  $O(\nu_3/6)$ , with  $\nu_3$  the third conditional absolute*

central moment. When both conditional laws are approximately symmetric, the leading third-order terms cancel and the error falls to fourth order.

### 3.4 Convergence rate

Theorem 6 gives a bias bound but does not characterise the full convergence rate. The following theorem provides the complete mean squared error decomposition.

**Theorem 9** (Mean squared error). *Under Assumptions 1 and 2, if  $P^*$  matches conditional moments of  $h(Y)$  to order  $k \geq 2$  exactly, then:*

- (a)  $\hat{\beta}_n^* - \beta_0 = b_k + O_p(n^{-1/2})$ , where  $b_k$  solves  $E^*[U_1^*(\beta_0 + b_k)] = \mathbf{0}$ , equals  $A^{-1} E^*[U_1^*(\beta_0)]$  to first order, and satisfies the bound (3) up to the factor  $\|A^{-1}\|$ .
- (b)  $\sqrt{n}(\hat{\beta}_n^* - \beta_0 - b_k) \xrightarrow{d} N(\mathbf{0}, V^*)$ , where  $V^* = A^{*-1} B^* A^{*-1}$  with  $A^* = E^*[-\partial U_1^*/\partial \beta^\top]$  and  $B^* = E^*[U_1^* U_1^{*\top}]$ , both evaluated at  $\beta_0 + b_k$ .
- (c) The mean squared error admits the decomposition

$$E[\|\hat{\beta}_n^* - \beta_0\|^2] = \underbrace{\|b_k\|^2}_{\text{bias}^2} + \underbrace{\frac{1}{n} \text{tr}(V^*)}_{\text{variance}} + o(n^{-1}). \quad (5)$$

*Proof.* Part (a) follows from the standard linearisation of  $M$ -estimators [36]: writing  $\hat{\beta}_n^* = \beta_0 + b_k + A^{*-1} n^{-1} \sum_i \{U_i^*(\beta_0 + b_k) - E^*[U_1^*(\beta_0 + b_k)]\} + o_p(n^{-1/2})$ , and noting that  $E^*[U_1^*(\beta_0 + b_k)] = \mathbf{0}$  by definition of  $b_k$  and the continuity of the score. Part (b) is the central limit theorem applied to the centred sum. Part (c) follows from  $E[\|b_k + n^{-1/2} W_n\|^2] = \|b_k\|^2 + n^{-1} E[\|W_n\|^2] + 2n^{-1/2} b_k^\top E[W_n]$ , where  $E[W_n] \rightarrow \mathbf{0}$ .  $\square$

**Remark 5** (Bias dominance at fixed  $k$ ). *The bias<sup>2</sup> term  $\|b_k\|^2$  does not vanish with  $n$ . For the synthetic data estimator to achieve the parametric rate  $O(n^{-1})$ , the bias must satisfy  $\|b_k\| = o(n^{-1/2})$ . By (3), this holds whenever*

$$\frac{E[\nu_{k+1}(\mathbf{X}) w(\mathbf{X}, \beta_0)]}{(k+1)!} = o(n^{-1/2}).$$

*Since the left side depends on  $k$  but not  $n$ , this is satisfied for all  $n$  when  $k$  is large enough that the factorial dominates the central moment. By Stirling's approximation,  $(k+1)! \sim \sqrt{2\pi(k+1)} ((k+1)/e)^{k+1}$ , so the bias decays factorially in  $k$ . In practice,  $k = 2$  typically suffices, matching the conditional mean and variance on the  $h$ -scale makes  $\|b_2\|$  negligible relative to  $n^{-1/2}$  for all realistic sample sizes.*

### 3.5 Tightness of the moment order

We now show that the moment order  $k$  in Theorem 6 cannot be reduced. Whenever the inverse support transform is not affine, matching  $k - 1$  conditional moments does not guarantee that the bias vanishes.

**Theorem 10** (Tightness). *Let  $h$  be a support transform with  $h^{-1}$  not affine (i.e.,  $(h^{-1})^{(j)} \neq 0$  for some  $j \geq 2$ ). For any  $k \geq 2$  such that  $(h^{-1})^{(k)} \neq 0$ , there exist distributions  $P$  and  $P^*$  such that:*

- (i)  $E^*[h(Y^*)^j | \mathbf{X}^* = \mathbf{x}] = E[h(Y)^j | \mathbf{X} = \mathbf{x}]$  for  $j = 1, \dots, k-1$ ,
- (ii)  $E^*[h(Y^*)^k | \mathbf{X}^* = \mathbf{x}] \neq E[h(Y)^k | \mathbf{X} = \mathbf{x}]$  for some  $\mathbf{x}$ ,
- (iii)  $\|b_{k-1}\| > 0$ .

*Proof.* Let  $Z = h(Y)$  and choose  $P$  such that  $Z | \mathbf{X} = \mathbf{x} \sim N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$  with  $\mu(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_0$ . Since  $Z$  is Gaussian, all cumulants of order  $\geq 3$  are zero:  $\kappa_j(\mathbf{x}) = 0$  for  $j \geq 3$ .

Now construct  $P^*$ : let  $Z^* | \mathbf{X}^* = \mathbf{x}$  have the same first  $k-1$  moments as  $Z | \mathbf{X} = \mathbf{x}$ , but with a  $k$ -th cumulant differing from that of  $Z | \mathbf{X} = \mathbf{x}$  by  $\kappa_k^*(\mathbf{x}) - \kappa_k(\mathbf{x}) = \delta(\mathbf{x}) \neq 0$ , where  $\kappa_k(\mathbf{x}) = 0$  for  $k \geq 3$  and  $\kappa_2(\mathbf{x}) = \sigma^2(\mathbf{x})$  under the Gaussian  $P$ . Add a scaled bounded perturbation to the Gaussian density:  $f^*(z | \mathbf{x}) \propto \phi_{\mu, \sigma}(z) \{1 + \varepsilon g_k((z - \mu)/\sigma)\}$  for small  $\varepsilon > 0$ , where  $g_k$  is bounded, orthogonal to  $1, u, \dots, u^{k-1}$  under the standard Gaussian weight, and satisfies  $E\{U^k g_k(U)\} \neq 0$ . Such a  $g_k$  exists for every  $k \geq 2$ , for example a truncated Hermite polynomial  $He_k$  re-orthogonalised against lower powers. Boundedness keeps  $f^*$  a density for small  $\varepsilon$ . An untruncated  $He_k$  is unbounded, so  $1 + \varepsilon He_k$  is negative somewhere for every  $\varepsilon > 0$  when  $k$  is odd. The perturbation preserves all moments of order  $< k$  while shifting the  $k$ -th cumulant by  $\delta(\mathbf{x})$  of order  $O(\varepsilon)$ .

The same perturbation also moves the cumulants of order above  $k$ , each by  $O(\varepsilon)$ , so they are not a second-order correction. Taking  $g_k$  with the parity of  $k$  places the next non-zero difference at order  $k+2$ , and with  $\sigma(\mathbf{x}) \equiv \sigma$  the  $j$ -th cumulant difference enters the score expansion with a factor  $\sigma^j$ , so the terms above order  $k$  are  $O(\sigma^2)$  relative to the leading one, with bounded derivative ratios on a compact covariate range. The bias is

$$b_{k-1} = A^{-1} E \left[ \frac{(h^{-1})^{(k)}(\mu_Z(\mathbf{X}))}{k!} \delta(\mathbf{X}) w(\mathbf{X}) \mathbf{X} \right] \{1 + O(\sigma^2)\}.$$

Choose  $\delta(\mathbf{x})$  to be a non-zero constant and  $\sigma$  small, and concentrate the covariate law where  $(h^{-1})^{(k)}(\mu_Z(\mathbf{x}))$  is non-zero of one sign. The displayed term then dominates, so  $\|b_{k-1}\| > 0$ .

Concretely, for  $h = \log$  and  $k = 2$ , take  $g_2 = He_2$ , which is bounded below, so no truncation is needed, and the Hermite generating identity  $E\{e^{\sigma U} He_2(U)\} = \sigma^2 e^{\sigma^2/2}$  evaluates the perturbed mean of  $h^{-1}(Z)$  exactly. A perturbation with  $\kappa_2^*(\mathbf{x}) \neq \kappa_2(\mathbf{x}) = \sigma^2(\mathbf{x})$  gives, exactly and with no remainder,

$$b_1 = A^{-1} E \left[ \frac{e^{\mu(\mathbf{X}) + \sigma^2(\mathbf{X})/2}}{2} (\kappa_2^*(\mathbf{X}) - \sigma^2(\mathbf{X})) w(\mathbf{X}) \mathbf{X} \right] \neq \mathbf{0},$$

the expectation vector having a strictly positive intercept coordinate when  $\kappa_2^* > \kappa_2$ , and  $A$  being invertible. Hence matching only the first conditional moment ( $k = 1$ ) is insufficient for the log transform, since the second conditional moment (variance) is required.  $\square$

**Corollary 11.** *For the Gamma family with log link ( $h = \log$ ):*

- (i) *Matching  $k = 1$  conditional moment of  $\log Y$  is insufficient, since the conditional variance of  $\log Y$  contributes bias of order  $|\kappa_2^* - \kappa_2|/2$ .*

- (ii) Matching  $k = 2$  conditional moments of  $\log Y$  (mean and variance) is sufficient up to third-moment effects, with bias  $O(v_3/6)$ .
- (iii) If  $\log Y \mid \mathbf{X}$  is Gaussian, so that all cumulants above the second vanish, then matching  $k = 2$  determines the conditional law and gives exact consistency. Cumulants that are negligible rather than vanishing leave the bias of (ii), small but not zero.

### 3.6 Insufficiency of marginal moment matching

The preceding results show that *conditional* moments drive inferential fidelity. We now prove the converse, that matching *marginal* moments of  $Y$  to any finite order does not guarantee consistency, even when the marginal fit is arbitrarily accurate.

**Theorem 12** (Insufficiency of marginal moments). *For any  $k \geq 1$ , there exist distributions  $P$  and  $P^*$  such that:*

- (i)  $P^*(\mathbf{X} \in \cdot) = P(\mathbf{X} \in \cdot)$  (covariate distribution preserved),
- (ii)  $E^*[Y^j] = E[Y^j]$  for  $j = 1, \dots, k$  (marginal moments of  $Y$  matched to order  $k$ ),
- (iii) but  $\hat{\beta}_n^* \xrightarrow{P} \beta_0 + \delta$  for some  $\delta \neq 0$ .

*Proof.* We construct an explicit counterexample. Let  $X \in \{0, 1\}$  with  $P(X = 1) = \pi$  and  $Y \mid X \sim N(\mu_X, \sigma^2)$  with  $\mu_0 \neq \mu_1$ . The regression coefficient is  $\beta = \mu_1 - \mu_0$ .

Consider the marginal distribution of  $Y$ :  $F_Y = \pi \Phi((y - \mu_1)/\sigma) + (1 - \pi) \Phi((y - \mu_0)/\sigma)$ , a Gaussian mixture. Let  $P^*$  be any distribution that preserves  $P(X)$  and matches the first  $k$  marginal moments of  $Y$ , but generates  $Y^*$  independently of  $X^*$ :  $P^*(Y \mid X) = F_Y$  for both values of  $X$ .

Since  $Y^*$  is independent of  $X^*$  under  $P^*$ :

$$E^*[Y^* \mid X^* = 1] = E^*[Y^* \mid X^* = 0] = E^*[Y^*] = E[Y].$$

The synthetic data regression coefficient converges to  $\beta^* = E^*[Y \mid X = 1] - E^*[Y \mid X = 0] = 0 \neq \beta$ , regardless of how many marginal moments are matched.

Note that the joint cross-moments  $E^*[X^a Y^j]$  for  $a \geq 1$  are *not* matched:  $E^*[X^a Y^j] = \pi E[Y^j]$  under  $P^*$ , while  $E[X^a Y^j] = \pi E[Y^j \mid X = 1]$  under  $P$ , which differ when  $\mu_1 \neq E[Y]$ . The theorem shows precisely that matching all pure marginal moments of  $Y$  to any finite order, while preserving  $P(\mathbf{X})$ , is insufficient for inferential fidelity when the conditional structure is destroyed.

The construction generalises. For any predictor configuration  $\mathbf{X}$ , one can choose  $P^*(Y \mid \mathbf{X}) = F_Y$  (the marginal) to match all marginal moments of  $Y$  while destroying the conditional structure.  $\square$

**Remark 6** (Marginal fit versus conditional structure). *Theorem 12 exposes a fundamental gap between distributional and inferential fidelity. A synthetic data generator that optimises marginal fit (e.g., by minimising the Wasserstein distance between  $F_Y$  and  $F_{Y^*}$ ) can achieve perfect marginal fidelity while*

having zero inferential fidelity. The conditional structure  $F(Y \mid \mathbf{X})$  is invisible to any criterion that evaluates  $F_Y$  alone.

## 4 Failure of distributional objectives for explanatory modelling

No training objective defined on the joint distribution guarantees inferential fidelity (Proposition 13), even when distributional fidelity is high. Inferential fidelity requires preserving conditional moments (Section 3), and the standard training objectives of machine learning generators do not target them.

### 4.1 The GAN objective targets distributional fidelity

A generative adversarial network [23] trains a generator  $G$  and discriminator  $D$  to solve

$$\min_G \max_D E_P[\log D(\mathbf{X}, Y)] + E_{P^*}[\log(1 - D(\mathbf{X}^*, Y^*))].$$

At the Nash equilibrium,  $P^* = P$  and  $D = 1/2$  everywhere. In principle, a perfect GAN recovers the full joint distribution and hence achieves inferential fidelity (since  $P^* = P$  implies exact conditional preservation).

In practice, three mechanisms cause the GAN to match marginal and low-order joint features while distorting the conditional structure:

- (i) *Mode averaging in sparse subgroups.* The discriminator evaluates the generator on the full sample. For a categorical predictor with a rare level (prevalence  $\pi \ll 1$ ), the discriminator receives  $O(n\pi)$  examples from that subgroup. The generator can satisfy the discriminator by producing outputs near the marginal mean  $E[Y]$  for the rare subgroup, rather than near the conditional mean  $E[Y \mid X = x_{\text{rare}}]$ . This mode averaging attenuates the regression coefficient for the rare level by a factor proportional to  $\pi$ .
- (ii) *Implicit marginalisation.* The discriminator's loss decomposes as

$$E_P[\log D] = \sum_s P(S = s) E_P[\log D \mid S = s],$$

where  $S$  indexes strata. Each stratum contributes to the total loss in proportion to its prevalence  $P(S = s)$ . The generator's gradient is dominated by the majority strata, and the conditional distribution in minority strata is undertrained.

- (iii) *Symmetric treatment of variables.* The GAN generator maps a latent vector  $\mathbf{z}$  to  $(\mathbf{X}^*, Y^*)$  jointly. It does not distinguish between predictors and response, nor between categorical and continuous variables. This symmetric treatment is analogous to the Gaussian copula treating all variables symmetrically (Section 1), which we showed empirically leads to coefficient attenuation.

## 4.2 The VAE objective

A VAE [29] maximises the evidence lower bound (ELBO):

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{X}, Y)}[\log p_\theta(\mathbf{X}, Y | \mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{X}, Y) \| p(\mathbf{z})).$$

The reconstruction term  $\mathbb{E}[\log p_\theta(\mathbf{X}, Y | \mathbf{z})]$  penalises the generator for deviating from the data. Under a factorised decoder  $p_\theta(\mathbf{X}, Y | \mathbf{z}) = \prod_j p_\theta(X_j | \mathbf{z})$ , each variable is reconstructed independently given the latent code. The conditional structure  $F(Y | \mathbf{X})$  is not explicitly represented. It emerges only implicitly through the shared latent space. The fidelity of this implicit conditioning depends on the capacity of the latent space and the expressiveness of the decoder, neither of which is controlled by the training objective.

## 4.3 Tabular data generators

CTGAN [57] addresses mixed tabular data by using mode-specific normalisation for continuous columns and training-by-sampling to balance categorical levels. The training-by-sampling mechanism ensures that each categorical value appears with equal frequency in training batches, which mitigates the mode-averaging problem (i) above. However, it does not directly target the conditional distribution  $F(Y | \mathbf{X})$ , and the generator still learns a joint model  $P^*(\mathbf{X}, Y)$ , and the conditional fidelity is a by-product, not a guarantee.

**Proposition 13** (No inferential guarantee from distributional objectives). *Let  $d$  be an  $f$ -divergence or a transport distance between the joint distributions  $P$  and  $P^*$  (e.g., KL divergence, Wasserstein distance, total variation). For any  $\varepsilon > 0$ , there exist  $P$  and  $P^*$  with  $d(P, P^*) < \varepsilon$  but  $\|\beta^* - \beta_0\| > c$  for a constant  $c > 0$  independent of  $\varepsilon$ .*

*Proof.* Take  $c = 1/2$  and fix  $\varepsilon > 0$ . Let  $X \in \{0, 1\}$  with  $P(X = 1) = \pi$  and  $Y | X \sim N(\mu_X, 1)$  with  $\mu_0 = 0$  and  $\mu_1 = 1$ , so that the regression coefficient is  $\beta = \mu_1 - \mu_0 = 1$ . Let  $P^*$  agree with  $P$  except that  $P^*(Y | X = 1) = N(0, 1)$ , so the conditional distribution for the rare class is replaced by that of the majority class.

The two laws share the covariate marginal and the  $X = 0$  conditional, so every  $f$ -divergence between them is  $\pi$  times the corresponding divergence between  $N(1, 1)$  and  $N(0, 1)$ , while a transport distance vanishes with  $\pi$  as well, through the coupling that moves only the rare-class conditional by the unit mean shift (at most  $\pi^{1/p}$  for the Wasserstein- $p$  distance). Explicitly,  $d_{\text{TV}}(P, P^*) = \pi d_{\text{TV}}(N(1, 1), N(0, 1)) = 0.383 \pi$ ,  $D_{\text{KL}}(P \| P^*) = \pi D_{\text{KL}}(N(1, 1) \| N(0, 1)) = \pi/2$ , and the Wasserstein-1 distance is  $W_1(P, P^*) = \pi$ . Setting  $\pi = \varepsilon/2$  makes all three smaller than  $\varepsilon$  simultaneously. For an  $f$ -divergence whose value on the pair  $\{N(1, 1), N(0, 1)\}$  is infinite, replace the substituted conditional  $N(0, 1)$  by the bounded tilt with density  $n(y-1)\{1 - \lambda \operatorname{sgn}(y-1)\}$ , with  $n$  the standard normal density and  $\lambda \in (0, 1)$  chosen to lower the substituted conditional mean by  $3/4$ , attainable since the fall equals  $\lambda E|Z|$  for  $Z$  standard normal with  $E|Z| = 0.798$ . The likelihood ratio then takes the two values  $(1 \mp \lambda)^{-1}$ , a compact subset of  $(0, \infty)$ , so the divergence is again  $\pi$  times a finite constant, and the coefficient gap in that branch is  $3/4 > c$ . In every case the divergence falls to zero as  $\pi$  does while the coefficient gap stays fixed, so a sufficiently small  $\pi$  gives  $d(P, P^*) < \varepsilon$ .

The synthetic regression coefficient is  $\beta^* = E^*[Y | X = 1] - E^*[Y | X = 0] = 0$ , so  $|\beta^* - \beta| = 1 > c$ , a positive constant independent of  $\varepsilon$ . Distributional closeness, whether measured by total variation, Kullback–Leibler, or Wasserstein divergence, therefore does not imply inferential closeness. The coupled choice  $\mu_1 = c/\varepsilon$  with  $P(X = 1) = \varepsilon$  would also separate the coefficients, but inflates the Kullback–Leibler and Wasserstein distances to  $c^2/(2\varepsilon)$  and  $c$ . Fixing the conditional-mean gap and shrinking the rare-class probability instead keeps all three divergences below  $\varepsilon$ .  $\square$

**Remark 7** (Rare subgroups and the relative savings loss). *Proposition 13 formalises the failure mode observed empirically, since a rare categorical level can carry a large regression coefficient, but because it contributes little to the joint distribution, a generator optimising distributional closeness has no incentive to model its conditional distribution accurately. The attenuation is structural, not a finite-sample artefact. A regression coefficient is not a Lipschitz functional of the joint law under any of these divergences, so closeness cannot control it, and the proof supplies the rate. Its two laws differ only on a set of probability  $\pi$ , so the divergence falls to zero with  $\pi$  while the coefficient gap stays constant.*

*The mechanism has a precise shrinkage counterpart. A generative model optimising  $P^*(\mathbf{X}, Y)$  implicitly shrinks rare-stratum conditional distributions toward the marginal. By the relative savings loss analysis of Efron and Morris [18], the shrinkage bias for stratum  $s$  at fixed stratum means is  $B_s^2(\theta_s - \bar{\theta})^2$ , where  $B_s = (\sigma^2/n_s)/(\tau^2 + \sigma^2/n_s)$  is the shrinkage factor toward the mean of the stratum means, with  $\tau^2$  the between-stratum and  $\sigma^2$  the within-stratum variance, so small  $n_s$  and atypical  $\theta_s$  produce large loss. The toxoplasmosis data of Efron and Morris [19] illustrate this, since the city with the highest incidence was ranked 12th by James–Stein estimators because its sample size was small. A synthetic data generator that smooths across strata similarly attenuates the conditional signal of rare subgroups.*

**Remark 8** (VAE ELBO and the no-guarantee proposition). *The construction in Proposition 13 applies directly to variational autoencoders. The VAE objective is the evidence lower bound  $\mathcal{L} = E_q[\log p(x | z)] - KL(q(z | x) \| p(z))$ , which decomposes into a reconstruction term (measuring distributional fit) and a regularisation term (enforcing a smooth latent space). Neither term constrains  $F(Y | X)$  separately from  $F(X)$ . The pathological construction shows that any generator minimising a joint divergence (including the VAE’s reconstruction loss) can leave a coefficient bias bounded away from zero for rare subgroups, because the divergence is dominated by the majority subgroup.*

#### 4.4 Connection to pseudo-true parameters

The framework of White [54] provides a unifying perspective. Under a misspecified synthetic distribution  $P^* \neq P$ , the quasi-likelihood estimator applied to synthetic data converges not to  $\beta_0$  but to the *pseudo-true parameter*

$$\beta^* = \arg \min_{\beta} D_{\text{KL}}(P^*(Y | \mathbf{X}) \| f(Y | \mathbf{X}; \beta)),$$

where  $f$  is the analyst’s model. Exact preservation,  $P^*(Y | \mathbf{X}) = P(Y | \mathbf{X})$  within the model class, is sufficient for  $\beta^* = \beta_0$  but not necessary, since  $\beta^*$  depends on  $P^*$  only through the conditional moments the criterion weights. Theorem 4 uses exactly that slack, fixing  $\beta^*$  from the first conditional moment alone.

For the specific case of a generator that adds independent noise to the covariates, a mechanism that approximates differentially private perturbation of the design, the bias is the classical errors-in-variables attenuation [20, Section 1.1.1]. For linear regression:

$$\beta^* = (\Sigma_{\mathbf{xx}} + \sigma^2 I)^{-1} \Sigma_{\mathbf{xx}} \beta_0 = (I + \sigma^2 \Sigma_{\mathbf{xx}}^{-1})^{-1} \beta_0, \quad (6)$$

where  $\sigma^2$  is the noise variance. This is ridge-type shrinkage toward zero. The matrix has eigenvalues in  $(0, 1)$ , so it contracts  $\beta_0$  in norm, and for orthogonal predictors it acts coordinate-wise, attenuating the  $j$ th coefficient by the reliability ratio  $v_j/(v_j + \sigma^2)$  with  $v_j$  the predictor variance, most strongly for the small  $v_j$  of rare categorical levels. Correlated predictors mix the coordinates, and an individual coefficient can then be inflated. The formula quantifies the mechanism identified qualitatively in our Proposition 13.

Wilde et al. [56] show that Bayesian inference from synthetic data is valid only if the generator equals the true data-generating process. A weaker but still restrictive condition is *congeniality*: the synthesis model must encompass the analyst’s model [34, 42]. Machine learning generators provide no such guarantee, because their training objectives are agnostic to the downstream analysis.

**Remark 9** (Scope: ML as generator versus ML as nuisance estimator). *Proposition 13 and the preceding analysis apply to machine learning used as a generative model, learning the joint distribution  $P(\mathbf{X}, Y)$  and sampling synthetic data from it. They do not apply to machine learning used as a nuisance estimator within a semiparametric framework. In doubly robust and debiased machine learning estimators [5, 8], the machine learning component estimates a nuisance function (an outcome regression or a propensity score), while a separate correction term, derived from the efficient influence function, targets the causal or inferential estimand directly. The correction term controls the conditional structure that Proposition 13 shows is uncontrolled by distributional objectives. Bruns-Smith et al. [5] make this precise. Augmented balancing weights that use ridge regression for both the outcome model and the weighting model reduce to a single undersmoothed ridge regression, and naive hyperparameter tuning can collapse the estimator to ordinary least squares, offering no improvement over the classical approach. This parallels the present paper’s conclusion that machine learning sophistication not directed at the relevant conditional functional provides no inferential benefit. Within the generative setting the proposition’s reach is empirical as well as theoretical. A reanalysis of 128 treatment comparisons from 115 phase 3 oncology trials under sixteen generators finds the excess variability of synthesis rising from parametric through tree-based to deep architectures, with naive analyses of deep-generator output declaring falsely significant treatment effects in up to half of the trials [11]. The simulation of Section 6 reproduces that ordering on a controlled grid, and the estimand in that reanalysis is a hazard ratio rather than a generalised linear model coefficient, so the agreement is across estimand classes rather than a repetition of the same experiment.*

## 5 Achieving inferential fidelity

The preceding sections established the conditional moment conditions for a known model with exact moment information, and showed that distributional objectives do not control these conditions. Three practical concerns remain. First, when strata are sparse, the conditional moments must be estimated



with a bias–variance trade-off (Section 5.1). Second, when the response variable is unknown to the data custodian, the conditions must hold in every direction simultaneously (Section 5.2). Third, for canonical-link models, exact inferential fidelity is achievable via a sufficient statistics projection that sidesteps both the sparsity and the moment-approximation problems (Section 5.3) and composes with a differentially private release (Section 5.4).

## 5.1 Sparse strata and the bias–variance trade-off

The results of Section 3 assume that  $P^*$  matches the conditional moments exactly. In practice, the conditional moments must be estimated from the data, and the accuracy of these estimates depends on the number of observations available in each conditioning stratum.

When the predictors include categorical variables with many levels, the strata defined by the cross-classification of all categorical predictors can be sparse. Let  $n_s$  denote the number of observations in stratum  $s$ . Two extreme strategies are available:

- (a) *Nonparametric (stratified bootstrap)*: within each stratum, sample from the empirical distribution of  $(\mathbf{X}_{\text{cnt}}, Y)$ . This preserves all conditional moments but has variance  $O(1/n_s)$  for each stratum-specific moment.
- (b) *Parametric (moment matching to order  $k$ )*: within each stratum, estimate the first  $k$  conditional moments and generate from the Cornish–Fisher quantile function [10]. This has bias  $O(\nu_{k+1}/(k+1)!)$  but variance  $O(1/n_s)$  only for the estimated moments, not for all functionals.

A shrinkage estimator interpolating between (a) and (b) offers the prospect of reducing the mean squared error of  $\hat{\beta}_n^*$ . The structure parallels the James–Stein problem [26], since we simultaneously estimate  $S$  stratum means  $\theta_s = E[h(Y) \mid S = s]$ , and the positive-part James–Stein estimator dominates the stratum-specific maximum likelihood estimator (MLE) when  $S \geq 3$ .

Concretely, let  $\hat{\theta}_s$  be the sample mean of  $h(Y)$  in stratum  $s$ , let  $\bar{\theta}$  be the mean of the stratum means, and let  $\hat{\sigma}^2$  be the average of the within-stratum variances. The empirical Bayes shrinkage estimator is

$$\tilde{\theta}_s = w_s \hat{\theta}_s + (1 - w_s) \bar{\theta}, \quad w_s = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \hat{\sigma}^2/n_s}, \quad (7)$$

where  $\hat{\tau}^2 = \max(0, \widehat{\text{Var}}(\hat{\theta}_s) - \hat{\sigma}^2 S^{-1} \sum_{s=1}^S n_s^{-1})$  estimates the between-stratum variance. Large  $n_s$  gives  $w_s \rightarrow 1$  and no shrinkage, and small  $n_s$  gives full shrinkage to  $\bar{\theta}$ .

The relative savings loss of the James–Stein estimator over the Bayes estimator is  $2/S$  uniformly in the between-stratum variance when the shrinkage centre is fixed, and  $3/S$  when that centre is itself estimated, the effective dimension dropping by one [18, Section 7]. Since (7) shrinks toward the estimated  $\bar{\theta}$ , it captures at least a fraction  $1 - 3/S$  of the possible improvement over the MLE.

**Theorem 14** (Sparse strata MSE). *Suppose there are  $S \geq 3$  strata with sizes  $n_1, \dots, n_S$  and stratum prevalences  $\pi_s = n_s/n$ . Let  $\theta_s = E[h(Y) \mid S = s]$  be the true stratum means,  $\hat{\theta}_s$  the sample means, and  $\tilde{\theta}_s$  the empirical Bayes estimates (7). If  $\tilde{\theta}_s$  is used in place of  $\hat{\theta}_s$  in the synthetic data generator, the MSE*

of  $\hat{\beta}_n^*$  satisfies

$$\mathbb{E}[\|\hat{\beta}_n^* - \beta_0\|^2] \leq 2\|A^{-1}b_k\|^2 + C_w\|A^{-1}\|^2 \sum_{s=1}^S \pi_s \left\{ (1 - w_s)^2(\theta_s - \bar{\theta})^2 + w_s^2 \frac{\sigma_s^2}{n_s} \right\} + \frac{\text{tr}(V^*)}{n} + o(n^{-1}), \quad (8)$$

where  $\sigma_s^2 = \text{Var}[h(Y) \mid S = s]$ ,  $\bar{\theta} = S^{-1} \sum_s \theta_s$  is the shrinkage centre of (7),  $b_k$  is the score bias of Theorem 6,  $C_w$  depends on the family, the link, and the design through the per-stratum score weights, and  $V^*$  is the sandwich variance of Theorem 9. The first term is the moment-truncation bias propagated through  $A^{-1}$ , the factor two paying for its separation from the shrinkage bias. The sum is the shrinkage contribution. The first term inside the braces is shrinkage bias<sup>2</sup> (pulling toward  $\bar{\theta}$ ), the second is estimation variance (reduced by the weight  $w_s < 1$ ). The trace term is the synthetic fit's own sampling variance.

*Proof.* The synthetic data score at  $\beta_0$  decomposes by stratum:

$$\frac{1}{n} \sum_{i=1}^n U_i^*(\beta_0) = \sum_s \hat{\pi}_s^* \bar{U}_s^*(\beta_0),$$

where  $\hat{\pi}_s^*$  is the synthetic stratum proportion and  $\bar{U}_s^*$  is the within-stratum mean score. The score bias in stratum  $s$  has two components: the truncation bias  $b_k$  (from matching only  $k$  moments) and the estimation bias from using  $\bar{\theta}_s$  instead of  $\theta_s$ .

The estimation bias in stratum  $s$  is  $\mathbb{E}[\bar{\theta}_s] - \theta_s = (1 - w_s)(\bar{\theta} - \theta_s)$ , contributing a score bias proportional to  $(1 - w_s)(\bar{\theta} - \theta_s)$ . The estimation variance is  $\text{Var}(\bar{\theta}_s) = w_s^2 \sigma_s^2 / n_s$ , up to the estimated centre's own noise, of the same order and absorbed into the constant.

Summing over strata with weights  $\pi_s$  bounds the shrinkage perturbation of the score by the braced sum, the per-stratum score weights entering the constant  $C_w$ . Propagating through  $A^{-1}$ , separating the truncation and shrinkage biases at the cost of the factor two, and adding the synthetic fit's own sampling variance  $\text{tr}(V^*)/n$ , whose cross terms with the biases are conditionally centred and fall into the remainder, gives (8).  $\square$

Under balanced designs, the crossover between the shrinkage and unshrunk regimes occurs at  $n \sim S\sigma^2/\tau^2$ . The optimal shrinkage rate, the connection to neural network training dynamics, and the high-dimensional strata asymptotics are given in Section . A simulation with  $S = 233$  strata ( $S/n = 0.12$ ) confirms that copula and calibrated generators degrade to 40–44% max deviation while the projection remains below 9% and reaches 0.0% at the highest  $S/n$  (Section ).

The five-step shrinkage-calibration pipeline that integrates the empirical Bayes shrinkage of Theorem 14 into the copula generator is detailed in the supplementary material (Section ).

## 5.2 Direction-agnostic fidelity

The preceding results assume a fixed response variable  $Y$  and a known conditioning set  $\mathbf{X}$ . In practice, the data custodian who generates the synthetic data may not know which variable the analyst will select as the response. We now extend the conditional moment framework to the *direction-agnostic* setting, where inferential fidelity must hold for any choice of response.

Let  $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$  denote the full vector of continuous variables, and let  $h_j$  denote the support transform for  $Z_j$ . Write  $\mathbf{W} = (h_1(Z_1), \dots, h_p(Z_p))^\top$  for the vector of transformed variables. Define the within-stratum moments

$$\boldsymbol{\mu}_s = E[\mathbf{W} \mid S = s], \quad \boldsymbol{\Sigma}_s = \text{Cov}(\mathbf{W} \mid S = s),$$

where  $S$  denotes the categorical stratum.

**Theorem 15** (Direction-agnostic fidelity). *Suppose Assumptions 1 and 2 hold for each choice of response variable  $Z_j$  with predictors  $\mathbf{Z}_{-j}$ , and that within each stratum the conditional law of  $W_j$  given  $\mathbf{W}_{-j}$  is elliptical under both  $P$  and  $P^*$ , so that its conditional mean and variance are determined by  $\boldsymbol{\mu}_s$  and  $\boldsymbol{\Sigma}_s$  (Remark 11; Section bounds the departure). If the synthetic distribution satisfies*

- (i)  $P^*(S \in \cdot) = P(S \in \cdot)$ ,
- (ii)  $E^*[\mathbf{W} \mid S = s] = E[\mathbf{W} \mid S = s]$  for all strata  $s$ , and
- (iii)  $\text{Var}^*(\mathbf{W} \mid S = s) = \text{Cov}(\mathbf{W} \mid S = s)$  for all strata  $s$ ,

*then for every generalised linear model of the form  $Z_j \sim g_j^{-1}(\mathbf{Z}_{-j}^\top \boldsymbol{\beta})$  the synthetic data estimator is consistent up to a bias controlled by the third conditional moment,*

$$\|\hat{\boldsymbol{\beta}}_n^{*(j)} - \boldsymbol{\beta}_0^{(j)}\| \leq C_j \cdot \frac{E[\nu_{3,j}(S) w_j(S)]}{3!} + O_p(n^{-1/2}) \quad \text{for each } j = 1, \dots, p, \quad (9)$$

where  $\nu_{3,j}(s) = \max(E[|W_j - \mu_{s,j}|^3 \mid S = s], E^*[|W_j^* - \mu_{s,j}|^3 \mid S = s])$  is the larger of the third conditional absolute central moments of the  $j$ -th transformed variable under the two laws. The full regression summary (standard errors, test statistics, significance) is controlled in each direction to the same order by Proposition 8.

*Proof.* Fix a direction  $j$  and write  $Y = Z_j$ ,  $\mathbf{X} = \mathbf{Z}_{-j}$ . The strata  $S$  determine the categorical part of  $\mathbf{X}$ . Within stratum  $s$ , the conditional moments of  $W_j = h_j(Y)$  given  $\mathbf{W}_{-j}$  are determined by the joint moments of  $\mathbf{W}$  if  $\mathbf{W} \mid S = s$  is elliptically distributed (which includes the Gaussian copula as a special case). Specifically:

$$E[W_j \mid \mathbf{W}_{-j}, S = s] = \mu_{s,j} + \boldsymbol{\Sigma}_{s,j,-j} \boldsymbol{\Sigma}_{s,-j,-j}^{-1} (\mathbf{W}_{-j} - \boldsymbol{\mu}_{s,-j}), \quad (10)$$

$$\text{Var}(W_j \mid \mathbf{W}_{-j}, S = s) = \Sigma_{s,jj} - \boldsymbol{\Sigma}_{s,j,-j} \boldsymbol{\Sigma}_{s,-j,-j}^{-1} \boldsymbol{\Sigma}_{s,-j,j}. \quad (11)$$

Conditions (ii) and (iii) ensure that  $\boldsymbol{\mu}_s$  and  $\boldsymbol{\Sigma}_s$  are matched under  $P^*$ . By (10)–(11), the conditional mean and variance of  $W_j$  given  $\mathbf{W}_{-j}$  are therefore matched, which gives moment matching to order  $k = 2$  in direction  $j$ . The bound (9) follows from Theorem 6 applied to each direction. Since this holds for all  $j$ , the result is direction-agnostic.  $\square$

**Corollary 16** (OLS invariance). *If all support transforms are the identity ( $h_j = \text{id}$  for all  $j$ ), conditions (ii)–(iii) of Theorem 15 reduce to matching the within-stratum mean vector and covariance matrix of  $\mathbf{Z}$ . In this case, the bias bound (9) is exactly zero for each direction, and  $\hat{\boldsymbol{\beta}}_n^{*(j)} = \hat{\boldsymbol{\beta}}_n^{(j)} + O_p(n^{-1/2})$  for all  $j = 1, \dots, p$ , so  $P^*$  has inferential fidelity in every direction in the sense of Definition 2.*

*Proof.* For  $h = \text{id}$ , the inverse transform  $h^{-1}$  is the identity, whose derivatives of order  $\geq 2$  all vanish. The Taylor remainder in the proof of Theorem 6 is therefore zero, giving exact moment matching at  $k = 2$  in every direction.  $\square$

The Gaussian copula suits the direction-agnostic setting: it models  $P^*(\mathbf{Z})$  symmetrically, its within-stratum correlation matrix already targets  $\Sigma_s$ , and a multivariate affine calibration matches the mean vector and covariance simultaneously (Section ).

**Remark 10** (Non-elliptical copulas and tail dependence). *The conditional moment formulas (10)–(11) hold exactly for the Gaussian copula and approximately for other copulas. If  $C$  is the true copula and  $C_{\text{ell}}$  is the best-fitting elliptical copula (matching the covariance matrix), the conditional expectation discrepancy enters the guarantee through the deviation bound  $\delta_s$  of Section , whose tail-driven component scales with the tail dependence parameters [35, Section 5.4]. For data without strong tail dependence ( $\lambda_U = \lambda_L = 0$ ) the tail-driven component vanishes, though a tail-independent copula can still depart from ellipticity in the body, which enters  $\delta_s$  directly. For data exhibiting strong tail dependence (e.g., financial returns, extreme environmental events), the covariance matrix alone does not determine the conditional moments, and the bound (9) may understate the actual bias. In such settings, a vine copula or a fully conditional specification may be needed to achieve direction-agnostic fidelity.*

**Remark 11** (Non-elliptical copulas). *The elliptical assumption underlying Theorem 15 fails for data with asymmetric tail dependence, sector-dependent correlation structure, or discrete responses. In such cases, the direction-agnostic guarantee does not hold, and the single-response projection of Theorem 17 should be preferred. A formal error bound quantifying the bias from non-elliptical copulas, in terms of the conditional-mean deviation with its tail-driven component scaled by the tail dependence coefficient, is given in Section .*

### 5.3 Sufficient statistics projection

The preceding results bound the bias from approximate moment matching. We now show that for canonical-link generalised linear models (GLMs), *exact* inferential fidelity can be achieved across all submodels simultaneously, using only  $p + 1$  constraints on the synthetic response vector.

Let  $\mathbf{X}$  be the  $n \times p$  model matrix (including the intercept) and  $g$  the canonical link for exponential family  $\mathcal{F}$ . The canonical-link score is  $U(\boldsymbol{\beta}) = \mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}(\boldsymbol{\beta}))/a(\phi)$ , so the MLE  $\hat{\boldsymbol{\beta}}_n$  is determined by the sufficient statistic  $\mathbf{X}^\top \mathbf{y}$ . The deviance is  $D = 2[\ell_{\text{sat}}(\mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}_n; \mathbf{y})]$ , where  $\ell_{\text{sat}}$  is the saturated log-likelihood [32, Section 2.3]. Only the kernel  $\sum_i \{y_i \theta(y_i) - b(\theta(y_i))\}$  of the saturated log-likelihood enters the deviance, where  $\theta(y)$  is the canonical parameter at which the family mean equals  $y$  and  $b$  is the cumulant function, the carrier terms  $c(y_i, \phi)$  cancelling. For the Gaussian family the full saturated log-likelihood is free of  $\mathbf{y}$  while its kernel is  $\mathbf{y}^\top \mathbf{y}/2$ , so the kernel is the object the projection must match.

**Theorem 17** (Sufficient statistics projection). *Let  $\mathcal{F}$  be an exponential family with canonical link  $g$ , and let  $\mathbf{X}$  be the  $n \times p$  model matrix for the full predictor set, of full column rank, so that  $\hat{\boldsymbol{\beta}}_n^{(J)}$  is unique for every  $J$ . If the synthetic response vector  $\mathbf{y}^*$  satisfies*

- (i)  $\mathbf{X}^\top \mathbf{y}^* = \mathbf{X}^\top \mathbf{y}$  (sufficient statistic matching), and

$$(ii) \sum_i \{y_i^* \theta(y_i^*) - b(\theta(y_i^*))\} = \sum_i \{y_i \theta(y_i) - b(\theta(y_i))\} \quad (\text{saturated kernel matching}),$$

then for every subset  $J \subseteq \{1, \dots, p\}$  and the corresponding submodel with design matrix  $\mathbf{X}_J$ :

- (a)  $\hat{\beta}_n^{*(J)} = \hat{\beta}_n^{(J)}$  (exact coefficient preservation),
- (b)  $\hat{\phi}_n^{*(J)} = \hat{\phi}_n^{(J)}$  for the deviance-based dispersion estimate  $D_J/(n - |J|)$ , which coincides with the Pearson estimate for the Gaussian family (exact dispersion preservation),
- (c) all standard errors, test statistics, confidence intervals, and  $p$ -values are identical.

The constraint set has  $p + 1$  equations in  $n$  unknowns, leaving  $n - p - 1$  degrees of freedom for generating novel values.

*Proof.* For part (a): since  $\mathbf{X}_J$  consists of columns of  $\mathbf{X}$ , condition (i) implies  $\mathbf{X}_J^T \mathbf{y}^* = \mathbf{X}_J^T \mathbf{y}$  for every  $J$ . The canonical-link MLE solves  $\mathbf{X}_J^T \boldsymbol{\mu}_J(\boldsymbol{\beta}) = \mathbf{X}_J^T \mathbf{y}$ , so the solution  $\hat{\beta}_n^{(J)}$  depends on  $\mathbf{y}$  only through  $\mathbf{X}_J^T \mathbf{y}$ . Since this is matched,  $\hat{\beta}_n^{*(J)} = \hat{\beta}_n^{(J)}$ .

For part (b): the deviance for subset  $J$  is  $D_J = 2[\ell_{\text{sat}}(\mathbf{y}) - \ell_J(\hat{\beta}_n^{(J)}; \mathbf{y})]$ . By part (a),  $\hat{\beta}_n^{*(J)} = \hat{\beta}_n^{(J)}$ . For canonical-link exponential families, the log-likelihood decomposes as  $\ell(\boldsymbol{\beta}; \mathbf{y}) = \sum_i [y_i \theta_i(\boldsymbol{\beta}) - b(\theta_i)]/a(\phi) + \sum_i c(y_i, \phi)$ . The first sum depends on  $\mathbf{y}$  only through  $\mathbf{X}_J^T \mathbf{y}$  and  $\boldsymbol{\beta}$ . The second sum  $\sum c(y_i, \phi)$  depends on individual  $y_i$ . In the deviance  $D_J = 2[\ell_{\text{sat}}(\mathbf{y}) - \ell_J(\hat{\beta}_n^{(J)}; \mathbf{y})]$ , the  $c(y_i, \phi)$  terms cancel. The remaining terms in  $\ell_J$  are matched by condition (i), and condition (ii) matches the surviving saturated kernel directly. Therefore  $D_J^* = D_J$  and  $\hat{\phi}_n^{*(J)} = D_J^*/(n - |J|) = D_J/(n - |J|) = \hat{\phi}_n^{(J)}$ .

Part (c) follows because standard errors, Wald statistics, and likelihood ratio statistics are functions of  $(\hat{\beta}_n^{(J)}, \hat{\phi}_n^{(J)}, \mathbf{X}_J)$ , all of which are matched.  $\square$

The saturated kernel condition (ii) takes a simple form for each exponential family:

| Family   | Canonical link | $\ell_{\text{sat}}$ kernel  | Constraint (ii)                                          |
|----------|----------------|-----------------------------|----------------------------------------------------------|
| Gaussian | identity       | $\frac{1}{2} \sum y_i^2$    | $\mathbf{y}^{*T} \mathbf{y}^* = \mathbf{y}^T \mathbf{y}$ |
| Poisson  | log            | $\sum (y_i \log y_i - y_i)$ | $\sum y_i^* \log y_i^* = \sum y_i \log y_i$              |
| Gamma    | inverse        | $-\sum (1 + \log y_i)$      | $\sum \log y_i^* = \sum \log y_i$                        |
| Binomial | logit          | 0 (binary)                  | automatic                                                |

The kernel column is  $\sum \{y_i \theta(y_i) - b(\theta(y_i))\}$ , the saturated log-likelihood with the  $c(y_i, \phi)$  terms dropped, which cancel in the deviance.

**Proposition 18** (Public-covariate cross-family invariance). *Suppose the covariates  $\mathbf{X}$  are released unchanged and have full column rank, and the synthetic response  $\mathbf{y}^*$  is produced by matching a canonical-link GLM with family  $\mathcal{F}_A$ , so that  $\mathbf{X}^T \mathbf{y}^* = \mathbf{X}^T \mathbf{y}$  (condition (i) of Theorem 17 on the original design). Then for any exponential family  $\mathcal{F}_B$  with canonical link, refitting  $\mathcal{F}_B$  on  $(\mathbf{X}, \mathbf{y}^*)$  yields  $\hat{\beta}_B^* = \hat{\beta}_B$ , the estimate on the original data. If  $\mathcal{F}_B$  has fixed dispersion (Poisson, binomial), all standard errors, test statistics, and  $p$ -values coincide.*

*Proof.* The canonical-link score for  $\mathcal{F}_B$  is  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}_B(\boldsymbol{\beta}))$ , so  $\hat{\boldsymbol{\beta}}_B$  solves  $\mathbf{X}^\top \boldsymbol{\mu}_B(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{y}$  and depends on  $\mathbf{y}$  only through  $\mathbf{X}^\top \mathbf{y}$ . This statistic is matched and  $\mathbf{X}$  is shared, so the estimating equation, and hence its root, is identical. Under fixed dispersion the information  $\mathbf{X}^\top \mathbf{W}_B(\hat{\boldsymbol{\beta}}_B) \mathbf{X}$  depends only on  $\hat{\boldsymbol{\beta}}_B$  and  $\mathbf{X}$ , both matched.  $\square$

**Remark 12.** When the covariates are also synthesised, the projection reconstructs  $\hat{\boldsymbol{\beta}}_A$  on a new design, so Proposition 18 no longer applies exactly, and a second analyst who fits a different family recovers the original coefficient only approximately. Table 1 shows the deviation is graceful for a canonical generator (worst cross-family deviation 5.8% for the Gaussian-identity generator, 12.9% for Poisson-log) and larger for a non-canonical log-link generator (up to 41.7%), because the latter reconstructs only a variance-weighted mean. Analysts fitting the generator’s own family retain the exact guarantee of Theorem 17 in every case.

Table 1: Cross-family refit under full synthesis: max |%| coefficient deviation (mean over 100 replications) when the generator matches family A (rows) and the analyst fits family B (columns). All six analyst families evaluated are shown. The analyst who fits the generator’s own family recovers the coefficients exactly (Theorem 17), zero for the continuous paths and the 2.4% probabilistic-rounding floor for Poisson. With public covariates the canonical-family cells are exact up to two measured artefacts (Proposition 18): the Poisson generator’s rounding floor (1.6%) and the boundary rows the gamma refit discards (0.4%).

| Generator A                     | Analyst family B, max  % |           |          |         |         |         |
|---------------------------------|--------------------------|-----------|----------|---------|---------|---------|
|                                 | Gauss/id                 | Gauss/log | Pois/log | Pois/id | Gam/inv | Gam/log |
| Sc1 Gaussian/identity (canon)   | 0.0                      | 4.8       | 1.5      | 5.8     | 3.1     | 4.8     |
| Sc3 Gamma/log (non-canon)       | 8.9                      | 18.4      | 10.6     | 11.2    | 24.5    | 0.0     |
| Sc4 Quasi-gamma/log (non-canon) | 14.7                     | 25.6      | 20.9     | 41.7    | 23.8    | 0.0     |
| Sc6 Poisson/log (canon)         | 6.0                      | 7.6       | 2.4      | 9.8     | 12.9    | 7.8     |

**Corollary 19** (Gaussian closed-form projection). *For the Gaussian family with identity link, the minimum-distance projection of a copula-generated vector  $\mathbf{y}_{\text{cop}}$  onto the constraint set of Theorem 17 is*

$$\mathbf{y}^* = \mathbf{X}\hat{\boldsymbol{\beta}}_n + c \cdot (\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}}, \quad c = \frac{\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|}{\|(\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}}\|}, \quad (12)$$

where  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is the hat matrix and  $\hat{\boldsymbol{\beta}}_n$  is the original OLS estimate. The fitted values are from the original data (exact fidelity), the residuals are from the copula (genuine novelty), and the scaling  $c$  matches the residual sum of squares. The scale is well defined whenever  $(\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}} \neq \mathbf{0}$ , an event of probability one for a continuously distributed copula draw. This preserves all  $2^p - 1$  OLS submodels simultaneously.

*Proof.* Condition (i):  $\mathbf{X}^\top \mathbf{y}^* = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}_n + c \cdot \mathbf{X}^\top (\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}} = \mathbf{X}^\top \mathbf{y} + \mathbf{0}$ , since  $\mathbf{X}^\top (\mathbf{I} - \mathbf{H}) = \mathbf{0}$ . Condition (ii):  $\mathbf{y}^{*\top} \mathbf{y}^* = \hat{\boldsymbol{\beta}}_n^\top \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}_n + c^2 \|(\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}}\|^2 = \|\mathbf{H}\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{y}$ , by the choice of  $c$  and orthogonality of  $\mathbf{H}$  and  $\mathbf{I} - \mathbf{H}$ .  $\square$

The projection rewrites only the response column of its base copula draw, so it defines a coupling of the projected sample to the base sample, and the distributional cost of exact inferential fidelity is

bounded by the mean per-observation displacement. Write  $d_i = y_i^* - y_i^{\text{cop}}$  for that displacement and  $\mu(\hat{\beta})$  for the fitted mean vector of the original fit.

**Lemma 20** (Displacement control). *Let  $\hat{\beta}$  denote the original fit and  $\hat{\beta}^{\text{cop}}$  the fit of the base draw on its own design  $\mathbf{X}^*$ , with  $\mathbf{H}^*$  the hat matrix of  $\mathbf{X}^*$  and  $m = n$ . (i) For the Gaussian closed-form path,  $\mathbf{d} = \mathbf{X}^*(\hat{\beta} - \hat{\beta}^{\text{cop}}) + (c - 1)(\mathbf{I} - \mathbf{H}^*)\mathbf{y}^{\text{cop}}$ , so*

$$m^{-1} \sum_{i=1}^m |d_i| \leq m^{-1/2} \|\mathbf{X}^*(\hat{\beta} - \hat{\beta}^{\text{cop}})\| + m^{-1/2} \|\mathbf{I} - \mathbf{H}\| \|\mathbf{y} - \mathbf{y}^{\text{cop}}\|.$$

(ii) For the positive-support Newton and alternating paths exiting at score tolerance  $\tau$ , with iteration weights in  $[w_{\text{lo}}, w_{\text{hi}}]$  and  $\lambda_{\min}(\mathbf{X}^{*\top} \mathbf{W} \mathbf{X}^*) \geq \lambda > 0$  along the path,

$$m^{-1} \sum_{i=1}^m |d_i| \leq (w_{\text{hi}}/\lambda) m^{-1/2} \|\mathbf{X}^{*\top} \mathbf{w} \{\mathbf{y}^{\text{cop}} - \mu(\hat{\beta})\}\| + O(\tau).$$

(iii) The binomial box path satisfies (ii) with the box shrink factor  $s \in (0, 1]$  multiplying the copula residual, adding  $(1 - s) m^{-1} \sum_i |e_i^{\text{cop}}|$ . (iv) The zero-inflated two-part path has  $d_i = 0$  on the zero rows, whose pattern comes from the base draw, and clause (ii) on the positive rows.

*Proof.* (i) The path releases  $\mathbf{y}^* = \mathbf{X}^* \hat{\beta} + c(\mathbf{I} - \mathbf{H}^*)\mathbf{y}^{\text{cop}}$  with  $c = \|\mathbf{I} - \mathbf{H}\| \|\mathbf{y} - \mathbf{y}^{\text{cop}}\| / \|\mathbf{I} - \mathbf{H}^*\| \|\mathbf{y}^{\text{cop}}\|$ . Decomposing  $\mathbf{y}^{\text{cop}} = \mathbf{H}^* \mathbf{y}^{\text{cop}} + (\mathbf{I} - \mathbf{H}^*) \mathbf{y}^{\text{cop}}$  and using  $\mathbf{H}^* \mathbf{y}^{\text{cop}} = \mathbf{X}^* \hat{\beta}^{\text{cop}}$  gives the displayed decomposition of  $\mathbf{d}$ . The Cauchy–Schwarz inequality gives  $m^{-1} \sum_i |d_i| \leq m^{-1/2} \|\mathbf{d}\|$ , the triangle inequality splits the two terms, and the second term’s norm is  $|c - 1| \cdot \|\mathbf{I} - \mathbf{H}^*\| \|\mathbf{y}^{\text{cop}}\| = \|\mathbf{I} - \mathbf{H}\| \|\mathbf{y} - \mathbf{y}^{\text{cop}}\|$  by the definition of  $c$ . (ii) The Newton and alternating paths move  $\mathbf{y}^{\text{cop}}$  to an exit state whose weighted score satisfies  $\|\mathbf{X}^{*\top} \mathbf{w} \{\mathbf{y}^* - \mu(\hat{\beta})\}\| \leq \tau$ , through updates lying in the span of the design columns. By the mean value theorem applied coordinatewise to the score map, there is an intermediate weight matrix  $\tilde{\mathbf{W}}$  with entries in  $[w_{\text{lo}}, w_{\text{hi}}]$  such that the displacement solves  $\mathbf{X}^{*\top} \tilde{\mathbf{W}} \mathbf{d} = \mathbf{g}_0 + O(\tau)$ , where  $\mathbf{g}_0 = \mathbf{X}^{*\top} \mathbf{w} \{\mathbf{y}^{\text{cop}} - \mu(\hat{\beta})\}$  is the score gap of the base draw at the original fit. Solving through the weighted normal equations and bounding the operator norms by the weight bounds and  $\lambda_{\min}(\mathbf{X}^{*\top} \mathbf{W} \mathbf{X}^*) \geq \lambda$  gives  $\|\mathbf{d}\| \leq (w_{\text{hi}}/\lambda) \|\mathbf{g}_0\| + O(\tau)$ , and Cauchy–Schwarz converts to the mean form as in (i). (iii) The box path applies (ii) and then shrinks the copula residual by the largest  $s \in (0, 1]$  keeping the release in  $[0, 1]$ , which moves each coordinate by a further  $(1 - s)|e_i^{\text{cop}}|$ , added by the triangle inequality. (iv) The zero pattern is copied from the base draw, so  $d_i = 0$  there, and the positive block is a clause-(ii) projection on the positive rows.  $\square$

**Proposition 21** (Coupling bound). *For every index set  $S$  of columns containing the response,*

$$W_1(\hat{F}_S^*, \hat{F}_S^{\text{cop}}) \leq m^{-1} \sum_{i=1}^m |d_i|,$$

*and the distance is zero when  $S$  excludes the response. The inequality holds for every realised draw. On integer supports the released response is  $\text{round}(y^*)$  and the bound holds with the total displacement  $m^{-1} \sum_i |\text{round}(y_i^*) - y_i^{\text{cop}}|$ , whose rounding contribution obeys  $E |\text{round}(z) - z| = 2 \text{frac}(z) \{1 - \text{frac}(z)\} \leq 1/2$  per observation.*

*Proof.* The two empirical measures place mass  $m^{-1}$  at the  $S$ -coordinates of the projected and base rows respectively. The identity pairing defines a coupling  $\pi = m^{-1} \sum_i \delta_{(\mathbf{z}_{i,S}^*, \mathbf{z}_{i,S}^{\text{cop}})}$ , and  $W_1$  is the infimum of  $\int \|u - v\| d\pi$  over couplings, so  $W_1 \leq m^{-1} \sum_i \|\mathbf{z}_{i,S}^* - \mathbf{z}_{i,S}^{\text{cop}}\|$ . The projection rewrites only the response column, so the summand equals  $|d_i|$  when  $S$  contains the response and zero otherwise. For a rounded release the same coupling and the triangle inequality give the bound with the total displacement. Probabilistic rounding takes  $\lfloor z \rfloor$  with probability  $1 - \text{frac}(z)$  and  $\lceil z \rceil$  with probability  $\text{frac}(z)$ , so  $E |\text{round}(z) - z| = 2 \text{frac}(z)\{1 - \text{frac}(z)\} \leq 1/2$ , the final bound since  $t(1 - t) \leq 1/4$  on  $[0, 1]$ .  $\square$

**Corollary 22** (Inheritance of  $k$ -wise fidelity). *Suppose the base generator has  $k$ -wise distributional fidelity and its implied fit is consistent,  $\hat{\beta}^{\text{cop}} - \hat{\beta} = o_p(1)$  with matching dispersion. Then the projected generator has  $k$ -wise distributional fidelity for the same  $k$ .*

*Proof.* By Lemma 20 and the consistency hypothesis,  $m^{-1} \sum_i |d_i| = o_p(1)$ . For any margin  $S$  with  $|S| \leq k$  and any bounded Lipschitz  $f$ ,

$$\left| \int f d\hat{F}_S^* - \int f d\hat{F}_S^{\text{cop}} \right| \leq \|f\|_{\text{Lip}} m^{-1} \sum_{i=1}^m |d_i| \xrightarrow{p} 0,$$

directly from the coupling of Proposition 21. Weak convergence of probability measures is equivalent to convergence of the integrals of bounded Lipschitz functions, so along every subsequence on which the displacement converges almost surely the two empirical margins share their weak limits, and the base generator's  $k$ -wise fidelity transfers by the standard subsequence characterisation of convergence in probability.  $\square$

The calibrated copula satisfies the consistency hypothesis at rate  $n^{-1/2}$  under its  $k = 2$  conditional-moment matching, while the raw copula's finite- $n$  coefficient tier (14 to 49%) makes the corollary an asymptotic statement for that base.

**Remark 13** (Integer supports). *The rounding term does not vanish with  $n$  against the continuous base draw, and rounding is what restores the integer support of the target law, so on count responses Corollary 22 is read toward the rounded release. The fractional binomial release is the same trade in the opposite direction, since it keeps exact inference by staying fractional, so its displacement is structural rather than vanishing. Table 32 quantifies both directions: Poisson rounding pays joint energy distance 0.0035 against the base copula's 0.0014 to restore integer support, the fractional binomial release pays 0.0504 against 0.0015 to keep exact inference, and on five of the six continuous-support designs the projected release's energy distance sits at or below the base copula's, with scenario 8 above by a tenth of the base value on the response pairs (0.0011 against 0.0010) and within Monte Carlo error on the full joint.*

**Remark 14** (Privacy structure of the projection). *The decomposition (12) reveals the privacy structure. The fitted component  $\mathbf{X}\hat{\beta}_n$  embeds  $p$  aggregate statistics with the same disclosure risk as publishing the regression output itself. The residual component carries  $n - p - 1$  degrees of freedom of genuine novelty from the copula.*



The projection in its unperturbed form is not differentially private, since adding or removing one observation changes  $\mathbf{X}^\top \mathbf{y}$  and hence the entire fitted component. A direct application of the Gaussian mechanism to the raw-scale sufficient statistic destroys inferential fidelity at practical privacy budgets, because the unregularised solve through  $(\mathbf{X}^\top \mathbf{X})^{-1}$  amplifies the noise by the reciprocal of its smallest eigenvalue, the same eigenvalue-dependent vulnerability as the ridge bias formula (6). The amplification belongs to the scale and the solve rather than to the statistic, and a whitened, regularised release removes it (Proposition 28). Perturbing the coefficient estimate  $\hat{\beta}_n$  directly and composing with the projection likewise reconciles differential privacy (DP) with exact inferential fidelity. Both constructions are developed in Section 5.4.

The per-person disclosure can be quantified via the leave-one-out influence [DFBETA; 4, 9]. Pre-masking (collapsing rare categorical levels before synthesis) is effective, since reducing the 9 housing levels to 3 lowers the maximum single-person influence from 1103% to 29%. A detailed privacy analysis, including the Gaussian mechanism derivation, DFBETA results, and pre-masking evaluation, is given in Section .

**Remark 15** (High  $p/n$  regime). The projection requires  $p \ll n$ . As  $p/n \rightarrow 1$ , the residual component vanishes and the synthetic data collapses onto the fitted values, providing no privacy. For high-dimensional settings, the projection must be combined with regularisation to stabilise the hat matrix.

**Remark 16** (Numerical implementation on positive support). Theorem 17 assumes that the predicted values  $\hat{\mu}_i = g^{-1}(\mathbf{x}_i^\top \hat{\beta}_n)$  lie in the interior of the response support. For the inverse link ( $\mu = 1/\eta$ ), extrapolated covariate patterns can produce  $\eta \approx 0$  or negative  $\mu$ , violating the positive support. The log link avoids this issue entirely ( $\mu = e^\eta > 0$  for all  $\eta$ ).

For non-canonical links, the sufficient statistic is no longer  $\mathbf{X}^\top \mathbf{y}$ , and the projection becomes a constrained optimisation on  $n$  variables with  $p + 1$  smooth constraints. In the implementation, a Newton iteration on the  $z = \log(y)$  scale solves the score equation exactly for all link functions while maintaining  $y = e^z > 0$  by construction, achieving exact coefficient preservation for both canonical and non-canonical links on positive support.

When the covariates are synthesised as well, the refit information  $(\mathbf{X}^{*\top} \mathbf{W} \mathbf{X}^*)^{-1}$  is a fresh matrix, and the standard errors drift even though the coefficients do not, by 1.6 to 6.9% on the simulation grid and 19.2% on the application of Section 6.4 (Table 33). Pinning the design Gram removes this drift while the design remains fully synthetic.

**Proposition 23** (Design-Gram projection). Write  $\mathbf{X} = (\mathbf{D}, \mathbf{C})$  for the  $n \times p$  model matrix of the full main-effects predictor set, where  $\mathbf{D}$  collects the intercept and the indicator columns of the categorical predictors and  $\mathbf{C}$  the continuous columns, and suppose  $\mathbf{X}$  has full column rank with  $n - \text{rank}(\mathbf{D})$  at least the number of continuous columns. Let the synthetic design  $\mathbf{X}^* = (\mathbf{D}^*, \mathbf{C}^*)$  satisfy

- (i)  $\mathbf{D}^{*\top} \mathbf{D}^* = \mathbf{D}^\top \mathbf{D}$  (cross-classification counts),
- (ii)  $\mathbf{D}^{*\top} \mathbf{C}^* = \mathbf{D}^\top \mathbf{C}$  (per-stratum sums), and
- (iii)  $\mathbf{C}^{*\top} \mathbf{C}^* = \mathbf{C}^\top \mathbf{C}$  (continuous Gram),

so that  $\mathbf{X}^{*\top}\mathbf{X}^* = \mathbf{X}^\top\mathbf{X}$ , and let the synthetic response  $\mathbf{y}^*$  satisfy conditions (i) and (ii) of Theorem 17 imposed on  $\mathbf{X}^*$ , the sufficient-statistic condition taken as the score identity  $\mathbf{X}^{*\top}\mathbf{y}^* = \mathbf{X}^{*\top}\boldsymbol{\mu}(\mathbf{X}^*\hat{\boldsymbol{\beta}}_n)$ , which reduces to  $\mathbf{X}^{*\top}\mathbf{y}^* = \mathbf{X}^\top\mathbf{y}$  for the Gaussian family. Then:

- (a) For the Gaussian family with identity link, the refitted coefficients, dispersion, standard errors,  $t$  statistics, confidence intervals, coefficient of determination,  $F$  statistic, and information criteria coincide with the original fit, simultaneously for every submodel  $J \subseteq \{1, \dots, p\}$ .
- (b) For an exponential family with canonical link and Fisher information weight matrix  $\mathbf{W}(\cdot)$ , if in addition  $\|\mathbf{X}^{*\top}\mathbf{W}(\mathbf{X}^*\hat{\boldsymbol{\beta}}_n)\mathbf{X}^* - \mathbf{X}^\top\mathbf{W}(\mathbf{X}\hat{\boldsymbol{\beta}}_n)\mathbf{X}\|_2 \leq \tau$  with  $\lambda = \|\{\mathbf{X}^\top\mathbf{W}(\mathbf{X}\hat{\boldsymbol{\beta}}_n)\mathbf{X}\}^{-1}\|_2$  and  $\lambda\tau < 1$ , then the refitted coefficients are exact and, at fixed dispersion  $\phi$ , every entry of the estimated covariance matrix moves by at most  $\phi\lambda^2\tau/(1 - \lambda\tau)$ , the factor  $\phi$  being one for the binomial and Poisson families.

A synthetic design satisfying (i)–(iii) exists whenever the base draw's residual Gram is nonsingular, by the construction in the proof.

*Proof.* Construction. Draw the stratum labels of the synthetic sample as a permutation of the original labels, so  $\mathbf{D}^*$  equals  $\mathbf{D}$  up to row order and (i) holds. Given any starting matrix  $\mathbf{C}_0$  (the base draw), set

$$\mathbf{C}_1 = \mathbf{D}^*(\mathbf{D}^{*\top}\mathbf{D}^*)^{-1}\mathbf{D}^\top\mathbf{C} + (\mathbf{I} - \mathbf{H}_{\mathbf{D}^*})\mathbf{C}_0,$$

with  $\mathbf{H}_{\mathbf{D}^*}$  the orthogonal projector onto the column space of  $\mathbf{D}^*$ , so  $\mathbf{D}^{*\top}\mathbf{C}_1 = \mathbf{D}^\top\mathbf{C}$  and (ii) holds. Write  $\mathbf{R} = (\mathbf{I} - \mathbf{H}_{\mathbf{D}^*})\mathbf{C}_0$  and  $\mathbf{G} = \mathbf{C}^\top\mathbf{C} - (\mathbf{D}^\top\mathbf{C})^\top(\mathbf{D}^\top\mathbf{D})^{-1}(\mathbf{D}^\top\mathbf{C})$ , the original within-stratum Gram, positive definite by the rank assumption. With  $\mathbf{M} = \mathbf{L}_R^{-1}\mathbf{L}_G$  for the upper-triangular Cholesky factors  $\mathbf{L}_R^\top\mathbf{L}_R = \mathbf{R}^\top\mathbf{R}$  and  $\mathbf{L}_G^\top\mathbf{L}_G = \mathbf{G}$ , replace the residual by  $\mathbf{R}\mathbf{M}$ . Right-multiplication preserves  $\mathbf{D}^{*\top}\mathbf{R}\mathbf{M} = \mathbf{0}$ , and  $(\mathbf{R}\mathbf{M})^\top\mathbf{R}\mathbf{M} = \mathbf{G}$ , so (iii) holds and  $\mathbf{X}^{*\top}\mathbf{X}^* = \mathbf{X}^\top\mathbf{X}$ .

(a) For the Gaussian family the score identity of the response condition reads  $\mathbf{X}^{*\top}\mathbf{y}^* = \mathbf{X}^\top\mathbf{y}$ , and the saturated kernel condition gives equality of residual sums of squares. Fitted and residual parts are orthogonal, so  $\mathbf{y}^{*\top}\mathbf{y}^* = \hat{\boldsymbol{\beta}}_n^\top\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}}_n + \text{RSS} = \mathbf{y}^\top\mathbf{y}$ , and the augmented cross-product matrices of  $(\mathbf{X}^*, \mathbf{y}^*)$  and  $(\mathbf{X}, \mathbf{y})$  coincide. For any submodel  $J$ , the estimate  $\hat{\boldsymbol{\beta}}^{(J)} = (\mathbf{X}_J^\top\mathbf{X}_J)^{-1}\mathbf{X}_J^\top\mathbf{y}$ , the residual sum of squares  $\text{RSS}_J = \mathbf{y}^\top\mathbf{y} - 2\hat{\boldsymbol{\beta}}^{(J)\top}\mathbf{X}_J^\top\mathbf{y} + \hat{\boldsymbol{\beta}}^{(J)\top}\mathbf{X}_J^\top\mathbf{X}_J\hat{\boldsymbol{\beta}}^{(J)}$ , the total sum of squares, and every summary derived from them are functions of the augmented cross-product matrix and  $n$  alone, which proves (a).

(b) The refitted coefficient solves the canonical-link score equation on  $(\mathbf{X}^*, \mathbf{y}^*)$ , which the score identity of the response condition sets to zero at  $\hat{\boldsymbol{\beta}}_n$ , and strict concavity of the canonical log-likelihood on a full-rank design makes the root unique, so the coefficients are exact. At fixed dispersion  $\phi$  the estimated covariance is  $\phi(\mathbf{X}^{*\top}\mathbf{W}^*\mathbf{X}^*)^{-1}$  evaluated at the common  $\hat{\boldsymbol{\beta}}_n$ . For symmetric positive definite  $\mathbf{A}$  and a symmetric perturbation with  $\|\tilde{\mathbf{A}} - \mathbf{A}\|_2 \leq \tau < 1/\|\mathbf{A}^{-1}\|_2$ , the Neumann series gives

$$\|\tilde{\mathbf{A}}^{-1} - \mathbf{A}^{-1}\|_2 \leq \frac{\|\mathbf{A}^{-1}\|_2^2 \tau}{1 - \|\mathbf{A}^{-1}\|_2 \tau},$$

and every matrix entry is bounded in modulus by the spectral norm, the dispersion multiplying

through, which proves (b). □

**Corollary 24** (Two-part projection under full synthesis). *For a hurdle response, draw the synthetic sample so that the per-cell counts of the cross-classification joined with the zero indicator are exact, and write  $\mathbf{z}$  for the synthetic zero indicator and  $\hat{\beta}_0$  for the original zero-model fit. Impose the conditions of Proposition 23(b) on the zero-block binomial model over all rows together with the zero-model score condition  $\mathbf{X}^{*\top}\{\mathbf{z} - \pi(\mathbf{X}^*\hat{\beta}_0)\} = \mathbf{0}$  within the same tolerance, and the conditions of Proposition 23 on the positive rows for the positive block. Refitting the two-part model on the release then reproduces the original two-part summary, read with the block's estimated dispersion, the zero block holding within the clause (b) tolerances throughout. For the positive block the conclusion is threefold. It is exact when that block is Gaussian. It holds within the clause (b) tolerances when that block's link is canonical. When that block carries a log link with  $V(\mu) = \mu^2$ , the Fisher weight  $(\partial\mu/\partial\eta)^2/V(\mu)$  is identically one, so the pinned unweighted Gram is the information itself and the refit reproduces the block to the tolerance at which that Gram is pinned.*

*Proof.* The joint label draw makes the per-cell counts of the cross-classification joined with the zero indicator exact, so the positive-row count matches the original. The hurdle likelihood separates into the zero block and the positive block with block-diagonal information. The zero block is a binomial GLM of the indicator on the full design, its score condition pins the coefficient within the stated tolerance, and Proposition 23(b) applied to the binomial information bounds its covariance move. The positive block is an ordinary projection on the pinned positive design, exact by Proposition 23(a) for a Gaussian positive block and within the clause (b) tolerance when its link is canonical. For a log link with  $V(\mu) = \mu^2$  the Fisher weight is identically one, so pinning the unweighted positive-block Gram pins the information itself and the clause (b) bound is attained at  $\tau = 0$ . A configuration satisfying every constraint simultaneously exists, since the row permutation of the original sample attains each of them exactly. □

**Remark 17** (Interaction designs). *Conditions (i)–(iii) cover every main-effects submodel. An analyst model with categorical-by-continuous interactions involves the per-stratum continuous Grams, which the pooled condition (iii) leaves free. Pinning them stratum by stratum extends the construction whenever every stratum holds more observations than continuous columns plus one. Sparse strata make this restrictive, and the rarest housing stratum of the application in Section 6.4 holds a single observation.*

**Remark 18** (Distributional and disclosure cost of pinning the design). *The rotation in the construction is linear on the raw scale, so a pinned continuous covariate can leave a bounded or positive support even though the response projection remains support-respecting. The release also discloses more than the  $p + 1$  statistics of Remark 14. Condition (i) releases the full categorical contingency table, which at exact counts is the categorical microdata up to row order, and conditions (ii) and (iii) release the per-stratum continuous sums and the pooled continuous Gram, the descriptive covariance table. Each adjustment displaces the base draw by  $O_p(n^{-1/2})$  per observation under fixed strata, so the inheritance guarantee of Corollary 22 applies to the pinned generator unchanged, and the Poisson channel keeps its rounding floor. Table 33 reports the measured costs and the attained tolerances on the simulation grid and the application.*

## 5.4 Differentially private projection

The sufficient statistics projection of Theorem 17 preserves the analyst's estimator exactly, since the fitted coefficient depends on the data only through the  $p + 1$  score-zero constraints that the projection reproduces. This exactness raises a privacy question and exposes a tension. Exact inferential fidelity preserves those sufficient statistics, while differential privacy must perturb them, so the two objectives conflict. This subsection characterises that tension. It gives two mechanisms, states when exact fidelity and privacy coexist and when privacy requires relaxing fidelity, and identifies the release for which no joint guarantee holds.

The organising observation is that any mechanism releasing a DP version of the coefficient estimate, followed by the projection, inherits the guarantee by post-processing. The Gaussian mechanism on the raw-scale  $\mathbf{X}^\top \mathbf{y}$ , analysed in Section , is amplified through the unregularised solve by  $(\mathbf{X}^\top \mathbf{X})^{-1}$  and destroys inference at any practical budget. The noise is therefore placed either on the coefficient itself, where no amplification occurs, or on the sufficient statistics of a whitened public scale, where the amplification is removed at its source.

We state privacy in the  $\mu$ -Gaussian differential privacy calculus of Dong, Roth and Su [14], which gives a single privacy parameter  $\mu$ , tight composition, and an exact conversion to  $(\epsilon, \delta)$ .

**Lemma 25** ( $\mu$ -Gaussian differential privacy). *A Gaussian mechanism releasing a statistic of  $L_2$  global sensitivity  $\Delta$  with noise  $N(\mathbf{0}, \sigma^2 \mathbf{I})$  is  $\mu$ -Gaussian differential privacy (GDP) with  $\mu = \Delta/\sigma$  [14]. The guarantee is closed under post-processing, so any function of a  $\mu$ -GDP release and of quantities independent of the confidential data is itself  $\mu$ -GDP. Mechanisms that are  $\mu_1, \dots, \mu_k$ -GDP compose to  $\{\sum_j \mu_j^2\}^{1/2}$ -GDP, which is  $(\epsilon, \delta(\epsilon))$ -DP with*

$$\delta(\epsilon) = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - e^\epsilon \Phi\left(-\frac{\epsilon}{\mu} - \frac{\mu}{2}\right). \quad (13)$$

Three routes characterise the fidelity-privacy tension. The first, subsample-and-aggregate, is the general guarantee. It holds for any GLM, whatever the link, whether or not the covariates are public, at the cost of exact fidelity. The second, the public-covariate release, is the restrictive special case in which exact fidelity and privacy coexist. The third, the whitened sufficient-statistic release, covers the intermediate case, in which the covariates are sensitive but the response admits a public clip. Its standing against subsample-and-aggregate depends on the budget and on the width of the public coefficient box, and it is ahead of the central box at the smallest budget and at every budget on a structurally bounded response. All three rest on a global sensitivity from public bounds, not on the local sensitivity of  $\hat{\beta}_n$ , which is itself data-dependent and does not deliver DP.

**Proposition 26** (Differentially private projection by subsample-and-aggregate). *For a GLM with any link, partition the  $n$  rows into  $m$  blocks by an assignment of row positions chosen independently of the data, and let  $\hat{\beta}_j^{(b)}$  be the coefficient fitted on block  $b$ . Fix public bounds  $\ell_j \leq u_j$  for each coordinate, set  $a_j = m^{-1} \sum_{b=1}^m \text{clip}(\hat{\beta}_j^{(b)}, \ell_j, u_j)$  and  $s_j = (u_j - \ell_j)/m$ , and release  $\hat{\beta}_{\text{DP},j} = a_j + s_j \sigma_Q Z_j$  with noise multiplier  $\sigma_Q > 0$  and  $Z \sim N(\mathbf{0}, \mathbf{I}_p)$ . Then  $\hat{\beta}_{\text{DP}}$  is  $(\sqrt{p}/\sigma_Q)$ -GDP under one-row replacement, and any release depending on the data only through  $\hat{\beta}_{\text{DP}}$  inherits the guarantee by post-processing. The estimand*

is the block-aggregate coefficient, so the inferential fidelity is approximate.

*Proof.* One-row replacement changes the data of exactly one block  $b_0$ , since the partition uses row positions and not values, so only  $\hat{\beta}^{(b_0)}$  changes. Each clipped coordinate lies in  $[\ell_j, u_j]$ , so  $a_j$  moves by at most  $(u_j - \ell_j)/m = s_j$ . The scaled aggregate  $Q_j = a_j/s_j$  therefore changes by at most 1 per coordinate, giving  $\|Q(D) - Q(D')\|_2 \leq \sqrt{p}$ . By Lemma 25 the Gaussian mechanism on  $Q$  with noise  $\sigma_Q$  is  $(\sqrt{p}/\sigma_Q)$ -GDP. Un-scaling by  $s_j$  and the downstream synthetic generation are post-processing, which preserves GDP.  $\square$

Block fitting and clipping inflate the variance of this route, and the utility cost is reported in Section 6.5. The block count is set by the public rule  $m = \lfloor n_{\text{eff}}/(5p) \rfloor$ , the largest count leaving five observations per released coefficient per block. Each input is public:  $n$  is fixed under one-row replacement,  $p$  is the length of the required public clip box, and on a two-part positive channel  $n_{\text{eff}}$  scales  $n$  by the published positive fraction of the design, so the rule is an a priori cap rather than a quantity tuned on the analysis sample. At the designs reported below the rule gives  $m = 66$  on every full-sample channel, and on the two zero-inflated positive channels  $m = 26$  for scenario 4 and  $m = 44$  for scenario 8, at the published zero rates 0.600 and 0.338. When the covariates are public, a more accurate construction recovers the full-sample estimate exactly.

**Proposition 27** (Public-covariate exact release). *Assume  $\mathbf{X}$  is public and of full column rank with  $n > p$ , and  $y_i$  are bounded,  $|y_i| \leq B_y$ . Let  $\hat{\beta}_n$  and  $\hat{\sigma}_n^2$  denote the OLS estimators on  $(\mathbf{X}, \mathbf{y})$ , and let  $h_{ij}$  denote the entries of the hat matrix  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . Under one-row replacement of  $\mathbf{y}$ ,  $\hat{\beta}_n$  changes by at most  $\Delta_\beta = 2B_y \max_i \|(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i\|_2$  in  $L_2$  norm and  $\hat{\sigma}_n^2$  by at most  $\Delta_{\sigma^2} = 4B_y^2 K_X / (n - p)$ , where  $K_X = \max_i \max\{1 - h_{ii}, \sum_{j \neq i} |h_{ij}|\}$  depends only on the public  $\mathbf{X}$ . Define:*

1.  $\hat{\beta}_{n,\text{DP}} = \hat{\beta}_n + N(\mathbf{0}, \sigma_\beta^2 \mathbf{I}_p)$  with  $\sigma_\beta = \Delta_\beta / \mu_\beta$ .
2.  $\hat{\sigma}_{n,\text{DP}}^2 = \max\{\hat{\sigma}_n^2 + N(0, \sigma_v^2), 0\}$  with  $\sigma_v = \Delta_{\sigma^2} / \mu_v$ .
3.  $\mathbf{e} \sim N(\mathbf{0}, \hat{\sigma}_{n,\text{DP}}^2 \mathbf{I}_n)$  independent of the data.
4.  $\mathbf{e}_{\text{proj}} = c_n(\mathbf{I}_n - \mathbf{H})\mathbf{e}$  with  $c_n$  chosen so that  $\|\mathbf{e}_{\text{proj}}\|^2 / (n - p) = \hat{\sigma}_{n,\text{DP}}^2$ .
5.  $\mathbf{y}^* = \mathbf{X}\hat{\beta}_{n,\text{DP}} + \mathbf{e}_{\text{proj}}$ .

The release  $(\mathbf{X}, \mathbf{y}^*)$  is  $\{\mu_\beta^2 + \mu_v^2\}^{1/2}$ -GDP and satisfies  $\hat{\beta}_n(\mathbf{X}, \mathbf{y}^*) = \hat{\beta}_{n,\text{DP}}$  exactly.

*Proof.* Replacing one entry of  $\mathbf{y}$  with  $|y_i| \leq B_y$  changes  $\hat{\beta}_n$  by  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i (y'_i - y_i)$ , of  $L_2$  norm at most  $\Delta_\beta$ . For the variance, write the residual sum of squares as  $\mathbf{y}^\top (\mathbf{I}_n - \mathbf{H}) \mathbf{y}$ . The replacement  $\mathbf{y}' = \mathbf{y} + (y'_i - y_i) \mathbf{e}_i$  gives the exact identity

$$\mathbf{y}'^\top (\mathbf{I}_n - \mathbf{H}) \mathbf{y}' - \mathbf{y}^\top (\mathbf{I}_n - \mathbf{H}) \mathbf{y} = (y'_i - y_i) \left\{ (y'_i + y_i)(1 - h_{ii}) - 2 \sum_{j \neq i} h_{ij} y_j \right\}.$$

Over the bounded universe  $|y_j| \leq B_y$  the sum is at most  $B_y \sum_{j \neq i} |h_{ij}|$  in absolute value, and  $|y'_i + y_i| \leq 2B_y - |y'_i - y_i|$ . Writing  $u = |y'_i - y_i|$  and bounding both rates by their maximum, the change is at most  $\max\{1 - h_{ii}, \sum_{j \neq i} |h_{ij}|\} u (4B_y - u) \leq 4B_y^2 \max\{1 - h_{ii}, \sum_{j \neq i} |h_{ij}|\}$ . Maximising over  $i$  bounds

the change in  $\hat{\sigma}_n^2$  by  $\Delta_{\sigma^2} = 4B_y^2 K_X / (n - p)$ . The configuration  $y_j = -B_y \text{sign}(h_{ij})$  for  $j \neq i$  with  $y_i = -y'_i = B_y$  changes the residual sum of squares by  $4B_y^2 \sum_{j \neq i} |h_{ij}|$  in absolute value, so the row-sum factor in  $K_X$  cannot be dropped. Steps (1) and (2) are Gaussian mechanisms at these global sensitivities, so by Lemma 25 they are  $\mu_\beta$ - and  $\mu_v$ -GDP, and their composition is  $\{\mu_\beta^2 + \mu_v^2\}^{1/2}$ -GDP. The truncation at zero in step (2) is post-processing of the Gaussian release and leaves the guarantee unchanged. It also makes the variance used in steps (3) and (4) nonnegative, so both steps are well defined, with  $\mathbf{e}_{\text{proj}} = \mathbf{0}$  when the truncated value is zero. Steps (3)–(5) use only  $\hat{\beta}_{n,\text{DP}}$ ,  $\hat{\sigma}_{n,\text{DP}}^2$ , and the public  $\mathbf{X}$  with fresh randomness, so they are post-processing. Finally, the OLS estimator on  $(\mathbf{X}, \mathbf{y}^*)$  equals

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}^* = \hat{\beta}_{n,\text{DP}} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{e}_{\text{proj}},$$

and  $\mathbf{X}^\top (\mathbf{I}_n - \mathbf{H}) = \mathbf{0}$  makes the second term vanish exactly.  $\square$

Here exact fidelity and privacy coexist. The analyst refits OLS to  $(\mathbf{X}, \mathbf{y}^*)$  and recovers  $\hat{\beta}_{n,\text{DP}}$  exactly, so the released estimate is the genuine DP coefficient rather than a block-aggregate surrogate. The price is that the noise scales with the public bound  $B_y$ , so on designs with a wide  $y$  range the exact release is noisier per coefficient than the subsample-and-aggregate route, a comparison we quantify in Section 6.5.

When the covariates are sensitive but the response admits a public clip, a third route releases the sufficient statistics themselves, on a scale chosen to remove the amplification that defeats the raw release.

**Proposition 28** (Whitened sufficient-statistic release). *Let  $\mathbf{W}$  be a fixed public positive-definite matrix, and suppose each covariate row is clipped into a public support box on which  $\|\mathbf{W}\mathbf{x}_i\|_2 \leq B_x$ , and the response is publicly standardised and clipped so that  $|y_{s,i}| \leq B_y$ . Write  $\mathbf{S} = \sum_i (\mathbf{W}\mathbf{x}_i)(\mathbf{W}\mathbf{x}_i)^\top$ ,  $\mathbf{b} = \sum_i (\mathbf{W}\mathbf{x}_i)y_{s,i}$ , and  $q = \sum_i y_{s,i}^2$ . Under one-row replacement the  $L_2$  sensitivities of  $\text{vech}(\mathbf{S})$ ,  $\mathbf{b}$ , and  $q$  are at most  $\sqrt{2} B_x^2$ ,  $2B_x B_y$ , and  $B_y^2$ . The first is strict whenever the box contains no pair of one-hot rows of norm  $B_x$  with disjoint supports, and the last is attained. Releasing the three statistics under independent Gaussian noise at scales  $\sigma_k = \Delta_k / \mu_k$  is  $\{\mu_v^2 + \mu_b^2 + \mu_q^2\}^{1/2}$ -GDP, and any function of the released triple and public quantities, in particular the ridge solve  $(\mathbf{S}_{\text{DP}} + \lambda \mathbf{I})^{-1} \mathbf{b}_{\text{DP}}$  for any  $\lambda$  depending only on released and public quantities, inherits the guarantee by post-processing.*

*Proof.* Write  $\mathbf{u} = \mathbf{W}\mathbf{x}'_i$  and  $\mathbf{v} = \mathbf{W}\mathbf{x}_i$  for the whitened replaced and original rows, so  $\|\mathbf{u}\|_2, \|\mathbf{v}\|_2 \leq B_x$  over the clipped public universe. One-row replacement moves  $\mathbf{S}$  by the symmetric matrix  $\mathbf{A} = \mathbf{u}\mathbf{u}^\top - \mathbf{v}\mathbf{v}^\top$ , and for symmetric  $\mathbf{A}$ ,

$$2 \|\text{vech}(\mathbf{A})\|_2^2 = \|\mathbf{A}\|_F^2 + \sum_j A_{jj}^2,$$

since the Frobenius norm counts each off-diagonal entry twice where  $\text{vech}$  counts it once. The Frobenius part is  $\|\mathbf{A}\|_F^2 = \|\mathbf{u}\|^4 + \|\mathbf{v}\|^4 - 2(\mathbf{u}^\top \mathbf{v})^2 \leq 2B_x^4$ , and the diagonal part is  $\sum_j (u_j^2 - v_j^2)^2 \leq \sum_j u_j^4 + \sum_j v_j^4 \leq \|\mathbf{u}\|^4 + \|\mathbf{v}\|^4 \leq 2B_x^4$ , using  $(a - b)^2 \leq a^2 + b^2$  for  $a, b \geq 0$  and  $\sum_j u_j^4 \leq \|\mathbf{u}\|^4$ . Hence  $\|\text{vech}(\mathbf{A})\|_2 \leq \sqrt{2} B_x^2$ . Equality in the diagonal part forces  $\sum_j u_j^4 = \|\mathbf{u}\|^4$ , so  $\mathbf{u}$  is supported on a single coordinate at norm  $B_x$ , and likewise  $\mathbf{v}$ , while equality in the Frobenius part forces  $\mathbf{u}^\top \mathbf{v} = 0$ , so the two

supports are disjoint. A support box containing no such pair therefore makes the bound strict. The move of  $\mathbf{b}$  is  $\mathbf{u}y'_{s,i} - \mathbf{v}y_{s,i}$ , of norm at most  $2B_xB_y$ , and the move of  $q$  is  $|y'^2_{s,i} - y^2_{s,i}| \leq B_y^2$ , attained at  $y_{s,i} = 0, y'_{s,i} = B_y$ . By Lemma 25 the three Gaussian channels are  $\mu_v$ -,  $\mu_b$ - and  $\mu_q$ -GDP and compose in quadrature. The ridge solve, the recentred interval below, and any downstream synthetic generation read only the released triple and public quantities, so they are post-processing and leave the guarantee unchanged.  $\square$

The whitening  $\mathbf{W} = \Sigma^{-1/2}$ , with  $\Sigma$  the population second-moment matrix of the published data-generating process, conditions  $\mathbf{S}$  near  $n\mathbf{I}$ , which keeps the solve stable at the noise scales the budget dictates. Releasing the second-moment matrix under Gaussian noise is the analyze-Gauss mechanism [17], and the regularised solve of noised statistics with an adaptively released penalty is the AdaSSP estimator of Wang [51]. Here the penalty needs no budget, since the published process supplies a public error curve whose minimiser is deterministic. The released triple also carries the analyst's uncertainty. The ridge solve is biased toward zero, so the calibrated interval is centred at the unpenalised solve of the released statistics, whose plug-in covariance follows from the released matrix and the public noise scales, while the ridge solve remains the point estimate (Section ).

**Remark 19** (Where the bounds come from). *All three propositions require bounds that are fixed before the data are seen, and none survives a bound read off the sample. In Proposition 26 the box width sets  $s_j = (u_j - \ell_j)/m$  and hence the noise scale, in Proposition 27 the response bound sets  $\Delta_\beta$  and  $\Delta_{\sigma^2}$ , and in Proposition 28 the whitening matrix, the covariate support box behind  $B_x$ , and the response clip behind  $B_y$  play the same role, all functionals of the published process. A box taken from the fitted coefficients, or a  $B_y$  taken from  $\max_i |y_i|$ , makes the noise scale a function of the protected data themselves, so two neighbouring datasets receive different noise and no sensitivity bound holds. Such a release is not DP at any  $\epsilon$ . This is the absence of a guarantee, not a weakened one.*

For  $B_y$  the distinction is sharp. Where the response is bounded by its measurement scale, as for a physiological range or a bounded score,  $B_y$  is available a priori. Where it is not, as for a Gaussian response, no finite  $B_y$  exists and the release requires an explicit public clipping step  $y_i \mapsto \min\{\max(y_i, c_1), c_2\}$  applied before the mechanism. The estimand is then the clipped-OLS coefficient rather than  $\hat{\beta}_n$ . With  $[c_1, c_2]$  set at six standard deviations of the published response marginal the clip binds with probability below  $10^{-8}$  per row, so the two estimands coincide on any realised sample, though they remain distinct functionals and the guarantee attaches to the clipped one.

For the coefficient box we take  $[\ell_j, u_j] = [-\kappa|\beta_j|, \kappa|\beta_j|]$ , centred at zero, with  $\beta_j$  the population coefficient of the data-generating process and  $\kappa$  a public shrinkage factor. Centring at zero rather than at  $\beta_j$  leaves the width, and hence the noise, unchanged while assuming strictly less, since it posits a magnitude scale and neither a sign nor a location. The box therefore encodes a prior on the size of an effect, which is the form of knowledge a subject-matter literature routinely supplies, and Section 6.5 reports utility as a function of  $\kappa$  rather than fixing it, so that the effect of the assumption is explicit.

**Remark 20** (Extensions). *Three variants use the same  $\mu$ -GDP accounting. First, Proposition 27 extends to the ridge estimator  $\hat{\beta}_n^{(\lambda)} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{y}$ , though the construction is not identical. The coefficient sensitivity carries  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1}$  in place of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ . The residual sum of squares becomes the quadratic form  $\mathbf{y}^\top \mathbf{M} \mathbf{y}$  with  $\mathbf{M} = (\mathbf{I}_n - \mathbf{H}_\lambda)^\top (\mathbf{I}_n - \mathbf{H}_\lambda)$  and  $\mathbf{H}_\lambda$  the ridge hat matrix, so the variance sensitivity carries*

the row constant  $\max_i \max\{M_{ii}, \sum_{j \neq i} |M_{ij}|\}$  in place of  $K_X$ . The fitted component  $\mathbf{X}\{\mathbf{I}_p + \lambda(\mathbf{X}^\top \mathbf{X})^{-1}\} \hat{\boldsymbol{\beta}}_{n, \text{DP}}$  in step (5) makes the analyst's ridge refit return the DP-perturbed ridge estimator exactly. Second, when  $\mathbf{X}$  is itself sensitive, it can be released under a DP copula: a Gaussian mechanism on the marginal counts, whose  $L_2$  sensitivity is  $\sqrt{2}$  because a replace-one swap decreases one count and increases another, and a Gaussian mechanism on the covariate correlations. That channel releases  $\text{vech}(S)$ , the upper triangle including the diagonal of the second-moment matrix  $S = \mathbf{W}^\top \mathbf{W}/n$  of scores that a public, row-local map clamps to  $[-M, M]$ , with  $p_c$  the number of continuous covariates. Its  $L_2$  sensitivity is

$$\Delta_S = M^2 p_c / n,$$

and this is the sensitivity rather than merely a bound on it, since a replace-one swap between two rows that both lie on the clamp and are mutually orthogonal moves the released vector by exactly  $\Delta_S$ , and such a pair is realisable, since a record outside the public support clamps to  $\pm M$  exactly. It is the half-vectorisation that must be bounded, not the full matrix, whose Frobenius norm double-counts every off-diagonal entry and so exceeds  $\Delta_S$  by  $\sqrt{2}$  at that same pair. The channel releases the raw moment rather than the Fisher transform, whose sensitivity is unbounded as the correlation approaches  $\pm 1$ . Every channel is Gaussian, so all three compose with the coefficient release by Lemma 25, the three  $\mu$  values adding in quadrature,  $\mu^2 = \mu_{\text{marg}}^2 + \mu_{\text{corr}}^2 + \mu_{\text{coef}}^2$ . A Laplace histogram would leave the marginal channel outside this accounting, and the stated  $\mu$  would then omit its cost. The copula is mixed-margin, so the categorical covariates join it. A categorical covariate with  $K$  levels enters the same latent Gaussian through its  $K - 1$  non-reference level indicators, each mapped to a bounded normal score computed from its released DP proportions, which are already public, so the encoding spends no further budget. The score matrix then carries  $p_{\text{tot}} = p_c + \sum_c (K_c - 1)$  columns, every one clamped to  $[-M, M]$ , and the displayed sensitivity holds with  $p_{\text{tot}}$  in place of  $p_c$ , since the bound is a statement about the clamp alone. Attainment does not transfer. An indicator column takes one of  $K$  fixed winsorised values strictly inside the clamp, so no mixed pair attains  $M^2 p_{\text{tot}}/n$ , and for the mixed matrix  $\Delta_S$  is a strict upper bound that over-noises the categorical channel in the safe direction of a delivered  $\mu$  no larger than the claimed one. Synthetic levels are reconstructed from the sampled latents by a shifted argmax calibrated so the level proportions reproduce the released marginal. The fitted cross entries are read on the point-biserial scale of the two-point indicator scores, while the reconstruction needs the latent scale, so before sampling each categorical-continuous entry is divided by the public factor  $\phi(\Phi^{-1}(p_k))/\{p_k(1 - p_k)\}^{1/2}$ , the exact ratio between the two scales for a normal threshold at the released proportion  $p_k$ . The factor is a function of the released proportions alone, so the division is post-processing and the sensitivity statement above is unchanged. A softmax alternative that samples the level stochastically injects choice noise independent of the continuous latents and attenuates the recovered association by approximately half, so the argmax reconstruction is the default. The recovered joint is descriptive and carries three qualifications. Only the pairwise latent association transfers, the associations are encoded relative to each covariate's reference level, and the wider matrix raises the correlation channel's noise, a cost the default budget split absorbs since that channel is allocated little budget relative to the coefficient release. This copula preserves the covariate correlation structure that independent DP marginals discard, and at the budgets we evaluate it does so at no inferential cost. On a design with two continuous covariates at a true correlation of 0.606, and at  $\epsilon \geq 5$ , the released correlation has mean 0.602 with standard deviation 0.016 at  $\epsilon = 10$  and no release requiring projection back into  $[-1, 1]$ , while independent DP marginals return 0.000



by construction. At  $\epsilon = 10$  the analyst's joint decision agreement is 80.4% against the independent baseline's 79.6%, within Monte Carlo error of each other. At  $\epsilon = 1$  the release is still centred (0.602) but too dispersed to rely on from a single draw, so we claim the fidelity only for  $\epsilon \geq 5$ . The release is sensitive to one design choice, the score map. A box-uniform probability integral transform over the full public break range compresses the scores into a narrow band, so the second-moment signal falls below the noise the mechanism must add, whereas a public normal-score map at a mildly winsorised clamp spreads them and raises the signal at no privacy cost, since  $\Delta_S$  tracks the score bound and not the score spread. The mixed copula extends this recovery to the categorical-to-continuous associations that independent marginals also discard. We measure them by the correlation ratio  $\eta$ , the square root of the between-level share of the continuous covariate's variance, whose value under independence concentrates near the floor  $\{(K - 1)/(n - 1)\}^{1/2}$  rather than at zero. On the same design, the three-level covariate's association with income has  $\eta = 0.127$  in the data. Over 100 releases at  $\epsilon = 10$ , the mixed copula recovers a mean of 0.114 where independent marginals sit at their floor of 0.028, and the binary covariate's association with income,  $\eta = 0.046$  in the data, returns 0.053. At  $\epsilon = 5$  the recovered means bracket the data values, 0.137 on the nominal covariate against 0.127. The de-attenuation removes the systematic coarsening bias. The four recovered means run 86 to 116% of their targets at  $\epsilon = 10$  and 86 to 130% at  $\epsilon = 5$ , a residual deviation dominated by the metric's own floor, which inflates the small binary associations above their targets. The softmax reconstruction returns 0.059 on the nominal covariate at  $\epsilon = 10$ , about half the argmax recovery, so it is not the default. At  $\epsilon = 1$  the channel is noise-dominated and the deterministic reconstruction transfers spurious association, inflating the nominal covariate's  $\eta$  to 0.282, so we claim the categorical recovery, like the continuous one, only for  $\epsilon \geq 5$ . The continuous block pays for the wider matrix, its error at  $\epsilon = 10$  moving from 0.027 to 0.031 as  $p_{\text{tot}}$  rises from 2 to 5 on this design, and the de-attenuation leaves that error untouched, since only the categorical-continuous entries are rescaled. The analyst's refit is untouched throughout, since it recovers  $\hat{\beta}_{n,\text{DP}}$  exactly whatever joint structure the released covariates carry. Third, for a binary outcome, drawing  $y_i^* \mid \mathbf{x}_i^* \sim \text{Bernoulli}(\text{expit}(\mathbf{x}_i^{*\top} \hat{\beta}_{n,\text{DP}}))$  is GDP by post-processing when  $\mathbf{X}^*$  is public, though the analyst then recovers  $\hat{\beta}_{n,\text{DP}}$  only up to the irreducible Bernoulli sampling variance.

**Remark 21** (Calibrated standard errors). *The analyst fitting ordinary least squares to the synthetic data of Proposition 27 recovers  $\hat{\beta}_{n,\text{DP}}$  exactly, but the standard errors reported by the fit do not reflect the DP noise. Let  $\sigma_{\beta,j}$  denote the Gaussian noise scale on coordinate  $j$  of the released coefficient. The calibrated standard error is*

$$\text{SE}_{\text{corr},j} = \{\text{SE}_{\text{syn},j}^2 + \sigma_{\beta,j}^2\}^{1/2}. \quad (14)$$

*The noise scales  $\sigma_{\beta,j}$  are public under DP and can be released alongside the synthetic data without further privacy cost. Confidence intervals computed with  $\text{SE}_{\text{corr}}$  attain close to nominal coverage at the budgets we evaluate in Section 6.5.*

**Remark 22** (Membership-inference audit). *Formal  $(\epsilon, \delta)$ -DP, obtained here from  $\mu$ -GDP through Lemma 25, bounds the advantage of any membership-inference adversary: the false positive and false negative rates satisfy  $\text{FPR} + e^\epsilon \text{FNR} \geq 1 - \delta$  [28, 52]. That bound is too loose to audit against, since at  $\epsilon = 10$  it permits an adversary accuracy of 0.99995 and so rejects nothing. We therefore audit against the  $\mu$ -GDP trade-off directly. For a  $\mu$ -GDP mechanism the optimal distinguisher on a neighbouring pair has  $\text{AUC} \leq \Phi(\mu/\sqrt{2})$ , which is 0.566 at  $\epsilon = 1$  and 0.904 at  $\epsilon = 10$ . All five release routes lie within that*

bound at every budget (Section ). Empirical audits are not a substitute for the analytic guarantee, but they detect implementation errors the analytic composition cannot see. Omitting the variance release of Proposition 27 produces estimates that fail the audit even when the composition claims  $(\epsilon, \delta)$ -DP.

The three routes therefore delimit what the projection can promise under privacy. Exact inferential fidelity and DP coexist only when the covariates are public (Proposition 27). With sensitive covariates and a publicly clipped response, the whitened release (Proposition 28) keeps the coefficient and its calibrated interval while the estimate acquires a known ridge structure. Otherwise privacy requires relaxing exact fidelity to the block-aggregate estimand (Proposition 26), and the joint synthetic release is not itself DP, since the projection fits a copula whose marginal and correlation statistics depend on  $\mathbf{y}$  beyond  $\hat{\beta}_{n,DP}$ . In that case the guarantee attaches to the extracted coefficient, not to the joint table. This is the fidelity-privacy characterisation the projection affords.

## 6 Simulation study

We evaluate the theoretical results through a progressive simulation study with eleven scenarios of increasing complexity, each with known data-generating process and true regression coefficients.

### 6.1 Design

In each scenario,  $n = 2000$  observations are generated from a Gaussian copula whose  $5 \times 5$  correlation matrix is fixed within each scenario, with three predictor correlations strengthened from scenario 7 onwards, ensuring all regression coefficients are significant. The five variables are a continuous response, a continuous predictor (age), a second continuous predictor (income, skewed from scenario 2 onwards), a binary predictor (female), and a three-level categorical predictor (race). The scenarios add complexity progressively:

| Scenario | Feature added                            | Family   | Link     |
|----------|------------------------------------------|----------|----------|
| 1        | baseline (all normal)                    | Gaussian | identity |
| 2        | skewed predictor                         | Gaussian | identity |
| 3        | strictly positive response               | Gamma    | log      |
| 4        | zero-inflated response                   | Quasi    | log      |
| 5        | ceiling effect                           | Quasi    | log      |
| 6        | integer count response                   | Poisson  | log      |
| 7        | correlated predictors                    | Gamma    | log      |
| 8        | all combined                             | Quasi    | log      |
| 9        | bounded response $[0, 100]$              | Gaussian | identity |
| 10       | Gaussian response, correlated predictors | Gaussian | identity |
| 11       | binary response                          | Binomial | logit    |

Scenarios 1 and 2 test the canonical-link conditions (Theorem 4). Scenarios 3, 4, 5, 7, and 8 test the non-canonical partial moment bound (Theorem 6) with the log support transform, and scenarios 4, 5, and 8 carry zero-inflation. Scenario 6 tests the Poisson branch of the projection, where each entry of the projected response is rounded up with probability its fractional part and down otherwise,

a probabilistic rounding that preserves the sufficient statistic in expectation. Scenario 9 tests the bounded support transform. At the simulation seed, scenario 5’s ceiling at 100 does not bind, so it coincides with scenario 4 (Appendix S1). The projection column (Theorem 17) is evaluated across all eleven scenarios. Scenario 11 tests the binomial-logit canonical branch, the one exponential family in Theorem 17’s family table that the first ten scenarios do not exercise.

All simulations were conducted in R 4.6.1 [41], on Windows 11 and Linux (x86\_64). The copula-based generators use base R. The machine learning generators use version 1.35 of the Synthetic Data Vault (SDV) library [39], and the differentially private variants use SmartNoise (smartnoise-synth 1.0.8), both in Python 3.13. Monte Carlo summaries use 100 independent replications, reproducible from master seed 6489. The R implementation of all generators and the projection algorithm is described in Appendix S5. The projection generator is also distributed as the R package `synthexpln` (<https://github.com/li-ruijie/synthexpln>).

## 6.2 Methods compared

The machine learning generators are selected to cover the major categories of learning objectives, so that the comparison tests whether the failure is structural (as Proposition 13 predicts) rather than architecture-specific:

| Category             | Method         | Objective              |
|----------------------|----------------|------------------------|
| Adversarial          | CTGAN [57]     | Wasserstein distance   |
| Variational          | TVAE           | ELBO (reconstr. + KL)  |
| Copula (statistical) | GaussianCopula | Pairwise correlations  |
| Hybrid               | CopulaGAN      | Copula marginals + GAN |

*Ours: conditional CF quantile with empirical Bayes calibration on the  $h$ -scale (Section 5.1, pipeline in Appendix S5)*

The four methods represent the dominant paradigms for tabular synthetic data: adversarial, variational, copula-based, and hybrid. The machine learning generators used default hyperparameters from the SDV library, with 300 epochs and batch size 500 for CTGAN and TVAE and the default settings for GaussianCopula and CopulaGAN. No hyperparameter tuning was performed, as the purpose of the comparison is to test structural limitations predicted by Proposition 13, not to optimise any particular architecture. To confirm this, we also ran CTGAN with an aggressive tuning configuration (1,000 epochs, batch size 100,  $3 \times 256$  generator and discriminator, learning rate  $10^{-4}$ ). The tuned variant produces *larger* maximum deviations in two of four scenarios tested (e.g., 239% vs. 175% on zero-inflated data), consistent with the theoretical prediction that the failure is structural rather than architectural.

For each generator and scenario, synthetic data of the same size ( $n = 2000$ ) is generated and the same GLM is fitted. The primary metric is the absolute difference between the estimated and true regression coefficients.

### 6.3 Results

The results confirm a three-tier hierarchy, with all tiers read from the same 100 replications: the machine learning generators fail structurally (mean maximum deviation above 99%, Proposition 13), the calibrated copula achieves approximate fidelity controlled by residual skewness (3–29% mean maximum deviation, Theorem 6 and Theorem 14), and the projection achieves exact fidelity on the continuous scenarios (0%) and leaves a 2.3% residual on the count scenario under the probabilistic rounding that restores the integer support (Theorem 17).

Table 2 shows the estimated coefficient for the rare categorical predictor (race3, 15% prevalence) across all methods and a representative subset of scenarios. This coefficient is the most demanding test of inferential fidelity because its estimation depends entirely on the conditional distribution within a small subgroup.

Table 2: Estimated coefficient for race3 (true prevalence 15%) across generators and scenarios. “Proj” is the sufficient statistics projection. “Calib” is the calibrated copula with empirical Bayes shrinkage (Section 5, pipeline in Appendix S5). CopulaGAN is omitted for brevity. Its results follow the same pattern as CTGAN (see Table 19).

| Sc. | Features         | Orig  | Proj  | Calib | CTGAN | TVAE  | GausCop |
|-----|------------------|-------|-------|-------|-------|-------|---------|
| 1   | normal baseline  | 12.60 | 12.60 | 12.29 | 8.18  | 15.21 | −0.57   |
| 3   | positive (gamma) | 0.60  | 0.60  | 0.58  | 0.25  | 0.59  | −0.01   |
| 4   | zero-inflated    | 0.49  | 0.49  | 0.41  | 0.76  | 2.13  | −0.07   |
| 6   | count (Poisson)  | 0.30  | 0.30  | 0.29  | 0.06  | 0.34  | −0.02   |
| 8   | all combined     | 0.63  | 0.63  | 0.61  | 0.45  | 0.88  | 0.27    |
| 9   | bounded [0,100]  | 19.79 | 19.79 | 20.14 | 6.57  | 11.94 | 2.70    |

Three patterns emerge:

*CTGAN* shows systematic attenuation of categorical predictor coefficients and collapses the continuous predictor coefficient for age toward zero, reversing its sign in the process (−0.562 to 0.115 in scenario 1 and −0.029 to 0.002 in scenario 8). This is consistent with the mode-averaging mechanism described in Section 4, since the generator satisfies the discriminator by averaging over subgroups rather than preserving their conditional distributions.

*TVAE* exhibits high variance and inflates the magnitude of 28 of the 36 coefficients of the representative run (race3: 0.49 to 2.13 in scenario 4 and race2 0.28 to 1.06 in scenario 8). The unreliability makes it unsuitable for inference even when individual estimates happen to be close to the truth.

*SDV GaussianCopula* produces wrong-sign race3 coefficients in four of the six scenarios of Table 2 (last column). This confirms the finding of Section 3 that pairwise correlation matching, even with polychoric and polyserial estimation, does not preserve the conditional structure.

The calibrated copula recovers race3 within 3% in the identity-link scenarios, 1 and 9 of Table 2, consistent with  $k = 2$  matching being exact for identity  $h$  (Corollary 7(i)). For non-identity links the residual is 4–5% (log-link scenarios) and reaches 15% in the zero-inflated scenario, reflecting the conditional skewness bias of Theorem 6.

The sufficient statistics projection (“Proj”) recovers the original coefficient exactly in every continuous scenario, and to within the probabilistic-rounding draw of 0.1% on the count scenario, confirming the theoretical guarantee of Theorem 17. On the binary scenario, over 100 replications, the projection recovers the logistic coefficients exactly (0.0%, mean maximum standard-error deviation 2.2%), whereas the copula’s mean maximum deviation is 20.9%, so the exact guarantee extends to the binomial family. The implementation uses a Newton iteration on the log y scale, which maintains positivity by construction and achieves exact fidelity for both canonical and non-canonical links (see Remark 16).

Across all eleven scenarios the projection ranks first, exact on the ten continuous and binary scenarios and within the probabilistic-rounding residual on the count scenario. The calibrated copula is second on the ten scenarios for which the empirical Bayes calibration is defined, the binomial scenario having no calibrated arm. Over the 100 replications, no machine learning method brings its mean maximum deviation below 99% in any scenario. The replications carry this claim, since on the single run of Table 2 TVAE attains 39.7% on scenario 9, a draw the replications place at 100%. CopulaGAN is omitted from the main tables for brevity. Its single-run deviations (94–338%) follow the same pattern as CTGAN, consistent with their shared adversarial training objective (Table 19).

The standard errors move while the coefficients do not, and the released design is the reason. Theorem 17 fixes  $\mathbf{X}$  and synthesises only  $\mathbf{y}^*$ , so its part (c) makes the standard errors identical functions of  $(\hat{\beta}_n, \hat{\phi}_n, \mathbf{X})$ . The generator evaluated here instead releases a fully synthetic table, drawing  $\mathbf{X}^*$  from the copula before projecting the response. The projection still targets the original sufficient statistic, so  $\hat{\beta}_n^* = \hat{\beta}_n$  exactly, while the information matrix  $(\mathbf{X}^{*\top} \hat{W} \mathbf{X}^*)^{-1}$  is formed on a different design and the standard errors shift with it, by 1.0 to 6.9% across the scenarios.

Releasing  $\mathbf{y}^*$  on the original  $\mathbf{X}$ , the public-covariate regime of Proposition 18 and of Proposition 27, returns the standard errors of every scenario whose response is not zero-inflated to exactness or to the probabilistic-rounding residual (Table 3):  $< 10^{-5}\%$  for the Gaussian scenarios, 0.0002% for Gamma/log, 0.15% for Poisson/log. The two zero-inflated scenarios do not return, for a specific reason, since the response there is fitted as a two-part model on the positive observations and the copula resamples which observations are zero, so the positive-subset design differs even when the full design is held fixed. Theorem 17 governs a single-part fit on the full design. The two-part fit is governed by its own extension, stated after the table, which closes this case as well once the release preserves the zero pattern.

*The two-part projection.* A hurdle model for a semicontinuous response splits the likelihood into a zero block, a canonical-link GLM for the indicator  $Z_i = \mathbf{1}\{y_i > 0\}$  on the full design  $\mathbf{X}$ , and a positive block, a GLM for  $y \mid y > 0$  on the rows  $P = \{i : y_i > 0\}$  with design  $\mathbf{X}_P$ . The blocks are variation independent, so the log-likelihood is the sum  $\ell_Z(\boldsymbol{\gamma}) + \ell_P(\boldsymbol{\beta})$ , the maximisers are found separately, and the information matrix is block diagonal.

**Proposition 29** (Two-part projection). *Suppose the synthetic response  $\mathbf{y}^*$  preserves the zero pattern,  $\mathbf{1}\{y_i^* > 0\} = Z_i$  for every  $i$ , so the positive set  $P^* = P$  and the zero-block statistic  $\mathbf{X}^\top \mathbf{Z}^* = \mathbf{X}^\top \mathbf{Z}$ , and on the shared positive design matches the restricted sufficient statistic  $\mathbf{X}_P^\top \mathbf{y}_P^* = \mathbf{X}_P^\top \mathbf{y}_P$  and, for the standard errors, the saturated kernel condition of Theorem 17(ii) on  $\mathbf{y}_P$ . Then the two-part estimator is preserved*

Table 3: Projection standard-error deviation under full synthesis against the fixed design of Theorem 17,  $\max |\%|$  over coefficients, single run. The coefficients are machine-exact in both columns on the identity-link rows, within the solver tolerance (at most 0.005%) on the gamma and two-part rows, and carry the probabilistic-rounding draw on the Poisson row (2.0% under full synthesis, 3.3% on the fixed design).

| Scenario | Family / link                     | Full synthesis | Fixed design  |
|----------|-----------------------------------|----------------|---------------|
| 1        | Gaussian / identity               | 1.0%           | $< 10^{-5}\%$ |
| 3        | Gamma / log                       | 2.2%           | 0.0002%       |
| 6        | Poisson / log                     | 6.9%           | 0.15%         |
| 9        | Gaussian / identity (bounded)     | 2.0%           | $< 10^{-5}\%$ |
| 4        | quasi-gamma / log (zero-inflated) | 3.7%           | 9.6%          |
| 8        | quasi-gamma / log (zero-inflated) | 4.3%           | 7.1%          |

exactly,  $(\hat{\gamma}^*, \hat{\beta}^*) = (\hat{\gamma}, \hat{\beta})$ , and all standard errors, test statistics, confidence intervals, and  $p$ -values of both blocks coincide.

*Proof.* By separability the two blocks are fitted independently. Preserving the indicator makes  $\mathbf{Z}^* = \mathbf{Z}$ , so the zero block is fitted on the identical data  $(\mathbf{X}, \mathbf{Z})$  and every zero-block summary is unchanged. It also fixes  $P^* = P$ , so the positive block is fitted on the same design  $\mathbf{X}_P$ . There the hypotheses on the restricted sufficient statistic and the saturated kernel are exactly those of Theorem 17, which gives  $\hat{\beta}^* = \hat{\beta}$  and the positive-block summary invariance of its part (c). Block diagonality of the information carries the two blocks' standard errors to the joint fit.  $\square$

The condition is that the release preserves the zero pattern, the two-part analogue of releasing the design unchanged. The zero model is itself a canonical GLM, so under privacy it composes in quadrature through the same  $\mu$ -GDP accounting as the binomial channel of Proposition 26. Holding the zero pattern fixed and projecting the positive component returns the two zero-inflated cells of Table 3 to exactness, the positive-block standard-error deviation falling from 9.6% and 7.1% to a 100-replication mean of 0.0001% on scenario 4 and 0.0002% on scenario 8, the zero block exact by construction on every replication.

Table 4 shows the age coefficient, a continuous predictor present in all scenarios. The projection recovers it exactly, the count-scenario rounding draw moving it by less than 0.05%. The calibrated copula recovers it within 12% in every scenario shown, and reaches 16.5% on the zero-inflated scenario 4, which Table 4 does not include (Appendix S1). CTGAN collapses it to near zero in every one of the five scenarios shown, reversing its sign in three of them.

Table 4: Estimated coefficient for age (continuous predictor) across generators and scenarios.

| Sc. | Features         | Orig   | Proj   | Calib  | CTGAN  | TVAE   | GausCop |
|-----|------------------|--------|--------|--------|--------|--------|---------|
| 1   | normal baseline  | -0.562 | -0.562 | -0.558 | 0.115  | -0.705 | -0.552  |
| 3   | positive (gamma) | -0.026 | -0.026 | -0.024 | 0.003  | -0.036 | -0.025  |
| 6   | count (Poisson)  | -0.013 | -0.013 | -0.012 | -0.000 | -0.016 | -0.012  |
| 8   | all combined     | -0.029 | -0.029 | -0.025 | 0.002  | -0.048 | -0.043  |
| 9   | bounded [0,100]  | -0.967 | -0.967 | -1.010 | -0.019 | -1.005 | -0.929  |

These tables show a single representative run. To confirm the patterns are not seed-dependent, 100 independent replications were run for all seven methods, with each replication re-training the ML generators from scratch. Over the 100 runs the projection’s max absolute percentage deviation (mean  $\pm$  sd) is 0.0% (0.0) for all continuous scenarios and 2.3% (1.1) for Poisson (rounding). The copula spans 14–49% (7–23) and the calibrated copula 3–29% (1–9). The machine learning generators span 102–145% (3–55) for CTGAN, 100–603% (3–183) for TVAE, 100–141% (0–40) for GaussianCopula, and 102–270% (5–110) for CopulaGAN. Full results are in Section .

## 6.4 Real-data application

As a test on real data, we apply the projection and copula generators to the baseline survey of the Population Health Index (Population Health Index (PHI)) study [58], a longitudinal health survey on 1,942 community-dwelling adults in the Central region of Singapore conducted from November 2015 to January 2019. The analysis models waist circumference as a linear function of age, sex, race (4 levels), education (3 levels), and housing type (9 levels), giving  $p = 16$  parameters with  $n = 1,927$  complete cases. This specification is chosen solely to exercise the generators on a real dataset with four categorical predictors (16 parameters) and sparse cross-strata. It is not representative of any substantive research question addressed by the original study.

The PHI dataset presents two challenges absent from the simulation scenarios. First, the 9 housing categories create 216 possible cross-strata from the interaction of all categorical predictors, with 26 singleton and 15 doubleton strata. Second, the waist circumference model has  $p = 16$ , nearly three times the simulation models ( $p = 6$ ).

When the bootstrap produces a non-singular design matrix, the sufficient statistics projection achieves exact coefficient recovery (0.0% maximum deviation, consistent with Theorem 17). In the remaining runs (42 of 100, against an expected rate of 37%) the bootstrap omits the lone observation in the rarest housing category, leaving  $\mathbf{X}^T \mathbf{X}$  singular so that exact recovery is no longer possible. Pre-masking the 9 housing levels to 3 removes the sparse housing levels responsible for the singular design and lowers the maximum single-person influence from 1,103% to 29% (Section ). Without pre-masking, the copula distorts housing coefficients by up to 1,084% (including sign reversals). The copula’s failure on this dataset is consistent with the sparse strata analysis of Section 5, since the 26 singleton and 15 doubleton cross-strata leave the per-stratum conditional moments too noisy to estimate accurately.

As a second analysis on the same data, following the source study’s model of healthcare use on the PHI health index [58], we fit a two-part model of specialist-outpatient visits (`util_soc`, 74% zeros) on the health index and demographics. The projection reproduces the analyst’s two-part fit, the logistic zero block exactly, the positive-block health-index coefficient within 0.9%, and every remaining positive-block coefficient within 10.5%, that maximum falling on a race cell holding 29 positive-block observations. The copula moves the health-index coefficient by 6%, the zero-model health-index coefficient by 15%, and others by up to 112% (Table 22).

As a fully reproducible demonstration, we also apply the generators to the diamonds dataset from the `ggplot2` R package [55, Section 3.10] ( $n = 5,000$  subsample,  $p = 20$ ), modelling price as a gamma regression with log link on carat, cut (5 levels), color (7 levels), clarity (8 levels), and depth.

With 20 parameters and three ordinal categorical predictors, the copula distorts rare levels by up to 1,065% (e.g., the coefficient for cutGood is inflated from 0.004 to 0.049). The projection achieves 0.1% maximum deviation across all 20 coefficients.

## 6.5 Differentially private synthetic data

We evaluate the DP extension of Section 5.4 against both the analytic baseline of the Gaussian mechanism applied to the sufficient statistic  $\mathbf{X}^\top \mathbf{y}$  (Section ) and the machine learning generators of Section 4 operated under DP training.

Table 5 reports decision agreement at  $\varepsilon = 10$ ,  $\delta = 10^{-6}$  on scenario 1 ( $n/p = 333$ , Gaussian identity link, 200 replications for the three projection routes and 15 for the baseline). The Gaussian mechanism on  $\mathbf{X}^\top \mathbf{y}$  recovers the sign of only 43% of coefficients and the significance of 92%, and only 8% of coefficients, averaged over replications, have the original value inside the synthetic 95% confidence interval. Its sensitivity  $\Delta = 2B_x B_y$  is computed from the public covariate and response bounds of Remark 19, fixed before the data are seen, so the comparison is on equal terms, and the baseline is not disadvantaged by an assumption from which the routes are exempt.

Placing the noise on the coefficient instead, both the subsample-and-aggregate route (Proposition 26) and the public-covariate exact release (Proposition 27) recover the analyst’s decisions more faithfully, at 100% and 85% joint decision agreement respectively, the former under a public coefficient box of  $\pm 3|\beta_j|$  and the block-count rule. Releasing the whitened statistics instead of the raw ones (Proposition 28) recovers 100%, which locates the baseline’s failure in its scale and its unregularised solve rather than in the statistic.

Table 5: Decision agreement on scenario 1 at  $\varepsilon = 10$ ,  $\delta = 10^{-6}$ , 200 replications for the three projection routes and 15 for the Gaussian baseline. The projection routes are calibrated to  $\mu = 1.848$ -GDP, equivalent to  $(\varepsilon, \delta) = (10, 10^{-6})$  by (13). “Sign” is the mean fraction of coefficients whose sign the synthetic refit preserves, averaged over replications. “Sig.” is the analogous mean fraction for significance at  $\alpha = 0.05$ . “Both” is the mean fraction of coefficients agreeing on both. “Cov.” is the mean fraction of coefficients whose original value lies within the synthetic 95% confidence interval, computed for the projection routes with the calibrated standard error (14). The subsample-and-aggregate route uses the public coefficient box  $\pm 3|\beta_j|$  of Remark 19, and Table 6 reports all three widths. The baseline’s sensitivity  $\Delta = 2B_x B_y$  is computed from the same public bounds,  $B_x$  being the largest covariate norm the published design admits and  $B_y$  the response bound of Remark 19.

| Mechanism                                           | Sign | Sig. | Both | Cov. |
|-----------------------------------------------------|------|------|------|------|
| Gaussian on $\mathbf{X}^\top \mathbf{y}$ (baseline) | 43%  | 92%  | 41%  | 8%   |
| Subsample-and-aggregate (Proposition 26)            | 100% | 100% | 100% | 96%  |
| Public-covariate exact (Proposition 27)             | 85%  | 100% | 85%  | 96%  |
| Whitened statistics (Proposition 28)                | 100% | 100% | 100% | 98%  |

The three routes trade scope against fidelity as set out in Section 5.4: subsample-and-aggregate covers any GLM, whatever the link, at approximate fidelity, the public-covariate release attains exact fidelity when  $\mathbf{X}$  is public, and the whitened release runs wherever the response admits a public clip.

Table 6 contrasts the three routes at  $\varepsilon = 10$  on scenario 1, and reports the subsample-and-aggregate



route at three widths of the public box, so that the cost of the prior in Remark 19 is visible rather than absorbed into a single choice. Where the noise dominates, utility degrades linearly in  $\kappa$ , as the relative coefficient noise  $2\kappa p^{1/2}/(m\mu)$  predicts, so a box three times wider costs about three times the noise, which shows that the box supplies a magnitude scale and no more. At  $\varepsilon = 10$  under the block-count rule the noise is small enough that the block-aggregation error floors the deviation for  $\kappa \leq 3$ , and the narrowest box additionally clips enough block fits that its coverage erodes, a bias the wider boxes avoid. The ridge, sensitive-covariate, and binary-outcome variants use the same  $\mu$ -GDP accounting (Remark 20) and are not tabulated separately.

The public-covariate route’s worst-case coefficient deviation reflects the response it must protect more than the route itself. Its noise scales with  $B_y$ . On scenario 1 the response is Gaussian and therefore unbounded, a public bound exists only through the clipping step of Remark 19, and the resulting bound is wide. The route’s mean worst-case deviation at  $\varepsilon = 10$  is 9060%, against 21% for subsample-and-aggregate at  $\kappa = 3$  under the block-count rule. On scenario 9, whose response is structurally confined to  $[0, 100]$  by its measurement scale, the same quantity is 55% against 20%, so the bounded response reduces the gap from four hundredfold to below threefold. The public-covariate route is the right choice exactly when the bounded-response premise of Proposition 27 is true of the measurement rather than imposed on it.

Decision agreement, being bounded and monotone in the budget, is more stable than the worst-case deviation, and it is the quantity an analyst acts on. On the same two scenarios the public-covariate route preserves 85% and 88% of joint sign-and-significance decisions, against 100% on both for subsample-and-aggregate at  $\kappa = 3$ . The worst-case deviation moves by two orders of magnitude between the scenarios while decision agreement moves by three points, because an isotropic perturbation expressed as a percentage is dominated by whichever substantive coefficient is smallest, and is therefore a statement about the design as much as about the mechanism. We report decision agreement throughout for that reason.

For comparison, DP-SGD-trained CTGAN and PATE-CTGAN, the Private Aggregation of Teacher Ensembles GAN (PATE-GAN) construction [27] applied to CTGAN (SmartNoise implementations, 25 replications), give mean maximum coefficient deviations of 112–352% and 65–262% at  $\varepsilon = 10$ , at the library default  $\delta \approx 1.1 \times 10^{-5}$ , on the six scenarios of Table 2, and do not, even absent DP, satisfy inferential fidelity (Proposition 13). In two scenarios one PATE-CTGAN replication’s evaluation fit fails on the degenerate synthetic draw and is excluded from that cell’s mean. The non-private machine learning generators of Table 2, operated on the same simulation scenarios, have no DP guarantee and already show mean maximum deviations of at least 100%. Adding DP noise degrades them further.

Table 7 reports how the two coefficient-space routes depend on the privacy budget on scenario 1, with subsample-and-aggregate at the central box width  $\kappa = 3$ . Under the block-count rule the subsample-and-aggregate route holds every decision from  $\varepsilon = 5$  and 95% of joint decisions at  $\varepsilon = 2$ . At  $\varepsilon = 1$  its joint agreement is 80%, and its worst coefficient deviation rises from 21% at  $\varepsilon = 10$  to 158% there, so only the smallest budget defeats it. The public-covariate route degrades more gently but from a lower base, its joint agreement climbing from 68% at  $\varepsilon = 1$  to 85% at  $\varepsilon = 10$ . The whitened release follows the same pattern, its joint agreement climbing from 80% at  $\varepsilon = 1$  to 100% at  $\varepsilon = 10$ .

Table 6: The three  $\mu$ -GDP routes contrasted on scenario 1 at  $\varepsilon = 10$ ,  $\delta = 10^{-6}$  ( $\mu = 1.848$ ), 200 replications. “DP scope” is what the guarantee attaches to: the extracted coefficient for subsample-and-aggregate, whose fidelity is approximate, the joint synthetic release for the public-covariate route, whose fidelity is exact, or the released statistic triple for the whitened route. “Bounds” is the public coefficient box  $\pm\kappa|\beta_j|$  of Remark 19, the public response bound  $B_y$ , or the whitened row and response bounds  $B_x, B_y$  of Proposition 28. “Sign” and “Both” are the mean fractions of coefficients with sign agreement and with joint sign and significance agreement, averaged over replications. “Cov.” is the calibrated 95% coverage (14). The subsample-and-aggregate worst coefficient deviation is 29%, 21%, and 35% at the three box widths, the public-covariate route’s is dominated by a coefficient small on its native scale, and the whitened route’s is 28% under the public-dispersion mode that matches the subsample-and-aggregate rows. The narrowest box’s 85% coverage is clipping bias, visible once the block-count rule shrinks the noise that used to mask it. The ridge, sensitive-covariate, and binary-outcome variants use the same accounting (Remark 20).

| Route                   | DP scope                | Bounds         | Sign | Both | Cov. |
|-------------------------|-------------------------|----------------|------|------|------|
| Subsample-and-aggregate | extracted $\hat{\beta}$ | $\kappa = 1.5$ | 100% | 100% | 85%  |
|                         |                         | $\kappa = 3$   | 100% | 100% | 96%  |
|                         |                         | $\kappa = 5$   | 100% | 100% | 96%  |
| Public-covariate exact  | joint release           | $B_y$          | 85%  | 85%  | 96%  |
| Whitened statistics     | statistic triple        | $B_x, B_y$     | 100% | 100% | 98%  |

and its worst deviation falling from 134% to 28% (Table 23).

The remaining degradation is what the theory predicts. The relative coefficient noise  $2\kappa p^{1/2}/(m\mu)$  reaches 94% at  $\varepsilon = 1$ , and  $\Phi(1/0.94) = 0.86$  is close to the 84% sign agreement observed there.

The calibrated interval remains reliable. The standard errors of (14) attain 95 to 96% coverage of the nominal 95% intervals on both coefficient-space routes at every budget, including those at which the point estimates are destroyed. The whitened release holds 95 to 100% likewise, with its interval centred at the unpenalised solve of the released statistics (Proposition 28). An analyst working from the released data at  $\varepsilon = 1$  is therefore left uninformed rather than misled, which is the property one wants of a private release and the reason we report coverage alongside agreement throughout.

Table 7: Scaling of the two routes with the privacy budget on scenario 1 at  $\delta = 10^{-6}$ ,  $B = 200$  Monte Carlo replications. Here  $\varepsilon \in \{1, 2, 5, 10\}$  corresponds to  $\mu \in \{0.237, 0.448, 1.020, 1.848\}$  by (13). “Max |%|” is the worst substantive-coefficient percentage deviation, reported for subsample-and-aggregate only, the public route’s being dominated by a coefficient small on its native scale. “Cov.” is coverage of the nominal 95% intervals computed with the calibrated standard error (14). The subsample-and-aggregate route uses the block-count rule  $m = \lfloor n/(5p) \rfloor$ , which is 66 at this design.

| $\varepsilon$ | Subsample-and-aggregate |      |      |      |      | Public-covariate |      |      |
|---------------|-------------------------|------|------|------|------|------------------|------|------|
|               | Max  %                  | Sign | Sig. | Both | Cov. | Sig.             | Both | Cov. |
| 1             | 158%                    | 84%  | 93%  | 80%  | 95%  | 96%              | 68%  | 95%  |
| 2             | 84%                     | 98%  | 96%  | 95%  | 95%  | 96%              | 75%  | 95%  |
| 5             | 37%                     | 100% | 100% | 100% | 96%  | 98%              | 82%  | 96%  |
| 10            | 21%                     | 100% | 100% | 100% | 96%  | 100%             | 85%  | 96%  |

The coefficient-space routes inherit the design’s conditioning. The public-covariate global sensitivity

$$\Delta_\beta = 2B_y \max_i \|(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i\|_2$$

grows with the condition number of  $\mathbf{X}^\top \mathbf{X}$ , and the subsample-and-aggregate block fits lose precision as the block size  $n/m$  falls, so both shed utility on high-dimensional or near-collinear designs. The whitened release confronts the same geometry before the noise is added, since  $\mathbf{W}$  restores the conditioning of the released matrix itself. On the well-conditioned scenarios evaluated here ( $n = 2000$ ,  $n/p$  of order  $10^2$ ) the noise is controlled, in contrast to the high-dimensional and collinearity regimes of Remark 15, where DP noise is unavoidable regardless of the mechanism.

## 7 Discussion

Inferential fidelity for generalised linear models is a property of the conditional distribution  $F(Y \mid \mathbf{X})$ , not of the joint or marginal distributions. The conditional moment conditions derived here provide a transparent, verifiable bridge between the synthesis and analysis stages. For any generalised linear model family, the required moment order is determined by the smoothness of the inverse support transform, and the sufficient statistics projection gives a constructive proof that exact fidelity is achievable.

The practical implication is that when the downstream analysis is explanatory, synthetic data generators should target conditional moment matching rather than distributional matching. When the goal is purely predictive, distributional objectives remain appropriate (Remark 23). The Gaussian copula, when combined with a stratified conditioning strategy and support-respecting transformations, provides a computationally tractable framework for this. The sufficient moment order depends on the GLM family through the smoothness of the inverse support transform  $h^{-1}$ : for identity  $h$  (Gaussian response),  $k = 1$  suffices. For  $h = \log$  (positive response), the bias decays as  $1/(k + 1)!$ , making  $k = 2$  sufficient in most practical settings.

Since copulas are inherently bivariate constructions, a natural question is why we do not use vine copulas [1, 3], which decompose multivariate dependence into a hierarchy of bivariate copulas and are the standard extension to more than two variables. Vines can capture asymmetric, tail-dependent, and non-elliptical dependence that a Gaussian copula cannot, and the same argument applies to any method that uses copulas, including SDV’s GaussianCopula synthesiser. However, the copula choice affects the distributional realism of the synthetic data but not the inferential guarantees.

A comparison of eight generation frameworks on a randomised trial illustrates the distinction in practice. Its most effective framework draws the baseline covariates from a vine copula and every post-randomisation variable from a conditional regression model, the architecture the conditional moment conditions require, and the authors decline to claim that the released data support substantive conclusions [40]. The projection supplies the guarantee that comparison stops short of.

When combined with the sufficient statistics projection, the copula contributes only the residual component  $(\mathbf{I} - \mathbf{H})\mathbf{y}_{\text{cop}}$ , while the fitted component  $\mathbf{X}\hat{\boldsymbol{\beta}}_n$  is determined entirely by the projection. Replacing the Gaussian copula with a vine would improve the distributional fidelity of these residuals

(KS distance, tail dependence) without affecting coefficient recovery.

Without the projection, upgrading to a vine still does not resolve the inferential failures documented in Section 6, because those failures are structural, and distributional objectives, however flexible, do not control the conditional moments that determine regression coefficients. Moreover, the Gaussian copula combined with the projection already achieves average KS distances of 0.02–0.04 across all scenarios (Section 6), which is near the sampling noise floor for  $n = 2,000$ . The marginal distributional gains from a vine would be small relative to the additional computational and modelling complexity (vine structure selection, bivariate family selection at each edge).

**Remark 23** (Inferential and predictive fidelity). *Inferential fidelity implies predictive fidelity. If  $\hat{\beta}_n^* = \hat{\beta}_n$  and  $\hat{\phi}_n^* = \hat{\phi}_n$ , then all point predictions, prediction intervals, and model selection outcomes are identical. Information criteria such as AIC and BIC match exactly for the Gaussian family, where the normalising constant depends on the data only through  $\mathbf{y}^\top \mathbf{y}$ . For other families they match only up to a constant common to all candidate models, so the absolute values can differ while the model ranking is preserved. The converse does not hold, since a synthetic dataset can produce accurate predictions while distorting individual coefficients, if the distortions cancel in the linear predictor. For non-parametric prediction methods, inferential fidelity is undefined, and distributional fidelity may be more relevant.*

The connection to the synthetic data inference literature is instructive. Multiple-imputation synthesis requires the synthesis model to be congenial to the analyst’s model [34, 42]. Wilde et al. [56] require the generator to equal the true data-generating process for valid Bayesian inference. Our conditional moment conditions provide a middle ground. The generator need not specify the full conditional distribution correctly, but must preserve specific moments of it. The sufficient moments are determined by the analyst’s model family through the smoothness of  $h^{-1}$ , connecting the synthesis and analysis stages through a transparent, verifiable condition.

The present framework considers a single synthetic dataset. The multiple synthetic datasets approach of Raghunathan, Reiter and Rubin [42] and Reiter [43] uses combining rules across  $m$  synthetic copies to capture generative variability. For the projection (Theorem 17), each copy satisfies the sufficient statistic constraint exactly, so every copy returns the same point estimate. The between-imputation variance in the combining rules of Rubin [46, Section 3.1] is therefore zero, and the total variance reduces to the within-imputation component. A single release is fully informative for the fitted coefficients, and multiple copies add generative variability only in the residual structure, not in the estimates. The reproduced standard error is the frequentist uncertainty the original analysis reports, not an understatement of it, since the synthesis adds no variance to propagate.

A generator that matches no statistic exactly must instead inflate the reported standard error by a correction factor, and the reanalysis of Decruyenaere et al. [11] measures the cost. The correction lowers the false positive rate of the best parametric synthesiser from 15.3% to 4.2% while its true positive rate falls from 82.1% to 60.7%, and it leaves the best deep generator at 12.5%. The projection needs no correction of this kind, since it reproduces the original standard error exactly.

When the analyst’s model is unknown in advance, the public-covariate release (Proposition 18) still delivers the exact estimate to a second analyst fitting a different exponential-family model, so the

guarantee does not require anticipating the family.

The sparse strata problem (Section 5) is the main practical limitation. Theorem 14 shows that empirical Bayes shrinkage provides an optimal bias–variance trade-off with an explicit crossover rate at  $n \sim S\sigma^2/\tau^2$ . In the PHI application (9 housing levels creating sparse categorical cross-strata), this crossover is reached for the 26 singleton and 15 doubleton strata, and the shrinkage estimator provides meaningful improvement over the unshrunk alternative.

For machine learning generators, Proposition 13 and the pseudo-true parameter analysis (Section 4) show that distributional closeness provides no guarantee of inferential closeness. The ridge-type bias formula (6) quantifies this, since for orthogonal predictors the attenuation is strongest for those with small variance, namely the rare categorical levels that carry the most informative regression coefficients. This is not a correctable deficiency of any particular architecture but a structural consequence of optimising distributional rather than conditional objectives. The impossibility applies to machine learning as a *generative* model. When machine learning instead estimates nuisance functions within a doubly robust framework that explicitly targets the inferential estimand [5], the conditional structure is controlled by the correction term, not by the generator’s training objective (Remark 9).

A practical concern is that the downstream analyst receives only the synthetic dataset and cannot verify the generator’s assumptions against the original data. For direction-agnostic fidelity, the analyst can compute empirical tail dependence coefficients  $\hat{\lambda}_U$  and  $\hat{\lambda}_L$  on the synthetic bivariate margins. If  $\max\{\hat{\lambda}_U, \hat{\lambda}_L\}$  is large, it flags dependence that the Gaussian copula may not have captured, and the direction-agnostic guarantee of Theorem 15 may not hold for all model directions. That maximum is the copula-side factor in the error bound of Section and the quantity Remark 11 requires to vanish, whereas the asymmetry  $|\hat{\lambda}_U - \hat{\lambda}_L|$  alone would pass a copula with symmetric tail dependence. In such cases, the analyst should use the single-response projection (Theorem 17) for the specific model of interest, which provides exact fidelity regardless of the copula’s dependence structure.

The sufficient statistics projection (Section 5.3) resolves this gap when the analysis model is known. For canonical-link GLMs, the projection achieves exact coefficient preservation across all  $2^p - 1$  submodels from only  $p + 1$  constraints, leaving  $n - p - 1$  degrees of freedom for generating novel values. The privacy assessment (distance to closest record, membership inference [48]) confirms that the projected data is not individually identifiable, the membership-inference area under the receiver operating characteristic curve being  $\approx 0.49$ , indistinguishable from chance, since the  $p$  embedded summary statistics are aggregate quantities, while the  $n - p - 1$  residual degrees of freedom carry genuine novelty from the copula.

However, upgrading this empirical privacy to formal differential privacy reveals a fundamental tension. The  $L_2$  sensitivity of the sufficient statistic  $\mathbf{X}^\top \mathbf{y}$  is high, and the Gaussian mechanism noise on the raw scale is amplified through the unregularised solve by  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , destroying inferential fidelity at practical privacy budgets (Remark 14). On the raw sufficient statistic, exact inferential fidelity and differential privacy conflict, the one preserving  $\mathbf{X}^\top \mathbf{y}$  and the other perturbing it.

Section 5.4 reconciles them under  $\mu$ -GDP: the public-covariate exact release (Proposition 27) attains

fidelity and privacy together for Gaussian OLS, the whitened sufficient-statistic release (Proposition 28) removes the amplification at its source wherever the response admits a public clip, and the subsample-and-aggregate route (Proposition 26) covers any GLM, whatever the link, at approximate fidelity (see Section ).

When the covariates are themselves sensitive, the coefficient release composes with a differentially private mixed-margin copula on the covariates, in which the categorical covariates join the released correlation, so the whole table is released privately and the joint covariate structure is preserved approximately (Remark 20, Section ). The residual limitation is that this recovery is descriptive and partial. Only the pairwise latent association transfers, and we claim the recovery only for  $\epsilon \geq 5$ , while the refit coefficients are exact at every budget.

When the analysis is exploratory, the copula with conditional moment calibration provides approximate fidelity across a broad class of models. The sufficient statistics projection is strongest for pre-registered studies, clinical trials, and regulatory submissions, where the analysis plan is fixed before data release.

## Data availability

Simulation code and the projection algorithm are at <https://github.com/li-ruijie/synthetic-inference>, and the R package `synthexpln` at <https://github.com/li-ruijie/synthexpln>. The diamonds dataset is public, in the `ggplot2` R package. The Population Health Index dataset is restricted under the National Healthcare Group data governance framework, Singapore, and access requests go to the corresponding author.

## Acknowledgement

Funded by MOH-SEEDFD24jul-0001 from the National Medical Research Council, Singapore.

## References

- [1] Kjersti Aas et al. ‘Pair-Copula Constructions of Multiple Dependence’. In: *Insurance: Mathematics and Economics* 44.2 (Apr. 2009), pp. 182–198. ISSN: 0167-6687. DOI: 10 . 1016 / j . insmatheco . 2007 . 02 . 001. (Visited on 08/04/2026).
- [2] Martin Abadi et al. ‘Deep Learning with Differential Privacy’. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. Vienna Austria: ACM, Oct. 2016, pp. 308–318. ISBN: 978-1-4503-4139-4. DOI: 10 . 1145/2976749 . 2978318. (Visited on 16/01/2024).
- [3] Tim Bedford and Roger M. Cooke. ‘Vines - A New Graphical Model for Dependent Random Variables’. In: *The Annals of Statistics* 30.4 (Aug. 2002), pp. 1031–1068. ISSN: 0090-5364, 2168-8966. DOI: 10 . 1214/aos/1031689016. (Visited on 08/04/2026).

- [4] David A. Belsley, Edwin Kuh and Roy E. Welsch. *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. Wiley Series in Probability and Statistics. New York: Wiley, Jan. 1980. ISBN: 978-0-471-05856-4. DOI: 10.1002/0471725153.
- [5] David Bruns-Smith et al. ‘Augmented Balancing Weights as Linear Regression’. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 88.3 (July 2026), pp. 699–723. ISSN: 1369-7412. DOI: 10.1093/jrssb/qkaf019. (Visited on 18/08/2026).
- [6] Mark Bun and Thomas Steinke. ‘Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds’. In: *Theory of Cryptography*. Ed. by Martin Hirt and Adam Smith. Berlin, Heidelberg: Springer, Jan. 2016, pp. 635–658. ISBN: 978-3-662-53641-4. DOI: 10.1007/978-3-662-53641-4\_24.
- [7] Stef van Buuren and Karin Groothuis-Oudshoorn. ‘Mice: Multivariate Imputation by Chained Equations in R’. In: *Journal of Statistical Software* 45 (Dec. 2011), pp. 1–67. ISSN: 1548-7660. DOI: 10.18637/jss.v045.i03.
- [8] Victor Chernozhukov et al. ‘Double/Debiased Machine Learning for Treatment and Structural Parameters’. In: *The Econometrics Journal* 21.1 (Feb. 2018), pp. C1–C68. ISSN: 1368-4221, 1368-423X. DOI: 10.1111/ectj.12097. (Visited on 04/11/2025).
- [9] R. Dennis Cook. ‘Detection of Influential Observation in Linear Regression’. In: *Technometrics* 19.1 (Feb. 1977), pp. 15–18. ISSN: 0040-1706. DOI: 10.1080/00401706.1977.10489493. (Visited on 08/04/2026).
- [10] E. A. Cornish and Ronald Aylmer Fisher. ‘Moments and Cumulants in the Specification of Distributions’. In: *Revue de l’Institut International de Statistique / Review of the International Statistical Institute* 5.4 (Jan. 1938), pp. 307–320. ISSN: 0373-1138. DOI: 10.2307/1400905. JSTOR: 1400905. (Visited on 08/04/2026).
- [11] Alexander Decruyenaere et al. ‘Challenges of Analyzing Synthetic Tabular Data Generated from 115 Phase 3 Oncology Trials’. In: *Scientific Reports* (July 2026). ISSN: 2045-2322. DOI: 10.1038/s41598-026-62212-z. (Visited on 28/07/2026).
- [12] Alexander Decruyenaere et al. ‘Debiasing Synthetic Data Generated by Deep Generative Models’. In: *Advances in Neural Information Processing Systems*. Vol. 37. Dec. 2024, pp. 41539–41576. DOI: 10.52202/079017-1315. (Visited on 08/04/2026).
- [13] Alexander Decruyenaere et al. ‘The Real Deal Behind the Artificial Appeal: Inferential Utility of Tabular Synthetic Data’. In: *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*. Vol. 244. UAI ’24. Barcelona, Spain: JMLR.org, July 2024, pp. 966–996. (Visited on 08/04/2026).
- [14] Jinshuo Dong, Aaron Roth and Weijie J. Su. ‘Gaussian Differential Privacy’. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84.1 (Jan. 2022), pp. 3–37. DOI: 10.1111/rssb.12454.
- [15] Cynthia Dwork and Jing Lei. ‘Differential Privacy and Robust Statistics’. In: *Proceedings of the Forty-First Annual ACM Symposium on Theory of Computing*. STOC ’09. New York, NY, USA: Association for Computing Machinery, May 2009, pp. 371–380. ISBN: 978-1-60558-506-2. DOI: 10.1145/1536414.1536466. (Visited on 08/04/2026).

- [16] Cynthia Dwork and Aaron Roth. ‘The Algorithmic Foundations of Differential Privacy’. In: *Foundations and Trends in Theoretical Computer Science* 9.3-4 (Aug. 2014), pp. 211–487. ISSN: 1551-305X. DOI: 10.1561/04000000042. (Visited on 08/04/2026).
- [17] Cynthia Dwork et al. ‘Analyze Gauss: Optimal Bounds for Privacy-Preserving Principal Component Analysis’. In: *Proceedings of the Forty-Sixth Annual ACM Symposium on Theory of Computing (STOC 2014)*. New York, New York, USA: ACM, 2014, pp. 11–20. DOI: 10.1145/2591796.2591883.
- [18] Bradley Efron and Carl Morris. ‘Stein’s Estimation Rule and Its Competitors-An Empirical Bayes Approach’. In: *Journal of the American Statistical Association* 68.341 (1973), pp. 117–130. ISSN: 0162-1459. DOI: 10.2307/2284155. JSTOR: 2284155.
- [19] Bradley Efron and Carl Morris. ‘Stein’s Paradox in Statistics’. In: *Scientific American* 236.5 (1977), pp. 119–127. ISSN: 0036-8733. DOI: 10.1038/scientificamerican0577-119. JSTOR: 24954030. (Visited on 05/04/2023).
- [20] Wayne A. Fuller. *Measurement Error Models*. 1st ed. Wiley Series in Probability and Statistics. New York: Wiley, June 1987. ISBN: 978-0-471-86187-4. DOI: 10.1002/9780470316665. (Visited on 15/07/2026).
- [21] C. Genest, K. Ghoudi and L.-P. Rivest. ‘A Semiparametric Estimation Procedure of Dependence Parameters in Multivariate Families of Distributions’. In: *Biometrika* 82.3 (Sept. 1995), pp. 543–552. ISSN: 0006-3444, 1464-3510. DOI: 10.1093/biomet/82.3.543. (Visited on 08/04/2026).
- [22] Sahra Ghalebikesabi et al. ‘Mitigating Statistical Bias Within Differentially Private Synthetic Data’. In: *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*. Vol. 180. UAI ’22. Eindhoven, The Netherlands: PMLR, Aug. 2022, pp. 696–705. (Visited on 08/04/2026).
- [23] Ian Goodfellow et al. ‘Generative Adversarial Nets’. In: *Advances in Neural Information Processing Systems*. Vol. 27. Curran Associates, Inc., Jan. 2014, pp. 2672–2680.
- [24] Peter Hall. *The Bootstrap and Edgeworth Expansion*. Springer Series in Statistics. New York, NY: Springer, Jan. 1992. ISBN: 978-0-387-94508-8. DOI: 10.1007/978-1-4612-4384-7. (Visited on 08/04/2026).
- [25] Peter D. Hoff. ‘Extending the Rank Likelihood for Semiparametric Copula Estimation’. In: *The Annals of Applied Statistics* 1.1 (June 2007), pp. 265–283. ISSN: 1932-6157. DOI: 10.1214/07-AOAS107.
- [26] W. James and Charles Stein. ‘Estimation with Quadratic Loss’. In: *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*. Vol. 4.1. University of California Press, Jan. 1961, pp. 361–380. (Visited on 18/11/2024).
- [27] James Jordon, Jinsung Yoon and Mihaela van der Schaar. ‘PATE-GAN: Generating Synthetic Data with Differential Privacy Guarantees’. In: *International Conference on Learning Representations*. 2019.
- [28] Peter Kairouz, Sewoong Oh and Pramod Viswanath. ‘The Composition Theorem for Differential Privacy’. In: *IEEE Transactions on Information Theory* 63.6 (June 2017), pp. 4037–4049. ISSN: 0018-9448. DOI: 10.1109/TIT.2017.2685505.



- [29] Diederik P. Kingma and Max Welling. ‘Auto-Encoding Variational Bayes’. In: *International Conference on Learning Representations*. Jan. 2014. (Visited on 08/04/2026).
- [30] Akim Kotelnikov et al. ‘TabDDPM: Modelling Tabular Data with Diffusion Models’. In: *Proceedings of the 40th International Conference on Machine Learning*. PMLR, July 2023, pp. 17564–17579. (Visited on 08/04/2026).
- [31] Roderick J. A. Little. ‘Statistical Analysis of Masked Data’. In: *Journal of Official Statistics* 9.2 (1993), pp. 407–426.
- [32] P. McCullagh and John A. Nelder. *Generalized Linear Models*. 2nd ed. Chapman and Hall/CRC, Aug. 1989. ISBN: 978-0-412-31760-6.
- [33] Peter McCullagh. ‘Quasi-Likelihood Functions’. In: *The Annals of Statistics* 11.1 (Mar. 1983), pp. 59–67. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/aos/1176346056. (Visited on 08/04/2026).
- [34] Xiao-Li Meng. ‘Multiple-Imputation Inferences with Uncongenial Sources of Input’. In: *Statistical Science* 9.4 (Jan. 1994), pp. 538–558. ISSN: 0883-4237. JSTOR: 2246252. (Visited on 08/04/2026).
- [35] Roger B. Nelsen. *An Introduction to Copulas*. Springer Science & Business Media, 2007.
- [36] Whitney K. Newey and Daniel McFadden. ‘Large Sample Estimation and Hypothesis Testing’. In: *Handbook of Econometrics*. Vol. 4. Handbooks in Economics. Amsterdam Lausanne New York: Elsevier, 1994, pp. 2111–2245. ISBN: 978-0-444-88766-5. DOI: 10.1016/S1573-4412(05)80005-4. (Visited on 04/11/2025).
- [37] Kobbi Nissim, Sofya Raskhodnikova and Adam Smith. ‘Smooth Sensitivity and Sampling in Private Data Analysis’. In: *Proceedings of the Thirty-Ninth Annual ACM Symposium on Theory of Computing*. STOC ’07. New York, NY, USA: Association for Computing Machinery, June 2007, pp. 75–84. ISBN: 978-1-59593-631-8. DOI: 10.1145/1250790.1250803. (Visited on 08/04/2026).
- [38] Beata Nowok, Gillian M. Raab and Chris Dibben. ‘Synthpop: Bespoke Creation of Synthetic Data in R’. In: *Journal of Statistical Software* 74.11 (2016), pp. 1–26. ISSN: 1548-7660. DOI: 10.18637/jss.v074.i11.
- [39] Neha Patki, Roy Wedge and Kalyan Veeramachaneni. ‘The Synthetic Data Vault’. In: *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. Montreal, QC, Canada: IEEE, Oct. 2016, pp. 399–410. ISBN: 978-1-5090-5206-6. DOI: 10.1109/DSAA.2016.49. (Visited on 08/04/2026).
- [40] Niki Z. Petrakos, Erica E. M. Moodie and Nicolas Savy. ‘A Framework for Generating Realistic Synthetic Tabular Data in a Randomized Controlled Trial Setting’. In: *Statistics in Medicine* 44.18-19 (2025), e70227. ISSN: 1097-0258. DOI: 10.1002/sim.70227. (Visited on 28/07/2026).
- [41] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria, 2026.
- [42] Trivellore E. Raghunathan, Jerome P. Reiter and Donald B. Rubin. ‘Multiple Imputation for Statistical Disclosure Limitation’. In: *Journal of Official Statistics* 19.1 (Jan. 2003), pp. 1–16.
- [43] Jerome P. Reiter. ‘Inference for Partially Synthetic, Public Use Microdata Sets’. In: *Survey Methodology* 29.2 (2003), pp. 181–188.
- [44] Jerome P. Reiter. ‘Using CART to Generate Partially Synthetic Public Use Microdata’. In: *Journal of Official Statistics* 21.3 (2005), pp. 441–462.

- [45] Donald B. Rubin. ‘Discussion: Statistical Disclosure Limitation’. In: *Journal of Official Statistics* 9.2 (1993), pp. 461–468. ISSN: 0282-423X.
- [46] Donald B. Rubin. *Multiple Imputation for Nonresponse in Surveys*. New York: Wiley, 1987.
- [47] Galit Shmueli. ‘To Explain or to Predict?’ In: *Statistical Science* 25.3 (Aug. 2010), pp. 289–310. ISSN: 0883-4237, 2168-8745. DOI: 10.1214/10-STS330. (Visited on 01/07/2025).
- [48] Reza Shokri et al. ‘Membership Inference Attacks Against Machine Learning Models’. In: *2017 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2017, pp. 3–18. DOI: 10.1109/SP.2017.41.
- [49] A. Sklar. ‘Fonctions de répartition à n dimensions et leurs marges’. Trans. by Ben Van Vliet. In: *Publications de l’Institut de Statistique de l’Université de Paris* 8 (1959), pp. 229–231.
- [50] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press, 2000. ISBN: 978-0-521-78450-4. DOI: 10.1017/CB09780511802256.
- [51] Yu-Xiang Wang. ‘Revisiting Differentially Private Linear Regression: Optimal and Adaptive Prediction & Estimation in Unbounded Domain’. In: *Proceedings of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence (UAI 2018)*. Monterey, California, USA: AUAI Press, 2018, pp. 93–103.
- [52] Larry Wasserman and Shuheng Zhou. ‘A Statistical Framework for Differential Privacy’. In: *Journal of the American Statistical Association* 105.489 (2010), pp. 375–389. DOI: 10.1198/jasa.2009.tm08651.
- [53] R. W. M. Wedderburn. ‘Quasi-Likelihood Functions, Generalized Linear Models, and the Gauss-Newton Method’. In: *Biometrika* 61.3 (Dec. 1974), pp. 439–447. ISSN: 0006-3444. DOI: 10.1093/biomet/61.3.439. (Visited on 08/04/2026).
- [54] Halbert White. ‘Maximum Likelihood Estimation of Misspecified Models’. In: *Econometrica* 50.1 (Jan. 1982), pp. 1–25. ISSN: 0012-9682. DOI: 10.2307/1912526. JSTOR: 1912526. (Visited on 08/04/2026).
- [55] Hadley Wickham. *Ggplot2: Elegant Graphics for Data Analysis*. Use R! New York: Springer, 2016. ISBN: 978-0-387-98140-6. DOI: 10.1007/978-3-319-24277-4.
- [56] Harrison Wilde et al. ‘Foundations of Bayesian Learning from Synthetic Data’. In: *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*. PMLR, Mar. 2021, pp. 541–549. (Visited on 08/04/2026).
- [57] Lei Xu et al. ‘Modeling Tabular Data Using Conditional GAN’. In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc., Jan. 2019. (Visited on 08/04/2026).
- [58] Chun Wei Yap et al. ‘Development of a Scalable and Extendable Multi-Dimensional Health Index to Measure the Health of Individuals’. In: *PLOS ONE* 15.10 (Oct. 2020), e0240302. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0240302. (Visited on 22/03/2026).

## Supplementary material

### S1. Simulation study: full coefficient tables

Tables 8 to 17 report all estimated coefficients for the calibrated copula and projection generators across the ten continuous and count simulation scenarios, and Table 18 adds the binomial scenario. Scenarios 1, 3, 4, 6, 8, and 9 are the six used in the main comparison. Scenarios 2 and 10 add a skewed predictor and a continuous-only design without changing the qualitative pattern. Scenario 7 isolates the correlated-predictor effect with an uncontaminated Gamma response, and scenario 8 combines correlated predictors with covariate-dependent zero-inflation and the ceiling. The projection achieves 0.0% maximum deviation in every continuous scenario and 2.0% for the Poisson scenario (probabilistic rounding).

Scenario 5 adds a ceiling at 100 to Scenario 4, but the largest positive response in the simulated sample is 99.6, so the ceiling does not bind and the two scenarios share identical original data, calibrated copula output, and projection output. Scenario 5 is retained for completeness.

Scenario 11 (Table 18) tests the binomial family. The empirical Bayes calibration acts on the link scale of a continuous or count response, so it does not apply to a binary outcome, and the baseline shown there is the Gaussian copula. The projection again recovers the logistic coefficients exactly.

Table 8: Scenario 1: Gaussian (identity).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 44.807   | 44.684     | −0.3  | 0.0        |
| age         | −0.562   | −0.558     | +0.8  | 0.0        |
| female1     | 4.125    | 4.057      | −1.7  | 0.0        |
| race2       | 6.867    | 6.772      | −1.4  | 0.0        |
| race3       | 12.602   | 12.290     | −2.5  | 0.0        |
| income      | 0.013    | 0.013      | +0.1  | 0.0        |

$n = 2000$ .

Table 9: Scenario 2: skewed predictor (Gaussian/identity).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 58.423   | 58.200     | −0.4  | 0.0        |
| age         | −0.535   | −0.521     | +2.6  | 0.0        |
| female1     | 4.009    | 3.912      | −2.4  | 0.0        |
| race2       | 7.085    | 7.043      | −0.6  | 0.0        |
| race3       | 12.926   | 12.674     | −1.9  | 0.0        |
| income      | 0.005    | 0.005      | −3.1  | 0.0        |

$n = 2000$ .

### S2. Six-method comparison (including CopulaGAN)

The main text reports results for three representative ML generators (CTGAN, TVAE, GaussianCopula). Table 19 includes CopulaGAN for completeness. All ML generators use SDV default hyperparameters.

CopulaGAN performs comparably to the other ML generators. On this single run the four reach at least 94% maximum deviation in all but two cells, both TVAE's (29.8% on scenario 1 and 54.4% on

Table 10: Scenario 3: Gamma (log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 3.333    | 3.283      | −1.5  | 0.0        |
| age         | −0.026   | −0.024     | +4.2  | 0.0        |
| female1     | 0.199    | 0.212      | +6.8  | 0.0        |
| race2       | 0.316    | 0.327      | +3.6  | 0.0        |
| race3       | 0.601    | 0.577      | −4.0  | 0.0        |
| income      | 0.0003   | 0.0003     | −1.3  | 0.0        |

 $n = 2000$ .

Table 11: Scenario 4: ZI quasi-gamma (log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 3.276    | 3.302      | +0.8  | 0.0        |
| age         | −0.021   | −0.018     | +16.5 | 0.0        |
| female1     | 0.143    | 0.118      | −17.5 | 0.0        |
| race2       | 0.139    | 0.131      | −5.9  | 0.0        |
| race3       | 0.488    | 0.414      | −15.1 | 0.0        |
| income      | 0.0002   | 0.0001     | −31.2 | 0.0        |

 $n = 2000$ .

Table 12: Scenario 5: ceiling effect (quasi-gamma/log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 3.276    | 3.302      | +0.8  | 0.0        |
| age         | −0.021   | −0.018     | +16.5 | 0.0        |
| female1     | 0.143    | 0.118      | −17.5 | 0.0        |
| race2       | 0.139    | 0.131      | −5.9  | 0.0        |
| race3       | 0.488    | 0.414      | −15.1 | 0.0        |
| income      | 0.0002   | 0.0001     | −31.2 | 0.0        |

 $n = 2000$ .

Table 13: Scenario 6: Poisson (log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 2.277    | 2.272      | −0.2  | +0.1       |
| age         | −0.013   | −0.012     | +5.6  | 0.0        |
| female1     | 0.097    | 0.108      | +11.2 | +2.0       |
| race2       | 0.166    | 0.134      | −19.4 | +1.3       |
| race3       | 0.297    | 0.286      | −3.9  | −0.1       |
| income      | 0.0001   | 0.0001     | −7.4  | −0.3       |

 $n = 2000$ .

Table 14: Scenario 7: correlated predictors (Gamma/log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 3.587    | 3.522      | −1.8  | 0.0        |
| age         | −0.035   | −0.034     | +3.6  | 0.0        |
| female1     | 0.209    | 0.225      | +7.6  | 0.0        |
| race2       | 0.359    | 0.373      | +4.1  | 0.0        |
| race3       | 0.696    | 0.673      | −3.3  | 0.0        |
| income      | 0.0003   | 0.0003     | −0.5  | 0.0        |

 $n = 2000$ .

Table 15: Scenario 8: all combined (quasi-gamma/log).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 3.429    | 3.360      | −2.0  | 0.0        |
| age         | −0.029   | −0.025     | +11.0 | 0.0        |
| female1     | 0.169    | 0.170      | +0.6  | 0.0        |
| race2       | 0.277    | 0.285      | +2.5  | 0.0        |
| race3       | 0.635    | 0.605      | −4.6  | 0.0        |
| income      | 0.0002   | 0.0002     | −10.9 | 0.0        |

 $n = 2000$ .

Table 16: Scenario 9: bounded [0,100] (Gaussian/identity).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 68.560   | 68.152     | −0.6  | 0.0        |
| age         | −0.967   | −1.010     | −4.4  | 0.0        |
| female1     | 5.672    | 5.460      | −3.7  | 0.0        |
| race2       | 10.521   | 10.724     | +1.9  | 0.0        |
| race3       | 19.788   | 20.140     | +1.8  | 0.0        |
| income      | 0.009    | 0.010      | +12.6 | 0.0        |

 $n = 2000$ .

Table 17: Scenario 10: Gaussian response, correlated predictors (Gaussian/identity).

| Coeff       | Original | Calibrated | % dev | Proj % dev |
|-------------|----------|------------|-------|------------|
| (Intercept) | 64.071   | 62.384     | −2.6  | 0.0        |
| age         | −0.731   | −0.646     | +11.6 | 0.0        |
| female1     | 4.306    | 4.233      | −1.7  | 0.0        |
| race2       | 7.936    | 7.704      | −2.9  | 0.0        |
| race3       | 15.019   | 14.604     | −2.8  | 0.0        |
| income      | 0.007    | 0.006      | −15.4 | 0.0        |

 $n = 2000$ .

Table 18: Scenario 11: binary response (Binomial/logit).

| Coeff       | Original | Copula | % dev | Proj % dev |
|-------------|----------|--------|-------|------------|
| (Intercept) | 1.673    | 1.470  | −12.1 | 0.0        |
| age         | −0.106   | −0.098 | +7.1  | 0.0        |
| female1     | 0.639    | 0.646  | +1.2  | 0.0        |
| race2       | 1.089    | 1.134  | +4.2  | 0.0        |
| race3       | 1.930    | 1.928  | −0.1  | 0.0        |
| income      | 0.0012   | 0.0012 | −3.8  | 0.0        |

$n = 2000$ . Calibration is undefined for a binary response, so the baseline shown is the Gaussian copula.

Table 19: Maximum absolute percentage deviation of regression coefficients across all six generators and four scenarios.

| Method         | Sc. 1  | Sc. 3  | Sc. 4  | Sc. 6  |
|----------------|--------|--------|--------|--------|
| Projection     | 0.0%   | 0.0%   | 0.0%   | 2.0%   |
| Copula         | 25.7%  | 24.2%  | 47.9%  | 23.5%  |
| CTGAN          | 120.4% | 112.7% | 210.3% | 102.7% |
| TVAE           | 29.8%  | 103.7% | 935.0% | 54.4%  |
| GaussianCopula | 110.3% | 107.6% | 234.3% | 110.6% |
| CopulaGAN      | 94.0%  | 103.2% | 338.2% | 108.2% |

scenario 6), and the 100 replications place both cells above 100% on average. The failure pattern is consistent with Proposition 13, since the deviation is largest for the rare categorical level (race3, 15% prevalence) and for the zero-inflated scenario, where the conditional structure is most complex.

### S3. Diamonds dataset: full results

The diamonds dataset from ggplot2 [55, Section 3.10] ( $n = 5,000$  subsample,  $p = 20$ ) models price as a gamma regression with log link. Table 20 reports the full coefficient comparison.

Table 20: Coefficient comparison on the diamonds dataset (gamma/log,  $n = 5,000$ ,  $p = 20$ ).

| Coefficient  | Original | Copula % | Proj % |
|--------------|----------|----------|--------|
| (Intercept)  | 5.565    | +3.2     | 0.0    |
| carat        | 2.254    | −9.6     | 0.0    |
| cutGood      | 0.004    | +1065.0  | +0.1   |
| cutVery Good | 0.036    | +105.4   | 0.0    |
| cutPremium   | 0.020    | +231.2   | 0.0    |
| cutIdeal     | 0.076    | +34.0    | 0.0    |
| colorE       | −0.045   | +61.5    | 0.0    |
| colorF       | −0.052   | +86.8    | 0.0    |
| colorG       | −0.118   | +35.3    | 0.0    |
| colorH       | −0.240   | +16.7    | 0.0    |
| colorI       | −0.406   | +33.1    | 0.0    |
| colorJ       | −0.560   | +19.6    | 0.0    |
| claritySI2   | 0.405    | −64.6    | 0.0    |
| claritySI1   | 0.590    | −49.9    | 0.0    |
| clarityVS2   | 0.684    | −43.8    | 0.0    |
| clarityVS1   | 0.742    | −43.8    | 0.0    |
| clarityVVS2  | 0.771    | −34.2    | 0.0    |
| clarityVVS1  | 0.799    | −36.1    | 0.0    |
| clarityIF    | 0.864    | −35.2    | 0.0    |
| depth        | −0.001   | +432.6   | 0.0    |
| max  %       |          | 1065.0%  | 0.1%   |

### S4. PHI application: full coefficient table

The PHI dataset ( $n = 1,927$ ,  $p = 16$ ) models waist circumference as a function of age, sex, race, education, and housing type [58]. Table 21 reports the full coefficient comparison.

The utilisation analysis of Section 6.4 models specialist-outpatient visits (util\_soc, 74% zeros) on the PHI health index and demographics with a two-part model, the structure of the source study’s

Table 21: Coefficient comparison on the PHI waist circumference model (OLS,  $n = 1,927$ ,  $p = 16$ ).

| Coefficient | Original | Copula % | Proj % |
|-------------|----------|----------|--------|
| (Intercept) | 87.230   | +3.5     | 0.0    |
| age         | 0.054    | −28.9    | 0.0    |
| female1     | −7.362   | +2.8     | 0.0    |
| race2       | 6.638    | +0.7     | 0.0    |
| race3       | 8.093    | −9.3     | 0.0    |
| race4       | 4.448    | +27.7    | 0.0    |
| edu2        | −2.722   | +21.0    | 0.0    |
| edu3        | −3.181   | −13.3    | 0.0    |
| housing2    | 0.338    | −1083.6  | 0.0    |
| housing3    | 1.746    | −121.3   | 0.0    |
| housing4    | 1.399    | −185.8   | 0.0    |
| housing5    | 1.506    | −53.0    | 0.0    |
| housing6    | 1.803    | −73.4    | 0.0    |
| housing7    | −1.908   | +184.5   | 0.0    |
| housing8    | 2.353    | −131.1   | 0.0    |
| housing9    | 2.982    | +78.6    | 0.0    |
| max  %      |          | 1083.6%  | 0.0%   |

spending model [58]. Table 22 reports the two-part coefficient comparison. The projection reproduces the logistic zero block exactly, and every positive-block coefficient within 1% except female1 at 4.0% and the sparse race2 cell (29 positive-block observations, a small coefficient), where the non-canonical log-link approximation of Theorem 6 reaches 10.4%. The copula distorts both blocks, moving the zero-model health-index coefficient by 15% and the zero-model race3 cell by 112%, without reversing any sign.

## S5. Implementation details

**Shrinkage-calibration pipeline.** The empirical Bayes shrinkage of Theorem 14 is integrated into the copula generator in five steps. (i) Compute per-stratum sample means  $\hat{\theta}_s$  and variances  $\hat{\sigma}_s^2$  on the  $h$ -scale. (ii) Estimate the between-stratum variance  $\tau^2$  by method of moments. (iii) Compute shrinkage weights  $w_s = \tau^2 / (\tau^2 + \sigma^2 / n_s)$ . (iv) Replace the per-stratum calibration targets with the shrunk estimates  $\tilde{\theta}_s = w_s \hat{\theta}_s + (1 - w_s) \bar{\theta}$ . (v) Calibrate the copula output to match  $\tilde{\theta}_s$  (not  $\hat{\theta}_s$ ) within each stratum. Without this integration, the calibration targets the noisy empirical moments, undoing the shrinkage (an order-of-operations error that produces pathological behaviour in sparse real datasets).

**Copula joint calibration.** The Gaussian copula generator models the joint distribution  $P^*(\mathbf{Z})$  symmetrically. The copula’s within-stratum correlation matrix, combined with the per-variable Cornish–Fisher quantile approximation, targets  $\Sigma_s$ . The joint calibration step applies a multivariate affine transform  $\mathbf{W}_{\text{new}}^* = A_s(\mathbf{W}^* - \hat{\mu}_s^*) + \hat{\mu}_s$  with  $A_s = \tilde{L}_s^T (L_s^*)^{-T}$ , where  $L_s^*$  and  $\tilde{L}_s$  are the upper Cholesky factors of  $\hat{\Sigma}_s^*$  and  $\tilde{\Sigma}_s$ , so that  $A_s \hat{\Sigma}_s^* A_s^T = \tilde{\Sigma}_s$  and the transform matches both the mean vector and covariance in a single operation. This is the multivariate generalisation of the location-scale calibration used for the single-response case.

Table 22: Two-part model of PHI specialist-outpatient utilisation,  $n = 1942$ , with a logistic zero model and a quasi-gamma positive block with log link. Health index is phi.

| Coefficient                                   | Original | Copula % | Proj % |
|-----------------------------------------------|----------|----------|--------|
| <i>Zero model (logistic)</i>                  |          |          |        |
| (Intercept)                                   | −1.006   | −38.3    | 0.0    |
| phi                                           | −0.016   | +14.6    | 0.0    |
| age                                           | 0.010    | +21.6    | 0.0    |
| female1                                       | 0.202    | +67.1    | 0.0    |
| race2                                         | −0.479   | +83.6    | 0.0    |
| race3                                         | 0.022    | +111.7   | 0.0    |
| race4                                         | 0.776    | −5.4     | 0.0    |
| <i>Positive block (quasi-gamma, log link)</i> |          |          |        |
| (Intercept)                                   | 1.712    | −14.8    | −0.3   |
| phi                                           | −0.005   | −6.0     | +0.8   |
| age                                           | −0.009   | +39.3    | +0.6   |
| female1                                       | −0.043   | +43.2    | +4.0   |
| race2                                         | 0.158    | −25.6    | +10.4  |
| race3                                         | −0.385   | +65.5    | +0.1   |
| race4                                         | 0.200    | −7.3     | +0.1   |
| max  %                                        |          | 111.7%   | 10.4%  |

## S6. Privacy analysis

This section develops the privacy analysis in three parts: why the Gaussian mechanism on the sufficient statistic is not viable, the empirical disclosure profile of the non-private projection (leave-one-out influence and pre-masking), and the validation, extensions, and audits of the three  $\mu$ -GDP routes of Section 5.4.

**Gaussian mechanism on the sufficient statistic.** The natural route to formal differential privacy is output perturbation: replace  $\mathbf{X}^T \mathbf{y}$  with  $\mathbf{X}^T \mathbf{y} + \boldsymbol{\eta}$ , where  $\boldsymbol{\eta} \sim N(\mathbf{0}, \sigma_{\text{DP}}^2 \mathbf{I}_p)$  and  $\sigma_{\text{DP}}$  is calibrated to the desired  $(\epsilon, \delta)$ -guarantee via the Gaussian mechanism [16]. The perturbed estimator  $\hat{\boldsymbol{\beta}}_{\text{DP}} = (\mathbf{X}^T \mathbf{X})^{-1}(\mathbf{X}^T \mathbf{y} + \boldsymbol{\eta})$  is unbiased, but its per-release deviation has standard deviation  $\sigma_{\text{DP}} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}$  for coefficient  $j$ .

This route fails in two ways, and the first is fatal before any budget is chosen. Under one-row replacement the  $L_2$  sensitivity of  $\mathbf{X}^T \mathbf{y}$  is  $\Delta_2 = 2B_x B_y$ , where  $B_x$  bounds  $\|\mathbf{x}_i\|_2$  and  $B_y$  bounds  $|y_i|$ , both fixed before the data are seen. The mechanism inverts  $\mathbf{X}^T \mathbf{X}$  in the clear, so it already presumes  $\mathbf{X}$  public and  $B_x$  is available.  $B_y$  is not. Where the response is unbounded above, as in the gamma and zero-inflated scenarios, no finite  $B_y$  exists, so  $\mathbf{X}^T \mathbf{y}$  has no finite global sensitivity and no  $\epsilon$  calibrates the mechanism at all. The route is undefined on those scenarios rather than merely imprecise, which is the obstacle Remark 19 states for the public-covariate release. Setting  $B_y = \max_i |y_i|$  would make the bounded-universe assumption true by construction on the observed sample, and deliver no guarantee at any  $\epsilon$ .

Where a public bound does exist the mechanism is defined, and it still fails, for the second reason. The noise is amplified through  $(\mathbf{X}^T \mathbf{X})^{-1}$ , so a coefficient on a rare categorical level sits on a small eigenvalue of  $\mathbf{X}^T \mathbf{X}$  and deviates by orders of magnitude more than a well-represented predictor does.



Scenario 1 admits  $B_y = 140$  by clipping the published  $N(50, 15)$  marginal at six standard deviations, and waist circumference admits the physiological bound  $B_y = 250$  cm. Rescaling every continuous covariate and the response to  $[0, 1]$  by those public ranges gives the smallest sound sensitivity available,  $\Delta_2 = 2\sqrt{1 + c + f}$  for  $c$  continuous covariates and  $f$  factors, since treatment coding activates at most one dummy per factor. Even at that best case, at  $\varepsilon = 10$  and  $\delta = 10^{-6}$ , the expected maximum percentage coefficient deviation is 484% for scenario 1 and 53,969% for the PHI application, whose sparse housing categories carry its condition number to  $4.5 \times 10^3$ . The deviation exceeds 50% in at least 99.99% of replications at every budget evaluated.

Both failures are properties of the construction evaluated above, the unregularised solve of a raw or unit-box release, rather than of a statistic release as such. The first, the missing response bound, is structural, and no rescaling removes it. The second is a matter of scale. On the unit-box scale a covariate whose public box is far wider than its dispersion collapses to a small column variance, which drives the smallest eigenvalue down. Whitening the design by the population second-moment matrix of the published data-generating process conditions  $\mathbf{X}^T \mathbf{X}$  near the identity, and the joint release of the whitened statistics solved through a public ridge penalty (Proposition 28) is evaluated below, where at the smallest budget evaluated it carries the lowest worst-case deviation of the three routes at the central box width.

The remaining obstacle reflects a genuine tension between formal differential privacy and regression inference. The coefficient-space mechanisms of Section 5.4, in particular subsample-and-aggregate [37], place the noise on the coefficient rather than the sufficient statistic and avoid the amplification. Subsample-and-aggregate also requires no bound on  $|y|$  at all, so it is the only route, and the default, wherever no public response bound exists. Propose-test-release [15] and zero-concentrated differential privacy [6] are complementary directions.

**Per-person influence (DFBETA).** Although the projection does not provide formal differential privacy, the per-person disclosure can be quantified via the leave-one-out influence [DFBETA; 4, 9]: the maximum over all observations  $i$  and coefficients  $j$  of  $|\Delta \hat{\beta}_{ij}|/|\hat{\beta}_j|$ , where  $\Delta \hat{\beta}_{ij}$  is the change in  $\hat{\beta}_j$  when observation  $i$  is removed. For our simulation scenarios ( $p = 6$ ), the maximum single-person influence is 1.1% for the Gaussian design and 1.7% for the gamma design ( $n = 2000$ ), so shifting a coefficient by 5% there needs at least 3–5 colluding individuals. In the zero-inflated design, computed on its  $n = 841$  positive subset, the maximum single-person influence is 6.8%, and a single individual suffices. For the PHI application ( $n = 1927$ ,  $p = 16$ ), sparse housing categories produce singleton cross-strata with leverage approaching 1, yielding single-person influence exceeding 1000% on housing coefficients.

**Pre-masking.** Pre-masking, the standard practice of collapsing rare categorical levels before synthesis, is effective. Reducing the 9 housing levels to 3 (two largest categories plus “other”) lowers the maximum single-person influence from 1103% to 29% and reduces the singleton cross-strata from 26 to 6. As a practical guideline, the data custodian should ensure that every cross-stratum contains at least  $k \geq 5$  observations before applying the projection. Collapsing housing alone does not reach this threshold, so a minimum stratum size enforced across all factors is needed to remove the residual singletons.

**Multi-scenario validation of the two routes.** Tables 5 and 7 are taken at scenario 1. Table 24 reports the two  $\mu$ -GDP routes across the broader scenario set at  $\varepsilon = 10$ , with the subsample-and-aggregate route at the central box width  $\kappa = 3$  of Remark 19. Subsample-and-aggregate (Proposition 26) runs on all six covered scenarios, the Gaussian designs 1, 2, 9, 10 and the log-link designs 3 (gamma) and 6 (Poisson), preserving every sign and joint decision with near-nominal coverage. The public-covariate exact release (Proposition 27) is defined only for the identity-link Gaussian scenarios (1, 2, 9, 10), where it keeps significance and coverage intact while its joint agreement sits at 84 to 88%. The PHI real-data application was calibrated under the retired local-sensitivity mechanism and is not re-run here under  $\mu$ -GDP. Extending the public-covariate route to it requires external physiological bounds on the response and is left to future work.

**The whitened sufficient-statistic release (Proposition 28).** The route releases  $\text{vech}(\mathbf{S})$ ,  $\mathbf{b}$ , and  $q$  of the proposition on the whitened scale  $\mathbf{x} \mapsto \mathbf{W}\mathbf{x}$ ,  $\mathbf{W} = \Sigma^{-1/2}$ , with  $\Sigma$  the population second-moment matrix of the published data-generating process computed from an independent draw of  $10^6$  rows at a public seed. On scenario 1 the whitening improves the condition number of  $\mathbf{X}^T\mathbf{X}$  from 171.3 on the unit-box scale to 1.106, which keeps the solve stable at the noise scales the budget dictates. The three Gaussian channels compose in one  $\mu$  quadrature. The ridge penalty is public, chosen as the minimiser of the population error curve of the published process at a fixed public seed, and the dispersion is either carried by the  $q$  channel or declared public exactly as in the subsample-and-aggregate route, the mode reported here for comparability.

The released triple carries the analyst's uncertainty as well. The ridge solve is biased toward zero, and a first-order correction  $\lambda(\mathbf{S}_{\text{DP}} + \lambda\mathbf{I})^{-1}\hat{\boldsymbol{\beta}}_w$  under-reads the shrinkage by half at  $\lambda = n$ , which the first 200-replication run exposed as 66 to 76% coverage at the strong-shrinkage cells. The calibrated interval is therefore centred at the unpenalised solve of the released statistics,  $\mathbf{S}_{\text{DP}}^{-1}\mathbf{b}_{\text{DP}}$  with eigenvalues floored at a public constant, whose plug-in covariance follows from the released matrix and the public noise scales, while the ridge solve remains the point estimate. With that centring the coverage of the nominal 95% interval lies between 94.7 and 100% over every scenario, budget, and dispersion mode of Table 23.

At  $\varepsilon = 10$  on scenario 1 the worst-case deviation is 27.9%, against 21.5% for subsample-and-aggregate at the central box  $\kappa = 3$  under the block-count rule (Table 24) and 29.2% at  $\kappa = 1.5$ , since the rule removed the gap the fixed  $m = 20$  comparison showed, and the ordering of the two routes now depends on the budget on scenarios 1, 2, and 10, the whitened release ahead at  $\varepsilon = 1$ , comparable at  $\varepsilon = 2$ , and behind at  $\varepsilon \geq 5$ . The bounded scenario 9 is the exception. Its public response bound is the structural  $[0, 100]$  range rather than a six-standard-deviation clip, so the whitened release carries less noise there and stays ahead of the rule at every budget (Table 23 against Table 24).

Two structural advantages remain: no coefficient box to choose, and the exact recovery of the released estimator with calibrated intervals.

**Full zero-inflated privacy.** The positive-component guarantee above extends to a fully private zero-inflated release. Fit a logistic zero-model for  $Z = \mathbb{I}(\mathbf{y} > 0)$  on  $\mathbf{X}$  and privatise its coefficient to  $\hat{\boldsymbol{\gamma}}_{\text{DP}}$  by a Gaussian mechanism that is  $\mu_Z$ -GDP, privatise the positive-component coefficient to

Table 23: The whitened sufficient-statistic release across the bounded-response scenarios and the budget grid (public-dispersion mode,  $\delta = 10^{-6}$ ,  $B = 200$  replications on scenario 1 and  $B = 25$  otherwise). “Max |%|” is the worst substantive-coefficient percentage deviation. “Both” and “Cov.” are joint decision agreement and calibrated 95% coverage of the recentred interval at  $\varepsilon = 10$ . The private-variance mode carries the dispersion through the  $q$  channel and costs up to three points of worst-case deviation from  $\varepsilon = 2$  (scenario 1 at  $\varepsilon = 10$ , 29.1% against 27.9%) and up to eight points at  $\varepsilon = 1$ . Joint agreement at  $\varepsilon = 1$  is 80 to 95% across the four scenarios.

| Scenario       | Max  %  deviation |                   |                   |                    | $\varepsilon = 10$ |      |
|----------------|-------------------|-------------------|-------------------|--------------------|--------------------|------|
|                | $\varepsilon = 1$ | $\varepsilon = 2$ | $\varepsilon = 5$ | $\varepsilon = 10$ | Both               | Cov. |
| Sc.1 Gaussian  | 134.2%            | 87.3%             | 49.5%             | 27.9%              | 100%               | 98%  |
| Sc.2 Gaussian  | 110.5%            | 75.8%             | 40.0%             | 22.6%              | 100%               | 97%  |
| Sc.9 Gaussian  | 75.6%             | 51.0%             | 25.2%             | 13.8%              | 100%               | 99%  |
| Sc.10 Gaussian | 116.2%            | 77.0%             | 44.6%             | 25.5%              | 100%               | 97%  |

Table 24: The two  $\mu$ -GDP routes across scenarios at  $\varepsilon = 10$ ,  $\delta = 10^{-6}$  ( $\mu = 1.848$ ),  $B = 200$  replications for scenario 1 and  $B = 25$  otherwise. Subsample-and-aggregate uses the public coefficient box at  $\kappa = 3$  and the block-count rule ( $m = 66$  at these designs). “Max |%|” is the worst substantive-coefficient percentage deviation for subsample-and-aggregate. “Both” is the mean fraction of coefficients with joint sign and significance agreement, averaged over replications, “Cov.” the calibrated 95% coverage (14). The public-covariate route is defined only for the identity-link Gaussian scenarios, marked “—” elsewhere.

| Scenario                      | Subsample-and-aggregate |      |      | Public-covariate |      |
|-------------------------------|-------------------------|------|------|------------------|------|
|                               | Max  %                  | Both | Cov. | Both             | Cov. |
| Sc.1 Gaussian ( $n/p = 333$ ) | 21.5%                   | 100% | 96%  | 85%              | 96%  |
| Sc.2 Gaussian                 | 20.4%                   | 100% | 95%  | 84%              | 96%  |
| Sc.9 Gaussian                 | 19.9%                   | 100% | 95%  | 88%              | 97%  |
| Sc.10 Gaussian                | 20.0%                   | 100% | 95%  | 85%              | 96%  |
| Sc.3 Gamma (log)              | 26.1%                   | 100% | 91%  | —                | —    |
| Sc.6 Poisson (log)            | 25.0%                   | 100% | 93%  | —                | —    |

$\mu_P$ -GDP by the same construction, sample a private synthetic indicator  $Z^* \sim \text{Bernoulli}[\text{expit}\{\mathbf{X}^* \hat{\gamma}_{\text{DP}}\}]$  coordinatewise, and project the positive component on the  $Z^* = 1$  subset to reproduce  $\hat{\beta}_{\text{DP}}$ . By Lemma 25 the release composes to  $\{\mu_Z^2 + \mu_P^2\}^{1/2}$ -GDP, and the downstream synthetic data inherits the guarantee by post-processing. The zero indicator enters the composition through the single Gaussian step  $\mu_Z$  in quadrature, so the composition is sound.

The two coefficient channels run at per-channel block counts from the public rule,  $m = 66$  for the zero-model, which is fitted on every row, and  $\lfloor n(1 - \pi_0)/(5p) \rfloor$  for the positive component, with  $\pi_0$  the published zero rate of the design, which resolves to 26 on scenario 4 and 44 on scenario 8. Table 25 reports the release on the two zero-inflated designs across the budget range, and our assessment of it is mixed.

Privately releasing the zero indicator has a cost, and its decisions are usable from moderate budgets. The private zero-model preserves 76 to 100% of the original sign decisions and 48 to 92% of the significance decisions, with both channels' agreement at 81% or above from  $\varepsilon = 5$ . At  $\varepsilon = 1$  on scenario 8 its significance agreement is 48%, so small budgets remain inadequate.

The positive-component coefficient is off by 88% on scenario 4 and 48% on scenario 8 at  $\varepsilon = 10$ , and by 621% and 348% at  $\varepsilon = 1$ , so a point-estimate user pays heavily at every budget in the grid. Completion degrades with the budget, since between 9 and 18 of the 20 replications complete without a fit failure, worst where the budget is smallest. The reference column is the retired local-sensitivity mechanism, which carries no formal guarantee at any  $\varepsilon$ . The sound release is 1.9 to 2.4 times worse than that mechanism at  $\varepsilon = 10$ , which is the price of the guarantee rather than a defect of the construction.

We conclude that a fully private zero-inflated release at  $n = 2000$  supports decision-level use from  $\varepsilon = 5$  and point estimation at none of these budgets, and we report both readings so that the cost of the zero-model channel is explicit rather than concealed by a favourable choice of box.

**Worst-case log-link privacy.** Proposition 26 gives the log-link guarantee without a local-sensitivity argument, since its  $\mu$ -GDP bound holds by one-row replacement for any link, from public per-coordinate clip boxes and the block-count rule. On the log-link scenarios, gamma (3) and Poisson (6), it preserves every sign and joint decision at  $\varepsilon \in \{5, 10\}$  with coverage 91 to 93% (Table 24), and its worst coefficient deviation at  $\varepsilon = 10$  is 25.0 to 26.1% at  $\kappa = 3$ . Every replication converged under the public box, including the count model, and at  $\varepsilon = 1$  the joint agreement falls to 72 to 81%. The route is therefore practical for both continuous-positive and count responses from  $\varepsilon = 2$ .

**Membership-inference audit.** The  $(\varepsilon, \delta)$  bound  $\text{FPR} + e^\varepsilon \text{FNR} \geq 1 - \delta$  of Remark 22 is too loose to audit against, since at  $\varepsilon = 10$  it permits an adversary accuracy of 0.99995 and so rejects nothing. We audit against the  $\mu$ -GDP trade-off directly. For a  $\mu$ -GDP mechanism the optimal distinguisher on a neighbouring pair has  $\text{AUC} \leq \Phi(\mu/\sqrt{2})$ .

Table 26 reports a membership-inference distinguishing game on a neighbouring pair for all five release routes, with  $B = 200$  releases per world and the decision taken on a bootstrap lower confidence bound, Bonferroni-corrected across the channels tested. The whitened sufficient-statistic release is audited under the released-eigenmin penalty rule with its dispersion carried privately, so all of its

Table 25: Fully private  $\mu$ -GDP zero-inflated release on the two zero-inflated designs, scenarios 4 and 8 (quasi-gamma, log link), at  $\delta = 10^{-6}$ ,  $B = 20$  Monte Carlo replications, with the budget split evenly between the zero-model and the positive component ( $\mu_Z = \mu_P = \mu/\sqrt{2}$ ). The budgets  $\varepsilon \in \{1, 2, 5, 10\}$  correspond to  $\mu \in \{0.237, 0.448, 1.020, 1.848\}$  by (13). “Pos. max |%|” is the worst positive-component coefficient deviation of the private positive-only refit against the original fit, averaged over replications. “Z sign” and “Z sig.” are the sign and significance agreement of the private logistic zero-model against the original. “Ref. max |%|” is the positive-component deviation of the retired local-sensitivity mechanism, which spends the whole budget on the positive component and carries no formal guarantee at any  $\varepsilon$ , shown to price the guarantee rather than as a competitor. “n ok” counts the replications that completed without a fit failure.

| $\varepsilon$                                           | Full $\mu$ -GDP release |        |        |       | Reference (no formal DP) |       |
|---------------------------------------------------------|-------------------------|--------|--------|-------|--------------------------|-------|
|                                                         | Pos. max  %             | Z sign | Z sig. | n ok  | Max  %                   | n ok  |
| <i>Scenario 4: zero-inflated quasi-gamma (log link)</i> |                         |        |        |       |                          |       |
| 1                                                       | 621.3%                  | 80.6%  | 63.9%  | 12/20 | 203.1%                   | 15/20 |
| 2                                                       | 319.2%                  | 90.3%  | 76.4%  | 12/20 | 139.5%                   | 18/20 |
| 5                                                       | 140.3%                  | 100.0% | 90.6%  | 16/20 | 64.6%                    | 20/20 |
| 10                                                      | 87.6%                   | 98.1%  | 89.8%  | 18/20 | 36.1%                    | 20/20 |
| <i>Scenario 8: all combined (quasi-gamma, log link)</i> |                         |        |        |       |                          |       |
| 1                                                       | 347.8%                  | 75.9%  | 48.1%  | 9/20  | 147.3%                   | 16/20 |
| 2                                                       | 185.5%                  | 89.4%  | 71.2%  | 11/20 | 94.3%                    | 18/20 |
| 5                                                       | 78.0%                   | 97.8%  | 81.1%  | 15/20 | 43.7%                    | 18/20 |
| 10                                                      | 48.3%                   | 100.0% | 91.7%  | 18/20 | 24.8%                    | 18/20 |

channels pass through the noise. Every route lies within the bound at every budget. The sensitive-covariate route’s tabulated AUC rises monotonically in  $\varepsilon$ , from 0.536 to 0.701, which is the budget acting in the expected direction.

The tabulated value is the maximum over the channels. At  $\varepsilon = 1$  that maximum is a covariate-correlation channel whose Bonferroni-corrected lower bound is at chance, and from  $\varepsilon = 2$  it is the released dispersion, whose bound exceeds chance from  $\varepsilon = 5$  (0.519, then 0.639 at  $\varepsilon = 10$ ). At  $\varepsilon = 10$  the coefficient bounds (0.521, 0.504) and the marginal-histogram bound (0.518) also exceed chance, while the correlation, level-proportion, and range channels remain at chance (0.500 to 0.501) at every budget, so the leakage the audit detects is confined to channels the mechanism noises, rising as the budget increases, and no channel that bypasses the noise fires.

Table 26: Empirical membership-inference AUC against the  $\mu$ -GDP optimal-distinguisher bound  $\Phi(\mu/\sqrt{2})$ ,  $\delta = 10^{-6}$ ,  $B = 200$  releases per world. Higher is worse. Values below the bound are consistent with the claimed guarantee.

| Route                         | $\varepsilon = 1$ | $\varepsilon = 2$ | $\varepsilon = 5$ | $\varepsilon = 10$ |
|-------------------------------|-------------------|-------------------|-------------------|--------------------|
| Public-covariate exact        | 0.520             | 0.555             | 0.520             | 0.547              |
| Subsample-and-aggregate       | 0.558             | 0.555             | 0.541             | 0.580              |
| Whitened statistics           | 0.552             | 0.527             | 0.515             | 0.546              |
| Sensitive covariates (copula) | 0.536             | 0.549             | 0.585             | 0.701              |
| Binary outcome (logistic)     | 0.536             | 0.545             | 0.544             | 0.581              |
| <i>Bound</i>                  | <i>0.566</i>      | <i>0.624</i>      | <i>0.765</i>      | <i>0.904</i>       |

Empirical audits are not a substitute for the analytic guarantee, but they detect implementation errors the analytic composition cannot see. Releasing without the variance release of Proposition 27 produces coefficient estimates that fail the audit even when the composition claims  $(\epsilon, \delta)$ -DP.

**Residual limitation.** One qualification remains, and Remark 19 states it precisely. The public-covariate exact release assumes a public bound  $B_y$  on  $|y_i|$ , the subsample-and-aggregate route assumes a public coefficient box, and the whitened release assumes a public whitening matrix with public row and response bounds. A bound read off the sample makes the noise scale a function of the data being protected, so neighbouring datasets receive different noise and no sensitivity bound holds. That is the absence of a guarantee, not a weakened one, and not “empirical” differential privacy. The release is not DP at any  $\epsilon$ .

## S7. Technical extensions of the partial moment theorem

This section develops two refinements of Theorem 6: the boundary behaviour of the Taylor remainder and a fractional moment relaxation of Assumption 2.

**Taylor boundary behaviour.** The Taylor remainder in Theorem 6 involves the factor  $|(h^{-1})^{(k+1)}(\xi)|$  where  $\xi$  lies between  $\mu_Z(\mathbf{x})$  and  $Z$  by the Lagrange form. The behaviour of this factor near support boundaries depends on the inverse link:

- (a) *Bounded  $h^{-1}$*  (e.g., expit for bounded  $[0, 1]$  support): all derivatives of expit are bounded on  $\mathbb{R}$ , so the remainder is uniformly controlled regardless of how close  $\mu$  is to the boundary.
- (b) *Exponentially growing  $h^{-1}$*  (e.g., exp for positive support):  $\exp^{(k)}(z) = \exp(z)$ , so the bound involves  $E[\exp(\mu_Z(\mathbf{X})) \nu_{k+1}(\mathbf{X})]/(k+1)!$ . The MGF condition in Assumption 2 controls this term. For exponential family models, the natural parameter space ensures finiteness [32, Section 2.2.2].
- (c) *Polynomially growing  $h^{-1}$*  (e.g.,  $\Phi^{-1}$  for probit): derivatives involve Hermite polynomials and grow polynomially. Finite moments of sufficiently high order suffice.

In all cases, the boundary issue reduces to whether  $|(h^{-1})^{(k+1)}(\xi)| \cdot E[|\tilde{Z}|^{k+1}]$  remains finite, which is controlled by Assumption 2.

**Fractional moment extension.** When the MGF does not exist and only a fractional moment is available, Assumption 2 can be replaced by the weaker condition  $E[|Z|^{k+\delta} \mid \mathbf{X} = \mathbf{x}] \leq K$  under both laws, for some  $\delta \in (0, 1]$ , provided the inverse transform has derivatives of orders  $k$  and  $k+1$  bounded on all of  $\mathbb{R}$  by  $M_k$  and  $M_{k+1}$ , as for the expit of case (a). The score bias then satisfies

$$\|b_k(\beta_0)\| \leq C \cdot E[\nu_{k+\delta}(\mathbf{X}) \cdot w(\mathbf{X}, \beta_0)],$$

where  $\nu_{k+\delta}$  is defined as in Theorem 6 with exponent  $k + \delta$  and  $C$  also carries the two derivative bounds.

A pointwise replacement of the Lagrange bound by  $|Z - \mu_Z|^{k+\delta}/(k+1)!$  fails on  $|Z - \mu_Z| > 1$ , which is the heavy-tailed regime that motivates the extension, so the proof splits the remainder at  $|Z - \mu_Z| = 1$ . Near the centre the Lagrange form gives  $|R_k| \leq M_{k+1}|Z - \mu_Z|^{k+1}/(k+1)! \leq M_{k+1}|Z - \mu_Z|^{k+\delta}/(k+1)!$ , and in the tail the order- $(k-1)$  remainder plus the  $k$ th Taylor term give  $|R_k| \leq 2M_k|Z - \mu_Z|^k/k! \leq 2M_k|Z - \mu_Z|^{k+\delta}/k!$ , so  $|R_k| \leq \max\{2M_k/k!, M_{k+1}/(k+1)!\} \cdot |Z - \mu_Z|^{k+\delta}$  everywhere. The rest proceeds identically.

This extends Theorem 6 to distributions with polynomial tails on the transformed scale, at the cost of a constant carrying the derivative bounds. For  $h = \text{logit}$  and a conditional law of  $Z$  with Pareto tails of index  $\alpha$ , the condition requires  $\alpha > k + \delta$ , which holds for  $k = 2$  whenever  $\alpha > 2 + \delta$ . Setting  $\delta = 1$  recovers the  $(k+1)$ th-moment form of Theorem 6 up to the constant.

## S8. Sparse strata extensions

This section develops three consequences of Theorem 14: the optimal shrinkage rate, an interpretation of neural network training as implicit shrinkage, and the high-dimensional strata asymptotics.

**Optimal shrinkage rate.** Under the balanced design  $n_s = n/S$  and homoscedastic  $\sigma_s^2 = \sigma^2$ , with  $\tau^2 = \sum_s \pi_s(\theta_s - \bar{\theta})^2$  the between-stratum variance, the shrinkage contribution in Theorem 14 reduces to  $(1-w)^2\tau^2 + w^2S\sigma^2/n$  over a common weight  $w$ , minimised exactly at  $w = \tau^2/(\tau^2 + S\sigma^2/n)$ , the population value of the weights in (7). The resulting MSE is

$$\mathbb{E}[\|\hat{\beta}_n^* - \beta_0\|^2] \leq 2\|A^{-1}b_k\|^2 + C_w\|A^{-1}\|^2 \frac{\tau^2 S\sigma^2/n}{\tau^2 + S\sigma^2/n} + \frac{\text{tr}(V^*)}{n} + o(n^{-1}),$$

with  $C_w$  and  $V^*$  as in Theorem 14, and the minimised contribution  $\tau^2(S\sigma^2/n)/(\tau^2 + S\sigma^2/n)$  is of the order of  $\min\{\tau^2, S\sigma^2/n\}$ . When  $\tau^2 \gg \sigma^2 S/n$  (heterogeneous strata, large  $n$ ), the weight  $w \rightarrow 1$  and the contribution is  $O(S\sigma^2/n)$ , the unshrunk rate. When  $\tau^2 \ll \sigma^2 S/n$  (homogeneous strata, small  $n$ ), the weight  $w \rightarrow 0$  and the contribution is  $O(\tau^2)$ , the bias floor from shrinking toward  $\bar{\theta}$ . The crossover occurs at  $n \sim S\sigma^2/\tau^2$ .

**Neural network training as implicit shrinkage.** A machine learning generator that learns a joint model  $P^*(\mathbf{X}, Y)$  implicitly imposes a prior on the stratum-specific parameters  $\theta_s$ . If this implicit prior is misspecified, the resulting “shrinkage” pulls estimates toward a wrong centre. The variance term of Theorem 14 still contracts by the factor  $w_s^2$ , while the bias term, now measured about the displaced centre, can grow by more than the variance falls, in which case the net MSE rises. This is the “unrelated problems” caveat of Efron and Morris [19].

The training dynamics control this implicit prior. At a fixed epoch budget, larger batch sizes take fewer gradient steps and so shrink more aggressively toward the marginal, while smaller learning rates slow convergence toward the data-specific conditional structure. Both mechanisms map onto  $w_s$  in Theorem 14, since a generator that has not fully converged acts as if  $w_s < 1$ , with the shrinkage centre displaced from  $\bar{\theta}$  toward the marginal mean  $\sum_s \pi_s \theta_s$  of  $h(Y)$ .

**High-dimensional strata.** Under the balanced design above with  $\tau^2$  fixed, when  $S$  grows with  $n$  and  $S = o(n)$ , the shrinkage weights  $w_s \rightarrow 1$  and the shrinkage contribution is  $O(S\sigma^2/n)$ , the unshrunk rate. If  $S$  grows proportionally to  $n$  (i.e.,  $n_s = O(1)$ ), the weights remain bounded away from both 0 and 1, and the shrinkage contribution to the MSE is  $O(1)$ , reflecting the impossibility of estimating per-stratum means from  $O(1)$  observations each. The sufficient statistics projection sidesteps this, since it requires only the  $p$ -dimensional sufficient statistic  $\mathbf{X}^\top \mathbf{y}$ , not the  $S$ -dimensional stratum means.

## S9. Non-elliptical error bound

Let  $C$  be the true copula of  $(W_1, \dots, W_p) \mid S = s$  and  $C_{\text{ell}}$  the best-fitting elliptical copula with the same covariance matrix  $\Sigma_s$ . If the conditional expectations under  $C$  differ from those under  $C_{\text{ell}}$  by

$$|E_C[W_j \mid W_{-j}, S = s] - E_{C_{\text{ell}}}[W_j \mid W_{-j}, S = s]| \leq \delta_s$$

for all  $j$  and  $P$ -almost every  $w_{-j}$ , then the bias bound of Theorem 15 is augmented by an additive term:

$$\|\hat{\beta}_n^{*(j)} - \beta_0^{(j)}\| \leq C_j \cdot \frac{E[\nu_{3,j}(S) w_j(S)]}{3!} + \|A^{-1}\| \cdot E\{\delta_s w_j(S) m_j(S)\} + O_p(n^{-1/2}),$$

with  $m_j(s) = 1 + E(\|\mathbf{W}_{-j}\| \mid S = s)$  carrying the stratum's predictor norm and intercept coordinate. For the Gaussian copula,  $\delta_s = 0$  exactly. More concretely, let  $\lambda_U(C)$  and  $\lambda_L(C)$  denote the upper and lower tail dependence coefficients of  $C$  [35, Definition 5.4.1]. For a bivariate margin  $(W_j, W_\ell)$  with correlation  $\rho$ , the component of the deviation driven by the joint tails is of order

$$\delta_s^{\text{tail}} = O\left(\frac{\sigma_{s,j}}{\sqrt{1-\rho^2}} \cdot \max(\lambda_U(C), \lambda_L(C)) \cdot |\tilde{\kappa}_{3,j}(s)|\right),$$

where  $\sigma_{s,j}$  is the conditional standard deviation and  $\tilde{\kappa}_{3,j}(s)$  is the standardised third cumulant. This factor is zero when  $\lambda_U = \lambda_L = 0$  (as for the Gaussian, Frank, and Plackett copulas) and grows linearly in the tail dependence coefficient for copulas with asymptotic dependence (Clayton, Gumbel, Marshall–Olkin). It does not control departures from ellipticity in the body of the distribution, which a tail-independent copula can carry, so the hypothesis on  $\delta_s$  remains the operative condition and the display calibrates only its tail-driven part.

A vine copula is the practical route to a non-zero tail dependence, and on real trial data it improves distributional realism, outperforming an adversarial network on bivariate dependence as well as on the margins [40]. Two features of that comparison bear on the constructions used here. First, its selected framework draws the covariates from the copula and every post-randomisation variable, the outcome included, from a conditional regression, which is the factorisation Theorem 4 and Definition 5 require of any generator with inferential fidelity. Second, its adversarial random forest returned values outside the attainable support, health scores off the measurement lattice and negative counts among them, so the released table needed repair before use. The support transform of Definition 3 rules that class out by construction, at the cost recorded in Remark 18 when the design Gram is pinned on the covariate side.

*Proof.* Decompose the conditional expectation under  $C$  as  $E_C[W_j \mid W_{-j}] = E_{C_{\text{ell}}}[W_j \mid W_{-j}] +$



$r_C(W_{-j})$ , where  $r_C$  is the copula residual. The score expectation under  $P^*$  acquires the additive term  $E\{\mathbf{W}_{-j} \cdot w_j(S) \cdot r_C(\mathbf{W}_{-j})\}$ , whose norm is at most  $E\{\delta_S w_j(S) m_j(S)\}$  under the hypothesis, and the  $M$ -estimator linearisation carries  $\|A^{-1}\|$  onto it. For the tail-driven component, the deviation of the copula's conditional quantile function from the elliptical conditional quantile in the joint tail scales as  $\lambda \cdot \sigma / \sqrt{1 - \rho^2}$  times the asymmetry of the marginal (the third cumulant), which gives the displayed order. Body departures are not controlled by tail coefficients and enter the hypothesis on  $\delta_s$  directly.  $\square$

## S10. High-dimensional strata simulation

To evaluate the sparse strata asymptotics of Section , we simulate a dataset with  $n = 2000$  and 7 categorical predictors (5 binary, 2 three-level), creating  $S = 233$  cross-strata with median stratum size 4 and minimum 1. This places the design in the regime where  $S$  is proportional to  $n$ .

Table 27 reports the inferential fidelity summary for the full 11-coefficient model. Table 28 gives the per-coefficient comparison. Table 29 varies the number of categorical predictors included in the stratum definition, tracing the transition from  $S = o(n)$  to  $S$  proportional to  $n$ .

Table 27: High-dimensional simulation: inferential fidelity summary ( $n = 2000, S = 233, S/n = 0.12$ ). sig is the significance agreement at 0.05, dir the sign agreement, < 5% the proportion within 5%, and SE the max standard error deviation.

| Method     | sig  | dir  | < 5% | max % | SE   |
|------------|------|------|------|-------|------|
| Copula     | 100% | 100% | 18%  | 39.7% | 8.5% |
| Calibrated | 100% | 100% | 9%   | 43.6% | 3.9% |
| Projection | 100% | 100% | 100% | 0.0%  | 2.8% |

$n = 2000$ .

Table 28: High-dimensional simulation: per-coefficient comparison ( $n = 2000, S = 233$ ).

| Coeff         | Original | Copula | % dev | Calib % dev | Proj % dev |
|---------------|----------|--------|-------|-------------|------------|
| (Intercept)   | 80.16    | 77.28  | −3.6  | −2.1        | 0.0        |
| age           | −0.33    | −0.30  | +9.3  | +7.1        | 0.0        |
| female1       | −4.62    | −3.32  | +28.1 | +25.4       | 0.0        |
| race2         | 3.67     | 3.12   | −15.1 | −20.1       | 0.0        |
| race3         | 6.28     | 5.12   | −18.4 | −26.7       | 0.0        |
| edu2          | 1.54     | 2.16   | +39.7 | −15.8       | 0.0        |
| edu3          | 4.15     | 4.14   | −0.2  | −22.3       | 0.0        |
| married1      | −1.38    | −1.07  | +22.5 | +23.9       | 0.0        |
| smoking1      | −2.82    | −2.31  | +18.2 | +21.0       | 0.0        |
| diabetes1     | −3.08    | −2.31  | +24.8 | +43.6       | 0.0        |
| hypertension1 | −2.31    | −1.65  | +28.7 | +26.1       | 0.0        |

$n = 2000$ .

Copula and calibrated generators degrade as  $S/n$  increases, from 22–25% to 40–44% max deviation. The calibrated generator worsens sharply when median  $n_s$  drops below 20, consistent with the crossover at  $n \sim S\sigma^2/\tau^2$  (Section ). The projection remains below 9% across all  $S/n$  ratios and reaches 0.0% at the highest  $S/n$ , confirming that it operates on the  $p$ -dimensional sufficient statistic rather than the  $S$ -dimensional stratum means. The residual deviation at lower  $S/n$  reflects the copula's imperfect reproduction of categorical proportions in the bootstrapped data.

Table 29: High-dimensional simulation:  $\max|\%|$  coefficient deviation as the number of strata  $S$  increases.

| $S/n$ | median $n_s$ | Copula | Calibrated | Projection |
|-------|--------------|--------|------------|------------|
| 0.001 | 1000         | 24.7%  | 22.0%      | 4.5%       |
| 0.003 | 231          | 24.2%  | 31.0%      | 5.2%       |
| 0.009 | 71           | 34.8%  | 28.1%      | 7.2%       |
| 0.018 | 38           | 34.2%  | 27.6%      | 8.6%       |
| 0.036 | 21           | 34.2%  | 26.6%      | 8.2%       |
| 0.066 | 7            | 38.5%  | 41.6%      | 3.8%       |
| 0.117 | 4            | 39.7%  | 43.6%      | 0.0%       |

$n = 2000$ . Strata formed by progressively crossing categorical predictors.

### S11. Monte Carlo all-method comparison (100 replications)

Each replication is an independent full-pipeline run, in which ML generators are re-trained from scratch (capturing SGD initialisation and minibatch variability). R generators re-sample with a new seed. Table 30 reports the maximum-deviation grid and Table 31 the significance-agreement and within-5% grids.

Table 30: Monte Carlo  $\max|\%|$  coefficient deviation: mean (sd) over 100 replications.

| Method     | Sc. 1      | Sc. 3      | Sc. 4       | Sc. 6      | Sc. 8      | Sc. 9      |
|------------|------------|------------|-------------|------------|------------|------------|
| Projection | 0.0 (0.0)  | 0.0 (0.0)  | 0.0 (0.0)   | 2.3 (1.1)  | 0.0 (0.0)  | 0.0 (0.0)  |
| Calibrated | 2.8 (0.5)  | 6.8 (2.7)  | 29.3 (9.4)  | 15.4 (7.8) | 13.7 (3.6) | 9.4 (3.3)  |
| Copula     | 14.3 (7.9) | 15.3 (7.3) | 48.8 (23.1) | 16.7 (8.1) | 21.7 (9.5) | 17.6 (8.7) |
| CTGAN      | 117 (25)   | 110 (16)   | 145 (55)    | 102 (3)    | 105 (8)    | 107 (16)   |
| TVAE       | 103 (11)   | 101 (3)    | 603 (183)   | 101 (5)    | 218 (67)   | 100 (3)    |
| GausCop    | 102 (3)    | 102 (4)    | 141 (40)    | 102 (3)    | 113 (21)   | 100 (0)    |
| CopulaGAN  | 115 (30)   | 113 (27)   | 270 (110)   | 102 (5)    | 125 (35)   | 117 (29)   |

100 replications per cell, except GaussianCopula on scenario 4, where one of the 100 quasi-gamma refits ended in a fit failure and the mean is over 99.

Table 31: Monte Carlo significance agreement (%) and proportion of coefficients within 5%: means over 100 replications.

| Method     | Significance (%) |     |     |     |     |     | < 5% (%) |     |     |     |     |     |
|------------|------------------|-----|-----|-----|-----|-----|----------|-----|-----|-----|-----|-----|
|            | 1                | 3   | 4   | 6   | 8   | 9   | 1        | 3   | 4   | 6   | 8   | 9   |
| Projection | 100              | 100 | 100 | 100 | 100 | 100 | 100      | 100 | 100 | 100 | 100 | 100 |
| Calibrated | 100              | 100 | 93  | 100 | 100 | 100 | 100      | 79  | 24  | 56  | 47  | 70  |
| Copula     | 100              | 100 | 90  | 100 | 100 | 100 | 70       | 60  | 24  | 57  | 35  | 49  |
| CTGAN      | 79               | 75  | 63  | 51  | 65  | 77  | 3        | 3   | 0   | 1   | 4   | 1   |
| TVAE       | 100              | 100 | 100 | 100 | 100 | 100 | 3        | 5   | 1   | 14  | 1   | 16  |
| GausCop    | 81               | 81  | 78  | 81  | 86  | 100 | 18       | 22  | 1   | 23  | 1   | 3   |
| CopulaGAN  | 80               | 75  | 74  | 55  | 74  | 77  | 4        | 4   | 0   | 1   | 2   | 1   |

100 replications per cell, except GaussianCopula on scenario 4, where the failed quasi-gamma refit of Table 30 leaves both means over 99.

The single-run results in Tables 2 and 4 are representative. Over the replications every ML generator has a mean maximum deviation of at least 99% in each scenario, the lowest cell being Gaussian-Copula's 99.9% on scenario 9, while the projection achieves 0.0% for continuous scenarios. CTGAN training variability reaches a standard deviation of 55% on zero-inflated data, confirming the failure

is structural rather than seed-dependent.

Table 32 measures both sides of the coupling bound of Proposition 21 over 100 replications. The per-draw inequality is asserted by the measurement script and held on every one of the 800 draws, five of the six continuous-support designs show the projected release’s energy distance to the original at or below the base copula’s, scenario 8 sits above by a tenth of the base value on the response pairs and within Monte Carlo error on the full joint, and the two discrete supports carry the structural trade-off of Remark 13.

Table 32:  $k$ -wise fidelity over 100 replications: mean absolute displacement and response-margin  $W_1$  (both in units of the original response standard deviation), and energy distances to the original data for the base copula draw and the projected release (full joint, and mean over response-involving variable pairs). The  $X$ -block pairs are identical between arms by construction and are omitted. Seeds 6490 to 6589.

| Design           | $n$  | mean disp | $W_1$ resp | $E$ joint |        | $E$ response pairs |        |
|------------------|------|-----------|------------|-----------|--------|--------------------|--------|
|                  |      |           |            | base      | proj   | base               | proj   |
| Sc. 1 gaussian   | 2000 | 0.0298    | 0.0152     | 0.0015    | 0.0014 | 0.0009             | 0.0008 |
| Sc. 3 gamma      | 2000 | 0.0385    | 0.0190     | 0.0014    | 0.0013 | 0.0009             | 0.0008 |
| Sc. 4 ZI q-gamma | 2000 | 0.0446    | 0.0228     | 0.0015    | 0.0015 | 0.0009             | 0.0009 |
| Sc. 6 poisson    | 2000 | 0.1229    | 0.1132     | 0.0014    | 0.0035 | 0.0010             | 0.0053 |
| Sc. 8 ZI q-gamma | 2000 | 0.0416    | 0.0206     | 0.0015    | 0.0016 | 0.0010             | 0.0011 |
| Sc. 9 bounded    | 2000 | 0.0394    | 0.0162     | 0.0015    | 0.0015 | 0.0010             | 0.0009 |
| Sc. 11 binomial  | 2000 | 0.6894    | 0.5333     | 0.0015    | 0.0504 | 0.0009             | 0.1844 |
| PHI waist        | 1927 | 0.0785    | 0.0273     | 0.0020    | 0.0018 | 0.0015             | 0.0012 |

Table 33 measures the design-Gram projection of Proposition 23 over the same grid (the Gram identity is re-asserted at analysis time on every pinned draw and held on all 800). Pinning closes the full-synthesis standard-error drift, to machine zero on the Gaussian-identity channels and the application and to the fixed-point tolerance on the gamma channels, while the Poisson channel stops at 0.32%, since rounding the response moves the refit weights off the pinned Gram. Per-replication random submodel refits on the Gaussian-identity channels agree to machine zero in coefficients and standard errors. The zero-block columns report the maximum absolute deviation of the refitted zero-model coefficients, the quantity the score hypothesis of Corollary 24 bounds, and they stay non-zero because that hypothesis is a bounded refinement and not an exact projection. The pinned release’s joint energy distance to the original sits below the free release’s on every design, the measured form of the distributional account in Remark 18.

**Larger-sample robustness.** To confirm that the machine learning failure is not an artefact of the  $n = 2000$  design, the four generators were re-run on scenario 1 at  $n = 20000$ , ten times the study’s sample, over three replications. The generators are high variance and this check redraws the data-generating process each replication, so it is an independent robustness check rather than an extension of the Monte Carlo table above, and the robust statistic is the smallest deviation across runs. No replication at  $n = 20000$  produced a maximum coefficient deviation below 28.7% for GaussianCopula or 35.1% for the neural generators CTGAN, TVAE, and CopulaGAN, against the 5% an analyst needs for valid inference. The tenfold larger sample does not restore inferential fidelity, consistent with the structural failure of Proposition 13.

Table 33: Design-Gram projection over 100 replications: full-synthesis standard-error deviation (% , maximum over coefficients) for the free and pinned arms, zero-block coefficient deviation (maximum absolute, two-part channels), support-violation rate of the continuous predictors (% of values, pinned arm), and joint energy distance to the original data. Coefficients stay machine-exact on the Gaussian-identity channels and the application, within solver tolerance (at most 0.11%) on the gamma and two-part channels, and keep the 2.3% rounding floor on Poisson. The exact-count permutation carries the application’s lone housing observation into every draw, so the free arm’s refit is full rank in 58 of 100 replications, the pinned arm’s in all 100. Seeds 6490 to 6589, arms paired per seed.

| Design            | SE dev (%) |                       | zero-block dev |        | viol (%) | $E$ joint |        |
|-------------------|------------|-----------------------|----------------|--------|----------|-----------|--------|
|                   | free       | pinned                | free           | pinned |          | free      | pinned |
| Sc. 1 gaussian    | 1.61       | $1.1 \times 10^{-11}$ | –              | –      | 0.006    | 0.0014    | 0.0008 |
| Sc. 3 gamma       | 3.21       | $2.6 \times 10^{-4}$  | –              | –      | 0        | 0.0013    | 0.0008 |
| Sc. 4 ZI q-gamma  | 6.29       | $2.7 \times 10^{-4}$  | 0.238          | 0.030  | 0        | 0.0015    | 0.0008 |
| Sc. 6 poisson     | 6.87       | 0.32                  | –              | –      | 0        | 0.0035    | 0.0032 |
| Sc. 7 gamma corr. | 3.63       | $2.5 \times 10^{-4}$  | –              | –      | 0        | 0.0014    | 0.0008 |
| Sc. 8 ZI q-gamma  | 4.83       | $7.3 \times 10^{-5}$  | 0.302          | 0.079  | 0        | 0.0016    | 0.0009 |
| Sc. 9 bounded     | 3.66       | $1.4 \times 10^{-11}$ | –              | –      | 0        | 0.0015    | 0.0009 |
| PHI waist         | 19.15      | $4.6 \times 10^{-11}$ | –              | –      | 0.062    | 0.0018    | 0.0009 |